STOCHASTIC CALCULUS

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Preface

These lecture notes are for the University of Cambridge Part III course Stochastic Calculus, given Lent 2017. The contents are very closely based on a set of lecture notes for this course due to Christina Goldschmidt. Please notify silvestri@mat.uniroma1.it for comments and corrections.

1. Introduction

In ordinary calculus, one learns how to integrate, differentiate, and solve ordinary differential equations. In this course, we will develop the theory for the stochastic analogs of these constructions: the Itô integral, the Itô derivative, and stochastic differential equations (SDEs).

Let us start with an example. Suppose a gambler is repeatedly tossing a fair coin, while betting £1 on head at each coin toss. Let $\xi_1, \xi_2, \xi_3 \dots$ be independent random variables denoting the consecutive outcomes, i.e.

$$\xi_k = \begin{cases} +1 & \text{if head at } k^{th} \text{ coin toss,} \\ -1 & \text{otherwise,} \end{cases} \qquad (\xi_k)_{k \geq 1} \text{ i.i.d.}$$

Then the gambler's net winnings after n coin tosses are given by $X_n := \xi_1 + \xi_2 + \ldots + \xi_n$. Note that $(X_n)_{n\geq 0}$ is a simple random walk on $\mathbb{Z}$ starting at $X_0 = 0$, and it is therefore a discrete time martingale with respect to the filtration $(\mathcal{F}_n)_{n\geq 0}$ generated by the outcomes, i.e. $\mathcal{F}_n = \sigma(\xi_1, \ldots, \xi_n)$. Now suppose that, at the mth coin toss, the gambler bets an amount H_m on head (we can also allow H_m to be negative by interpreting it as a bet on tail). Take, for now, $(H_m)_{m\geq 1}$ to be deterministic (for example, the gambler could have decided in advance the sequence of bets). Then it is easy to see that

$$(H \cdot X)_n := \sum_{k=1}^{n} H_k(X_k - X_{k-1}) \tag{1.1}$$

gives the net winnings after n coin tosses. We claim that the stochastic process $H \cdot X$ is a martingale with respect to $(\mathcal{F}_n)_{n\geq 0}$. Indeed, integrability and adaptedness are clear. Furthermore,

$$\mathbb{E}\big((H \cdot X)_{n+1} - (H \cdot X)_n \big| \mathcal{F}_n\big) = H_{n+1}\mathbb{E}(X_{n+1} - X_n|\mathcal{F}_n) = 0 \tag{1.2}$$

for all $n \geq 1$, which shows that $H \cdot X$ is a martingale. In fact, this property is much more general. Indeed, instead of deciding his bets in advance, the gambler might want to allow its $(n+1)^{th}$ bet to depend on the first n outcomes. That is, we can take H_{n+1} to be random, as long as it is $\mathcal{F}_n$-measurable. Such processes are called *previsible*, and they will play a fundamental role in this course. In this more general setting, $(H \cdot X)_n$ still represents the net amount of winnings after n bets, and (1.2) still holds, so that $H \cdot X$ is again a discrete time martingale. For this reason, the process $H \cdot X$ is usually referred to as discrete martingale transform.

Remark 1.1. *This teaches us that we cannot make money by gambling on a fair game!*

Our goal for the first part of the course is to generalise the above reasoning to the continuous time setting, i.e. to define the process

$$\int_0^t H_s dX_s$$

for $(H_t)_{t\geq 0}$ suitable previsible[1] process and $(X_t)_{t\geq 0}$ continuous martingale (think, for example, of X as being standard one-dimensional Brownian Motion, the analogue to a simple random walk in the continuum). To achieve this, as a first attempt one could try to generalise the Lebesgue-Stieltjes theory of integration to more general integrators: this will lead us to talk about finite variation processes. Unfortunately, as we will see, the only continuous martingales which have finite variation are the constant ones, so a new theory is needed in order to integrate, for example, with respect to Brownian Motion. This is what is commonly called *Itô's theory of integration*. To see how one could go to build this new theory, note that definition (1.1) is

[1]At this point we have not yet defined the notion of previsible process in the continuous time setting: this will be clarified later in the course.

perfectly well-posed in continuous time whenever the integrand H is piecewise constant taking only finitely many values. In order to accommodate more general integrands, one could then think to take limits, setting, for example,

$$(H \cdot X)_t \text{ " := " } \lim_{\varepsilon \to 0} \sum_{k=0}^{\lfloor t/\varepsilon \rfloor} H_{k\varepsilon}(X_{(k+1)\varepsilon} - X_{k\varepsilon}). \tag{1.3}$$

On the other hand, one quickly realises that the r.h.s. above in general does not converge in the usual sense, due to the roughness of X (think, for example, to Brownian Motion). What is the right notion of convergence that makes the above limit finite? As we will see in the construction of Itô integral, to get convergence one has to exploit the cancellations in the above sum. As an example, take X to be Brownian Motion, and observe that

$$\begin{aligned}
\mathbb{E}\Big[\Big(\sum_{k=0}^{\lfloor t/\epsilon \rfloor} H_{k\varepsilon}(X_{(k+1)\varepsilon} - X_{k\varepsilon})\Big)^2\Big] = \\
&= \mathbb{E}\Big[\sum_{k=0}^{\lfloor t/\varepsilon \rfloor} H_{k\varepsilon}^2 \left(X_{(k+1)\varepsilon} - X_{k\varepsilon}\right)^2 + \sum_{j \neq k}^{\lfloor t/\varepsilon \rfloor} H_{j\varepsilon} H_{k\varepsilon}(X_{(k+1)\varepsilon} - X_{k\varepsilon})(X_{(j+1)\varepsilon} - X_{j\varepsilon})\Big] \\
&= \mathbb{E}\Big[\sum_{k=0}^{\lfloor t/\varepsilon \rfloor} H_{k\varepsilon}^2 \left(X_{(k+1)\varepsilon} - X_{k\varepsilon}\right)^2\Big] = \varepsilon \sum_{k=0}^{\lfloor t/\varepsilon \rfloor} H_{k\varepsilon}^2 \to \int_0^t H_s^2 ds \quad \text{as} \quad \varepsilon \to 0,
\end{aligned}$$

where the cancellations are due to orthogonality of the martingale increments. These type of considerations will allow us to provide a rigorous construction of the Itô integral for X continuous martingale (such as Brownian Motion), and in fact even for slightly more general integrands, namely local martingales and semimartingales.

Once the stochastic integral has been defined, we will discuss its properties and applications. We will look, for example, at iterated integrals, the stochastic chain rule and stochastic integration by part. Writing, to shorten the notation,

$$dY_t = H_t dX_t$$

to mean $Y_t = \int_0^t H_s dX_s$, we will learn how to express $df(Y_t)$ in terms of dY_t by mean of the celebrated Itô's formula. This is an extremely useful tool, of which we will present several applications. For example, we will be able to show that any continuous martingale is a time-changed Brownian Motion (this is the so-called Dubins-Schwarz theorem).

In the second part of the course we will then focus on the study of Stochastic Differential Equations (SDEs in short), that is equations of the form

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dB_t, \qquad X_0 = x_0, \tag{1.4}$$

for suitably nice functions a, σ. These can be thought of as arising from randomly perturbed ODEs. To clarify this point, suppose we are given the ODE

$$\begin{cases} \dfrac{dx(t)}{dt} = b(t, x(t)), \\ x(0) = x_0, \end{cases} \tag{1.5}$$

which writes equivalently as $x(t) = x_0 + \int_0^t b(s, x(s))ds$. In many applications we may wish to perturb (1.5) by adding some noise, say Brownian noise. This gives a new (stochastic) equation

$$X_t = x_0 + \int_0^t b(s, X_s)ds + \sigma B_t,$$

where the real parameter σ controls the intensity of the noise. In fact, we may also want to allow such intensity to depend on the current time and state of the system, to get

$$X_t = x_0 + \int_0^t b(s, X_s)ds + \int_0^t \sigma(s, X_s)dB_s,$$

where the last term is an Itô integral. This, written in differential form, gives (1.4). Of such SDEs we will present several notions of solutions, and discuss under which conditions on the functions b and σ such solutions exist and are unique.

Finally, in the last part of the course we will talk about diffusion processes, which are Markov processes characterised by martingales properties. We will explain how such processes can be constructed from SDEs, and how they can be used to build solutions to certain (deterministic) PDEs involving second order elliptic operators.

2. Preliminaries

2.1. **Finite variation integrals.** Recall that a function $a\colon [0, \infty) \to \mathbb{R}$ is said to be *càdlàg* if it is right-continuous and has left limits. In other words, for all $x \in (0, \infty)$ we have both

$$\lim_{y \to x^-} a(y) \quad \text{exists} \quad \text{and} \quad \lim_{y \to x^+} a(y) = a(x).$$

We write $a(x^-)$ for the left limit at x, and let $\Delta a(x) := a(x) - a(x^-)$. Assume that a is non-decreasing and càdlàg with $a(0) = 0$. Then there exists a unique Borel measure da on $[0, \infty)$ such that for all $s < t$, we have that

$$da((s, t]) = a(t) - a(s).$$

For any $f : \mathbb{R} \to \mathbb{R}$ measurable and integrable function we can define the Lebesgue-Stieltjes integral $f \cdot a$ by setting

$$(f \cdot a)(t) = \int_{(0,t]} f(s)da(s)$$

for all $t \geq 0$. It is easy to see that $f \cdot a$ is right-continuous. Moreover, if a is continuous then $f \cdot a$ is itself continuous. In this case, we can write

$$\int_{(0,t]} f(s)da(s) = \int_0^t f(s)da(s)$$

unambiguously. We are now interested in enlarging the class of functions a against which we can integrate. A first, trivial observation is that we can use linearity to integrate against functions a which write as $a = a^+ - a^-$, for some a^+, a^- satisfying the conditions above. Then we set

$$(f \cdot a)(t) := (f \cdot a^+)(t) - (f \cdot a^-)(t)$$

for all f measurable functions and $t \geq 0$, provided both terms in the r.h.s. are finite. How large is the class of such functions? It turns out to exactly coincide with the class of finite variation functions.

Definition 2.1. *Let $a : [0, \infty) \to \mathbb{R}$ be a càdlàg function. For any $n \in \mathbb{N}$ and $t \geq 0$, let*

$$v^n(t) = \sum_{k=0}^{\lceil 2^n t \rceil - 1} |a((k+1)2^{-n}) - a(k2^{-n})|. \tag{2.1}$$

Then $v(t) := \lim_{n\to\infty} v^n(t)$ exists for all t, and it is called total variation of a on $(0, t]$. If $v(t) < \infty$ then a is said to have finite variation on $(0, t]$. We say that a is of finite variation if it is of finite variation on $(0, t]$ for all $t \geq 0$.

To show that the above definition makes sense we have to show that $\lim_{n\to\infty} v^n(t)$ exists at all t. Indeed, let $t_n^+ = 2^{-n}\lceil 2^n t \rceil$, $t_n^- = 2^{-n}(\lceil 2^n t \rceil - 1)$, and note that $t_n^+ \geq t \geq t_n^-$ for all n. We have

$$v^n(t) = \sum_{k=0}^{2^n t_n^- - 1} |a((k+1)2^{-n}) - a(k2^{-n})| + |a(t_n^+) - a(t_n^-)|.$$

Now, the first term is increasing in n by the triangle inequality, and hence it has a limit as $n \to \infty$. Moreover, by the càdlàg property the second term converges to $|\Delta a(t)|$ as $n \to \infty$. In all, $v^n(t)$ has a limit as $n \to \infty$.

Lemma 2.2. *Let $a : [0, \infty) \to \mathbb{R}$ be a càdlàg function of finite variation, that is $v(t) < \infty$ for all $t \geq 0$. Then v is also càdlàg with $\Delta v(t) = |\Delta a(t)|$, and non-decreasing in t. In particular, if a is continuous then also v is continuous.*

The proof of this result is left to the reader as an exercise.

We can now state the aforementioned characterisation result.

Proposition 2.3. *A càdlàg function $a : [0, \infty) \to \mathbb{R}$ can be written as difference of two right-continuous, non-decreasing functions if and only if a is of finite variation.*

Proof. Assume that $a = a^+ - a^-$, for a^+, a^- non-decreasing, right-continuous functions. We now show that a is of finite variation. Indeed, for any $s \leq t$ we have

$$|a(t) - a(s)| \leq (a^+(t) - a^+(s)) + (a^-(t) - a^-(s)),$$

where both terms are non-negative since a^+, a^- are non-decreasing. Plugging this into (2.1), and observing that the summations defining the total variation of a^+, a^- are telescopic, we gather that

$$v^n(t) \leq a^+(t_n^+) - a^+(0) + a^-(t_n^+) - a^-(0).$$

Being a^+, a^- càdlàg , the r.h.s. converges to $a^+(t) - a^+(0) + a^-(t) - a^-(0)$, which is finite. Since t is arbitrary, this shows that $v(t) < \infty$ for all $t \geq 0$, i.e. a is of finite variation.

We now prove the reverse implication by constructing a^+ and a^- explicitly. Assume that $v(t) < \infty$ for all $t \geq 0$. Set

$$a^+ = \frac{1}{2}(v + a)\,, \qquad a^- = \frac{1}{2}(v - a)\,.$$

Then clearly, $a = a^+ - a^-$, and a^+, a^- are càdlàg since v and a are. It remains to show that they are non-decreasing. For $s < t$, define $t_n^\pm$ as above, and $s_n^\pm$ analogously. Then we have

$$\begin{aligned} a^+(t) - a^+(s) &= \lim_{n\to\infty} \frac{1}{2}\big[(v^n(t) - v^n(s)) + (a(t) - a(s))\big] \\ &= \lim_{n\to\infty} \frac{1}{2}\Bigg[\sum_{k=2^n s_n^+}^{2^n t_n^- - 1} \Big(\underbrace{|a((k+1)2^{-n}) - a(k2^{-n})| + \big(a((k+1)2^{-n}) - a(k2^{-n})\big)}_{\geq 0} \Big) \\ &\qquad + \underbrace{|a(t_n^+) - a(t_n^-)| + (a(t_n^+) - a(t_n^-))}_{\geq 0} \Bigg] \geq 0. \end{aligned}$$

The same argument can be used to show that a^- is non-decreasing. □

2.2. **Random integrators.** We now discuss integration with respect to random finite variation functions. Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ denote a filtered probability space. Recall that a process $X : \Omega \times [0, \infty) \to \mathbb{R}$ is adapted to the filtration $(\mathcal{F}_t)_{t\geq 0}$ if $X_t = X(\cdot, t)$ is $\mathcal{F}_t$-measurable for all $t \geq 0$. Moreover, X is càdlàg if $X(\omega, \cdot)$ is càdlàg for all $\omega \in \Omega$ (that is, if all realizations of X are càdlàg).

Definition 2.4. *Given a càdlàg adapted process $A : \Omega \times [0, \infty) \to \mathbb{R}$, its total variation process $V : \Omega \times [0, \infty) \to \mathbb{R}$ is defined pathwise by letting $V(\omega, \cdot)$ be the total variation of $A(\omega, \cdot)$, for all $\omega \in \Omega$.*

We say that A is of finite variation if $A(\omega, \cdot)$ is of finite variation for all $\omega \in \Omega$ (that is, if all realizations of A are of finite variation).

Lemma 2.5. *Let A be a càdlàg , adapted process of finite variation V. Then V is càdlàg , adapted and pathwise non-decreasing.*

Proof. By Lemma 2.2, for all $\omega \in \Omega$ the path $V(\omega, \cdot)$ is càdlàg and non-decreasing, so the process V is càdlàg and pathwise non-decreasing. To see that it is adapted, recall that for any $t \geq 0$ we define $t_n^- = 2^{-n}(\lceil 2^n t \rceil - 1)$, and introduce for all n the process

$$\tilde{V}_t^n = \sum_{k=0}^{2^n t_n^- - 1} |A_{(k+1)2^{-n}} - A_{k2^{-n}}|.$$

Then $\tilde{V}^n$ is adapted for all n, since $t_n^- \leq t$. Moreover,

$$V_t = \lim_{n\to\infty} \tilde{V}_t^n + |\Delta A_t|,$$

which shows that V_t is $\mathcal{F}_t$-measurable since both terms in the r.h.s. are $\mathcal{F}_t$-measurable. □

Our next goal is to be able to integrate random functions against A. Clearly, we can define the integral pathwise, but doing so does not guarantee that the resulting process is both càdlàg and adapted. To have this, we need to restrict our class of integrands to so called previsible processes.

2.2.1. *Previsible processes.* Recall from the introduction that a discrete time process $H = (H_n)_{n\geq 1}$ is previsible with respect to a filtration $(\mathcal{F}_n)_{n\geq 0}$ if H_{n+1} is $\mathcal{F}_n$-measurable for all $n \geq 0$. Here is the analogous definition in continuous time.

Definition 2.6. *The previsible σ-algebra $\mathcal{P}$ on $\Omega \times (0, \infty)$ is the σ-algebra generated by sets of the form $E \times (s, t]$, for $E \in \mathcal{F}_s$ and $s < t$. A process $H : \Omega \times (0, \infty)$ is said to be previsible if it is measurable with respect to $\mathcal{P}$.*

It is easy to see directly from the definition that, for example, any process of the form

$$H(\omega, t) = Z(\omega)\mathbb{1}_{(t_1,t_2]}(t), \qquad Z \ \mathcal{F}_{t_1}\text{-measurable},$$

is previsible. Thus clearly if we let

$$H(\omega, t) = \sum_{k=0}^{n} Z_k(\omega)\mathbb{1}_{(t_k,t_{k+1}]}(t) \tag{2.2}$$

for some $n \in \mathbb{N}$, $0 = t_0 < t_1 \cdots < t_{n+1} < \infty$ and Z_k $\mathcal{F}_{t_k}$-measurable for all k, then H is previsible. Processes of the form (2.2) are called *simple*, and they will play an important role in the construction of the Itô integral later in the course.

Remark 2.7. *Note that all simple processes are left-continuous and adapted. In general, it is possible to show that $\mathcal{P}$ is the smallest σ-algebra on $\Omega \times (0, \infty)$ such that all left-continuous, adapted processes are measurable.*

Although not all càdlàg adapted processes are previsible, the next result shows that their left-continuous modification is.

Proposition 2.8. *Let X be a càdlàg adapted process, and define for all t the process $H_t := X_{t^-}$. Then H is previsible.*

Proof. Since X is càdlàg and adapted, H is left-continuous and adapted. Now to see that H is previsible we express it as a limit of previsible processes. For $n \geq 1$, define the left-continuous, piecewise constant process

$$H_t^n := H_{t_n^-} = \sum_{k=0}^{\infty} H_{k2^{-n}} \mathbb{1}_{(k2^{-n},(k+1)2^{-n}]}(t).$$

Then the processes H^n are clearly previsible for all $n \geq 1$. Moreover, by left-continuity of H we have that $\lim_{n\to\infty} H_t^n = \lim_{n\to\infty} H_{t_n^-} = H_t$ for all t. Thus H is a limit of previsible processes, and it is therefore previsible. □

Remark 2.9. *By the above proposition, all continuous processes (such as Brownian Motion) are previsible.*

The following result provides a useful tool to determine that a process is not previsible.

Proposition 2.10. *If a process H is previsible, then H_t is measurable with respect to $\mathcal{F}_{t^-} = \sigma(\mathcal{F}_s : s < t)$, for all $t \geq 0$.*

Proof. Exercise. □

This shows, for example, that a Poisson process $(N_t)_{t\geq 0}$ is not previsible since N_t is not $\mathcal{F}_{t^-}$-measurable.

We can now show that integrating previsible processes against càdlàg , adapted processes of finite variation results in càdlàg , adapted processes of finite variation.

Theorem 2.11. *Let $A : \Omega \times [0,\infty) \to \mathbb{R}$ be a càdlàg , adapted process of finite variation V. Let H be a previsible process, and assume that for all $\omega \in \Omega$ it holds*

$$\int_{(0,t]} |H(\omega,s)| dV(\omega,s) < \infty, \qquad \forall t > 0. \tag{2.3}$$

Then the process $(H \cdot A) : \Omega \times [0,\infty) \to \mathbb{R}$ defined by

$$(H \cdot A)(\omega,t) := \int_{(0,t]} H(\omega,s) dA(\omega,s), \qquad (H \cdot A)(\omega,0) \equiv 0 \tag{2.4}$$

is càdlàg , adapted and of finite variation.

Proof. Note that the integral in (2.4) is well defined thanks to the integrability condition (2.3). Indeed, write $H^{\pm} := \max\{\pm H, 0\}$ and $A^{\pm} = \frac{1}{2}(V \pm A)$, so that $H = H^+ - H^-$ and $A = A^+ - A^-$. Then $|H| = H^+ + H^-$ and $V = A^+ + A^-$, so that (2.3) reads

$$\begin{aligned} |H| \cdot V &= (H^+ + H^-) \cdot (A^+ + A^-) \\ &= H^+ \cdot A^+ + H^- \cdot A^+ + H^+ \cdot A^- + H^- \cdot A^- < \infty, \end{aligned}$$

which implies that all terms in the above summation are finite, since non–negative. Then

$$\begin{aligned} H \cdot A &= (H^+ - H^-) \cdot (A^+ - A^-) \\ &= H^+ \cdot A^+ - H^- \cdot A^+ - H^+ \cdot A^- + H^- \cdot A^-, \end{aligned}$$

and all terms in the r.h.s. are finite.

Step 1: Càdlàg. Note that $\mathbb{1}_{(0,s]} \to \mathbb{1}_{(0,t]}$ as $s \searrow t$ and $\mathbb{1}_{(0,s]} \to \mathbb{1}_{(0,t)}$ as $s \nearrow t$. Recall that $(H \cdot A)_t = \int H_s \mathbb{1}(s \in (0,t]) dA_s$. By the dominated convergence theorem (with ω fixed), we have that

$$\begin{aligned} (H \cdot A)_t &= \int H_s \lim_{r \searrow t} \mathbb{1}(s \in (0,r]) dA_s \\ &= \lim_{r \searrow t} \int H_s \mathbb{1}(s \in (0,r]) dA_s \\ &= \lim_{r \searrow t} (H \cdot A)_r. \end{aligned}$$

Therefore $H \cdot A$ is right-continuous. A similar argument implies that $H \cdot A$ has left limits.

Moreover, we have that

$$\Delta(H \cdot A)_t = \int H_s \mathbb{1}(s = t) dA_s = H_t \Delta A_t.$$

Step 2: Adapted. We prove adaptedness by a monotone class argument. Suppose that $H = \mathbb{1}_{B \times (s,u]}$ where $B \in \mathcal{F}_s$ and $u > s$. Then we have that

$$(H \cdot A)_t = \mathbb{1}_B \left(A_{t \wedge u} - A_{t \wedge s} \right),$$

which is clearly $\mathcal{F}_t$-measurable. Let

$$\mathcal{A} = \{C \in \mathcal{P} : \mathbb{1}_C \cdot A \quad \text{is adapted to } (\mathcal{F}_t)_{t \geq 0}\},$$

we aim to show that $\mathcal{A} = \mathcal{P}$. To this end, let

$$\Pi = \{B \times (s,u] : B \in \mathcal{F}_s,\ s < u\}.$$

Then Π is a π-system and we have shown above that $\Pi \subseteq \mathcal{A}$. To prove the reverse inclusion, observe that $\mathcal{A}$ is a d-system, so by Dynkin's lemma we have that $\mathcal{P} = \sigma(\Pi) \subseteq \mathcal{A} \subseteq \mathcal{P}$. Therefore $\mathcal{A} = \mathcal{P}$, i.e. $H \cdot A$ is adapted for all $H = \mathbb{1}_C$, $C \in \mathcal{P}$.

Now to generalise, suppose that $H \geq 0$ is previsible. Set

$$\begin{aligned} H^n &= (2^{-n}\lfloor 2^n H \rfloor) \wedge n \\ &= \sum_{k=1}^{2^n n-1} 2^{-n} k \mathbb{1}(\underbrace{H \in [2^{-n}k, 2^{-n}(k+1))}_{\in \mathcal{P}}) + n\mathbb{1}(\underbrace{H > n}_{\in \mathcal{P}}), \end{aligned}$$

and note that H^n is a finite linear combination of functions of the form $\mathbb{1}_C$ for $C \in \mathcal{P}$. It therefore follows by linearity that $(H^n \cdot A)_t$ is $\mathcal{F}_t$-measurable for all t. Moreover, by the Monotone Convergence Theorem $(H^n \cdot A)_t \to (H \cdot A)_t$ as $n \to \infty$, and hence $(H \cdot A)_t$ is $\mathcal{F}_t$-measurable for all $t \geq 0$, as limit of $\mathcal{F}_t$-measurable functions. This extends in the usual way to the case of general H by writing $H = H^+ - H^-$ and observing that $H^+ \cdot A$ and $H^- \cdot A$ are both adapted, so such is $H \cdot A$.

Step 3: Finite variation. Simply observe that

$$H \cdot A = (H^+ - H^-) \cdot (A^+ - A^-) = (H^+ \cdot A^+ + H^- \cdot A^-) - (H^- \cdot A^+ + H^+ \cdot A^-)$$

is a difference of non-decreasing processes, hence of finite variation. □

Our next goal is to integrate against continuous martingales, such as Brownian Motion. By the previous theorem, were we able to identify a class of continuous martingales with finite variations we would be done, as then we would be able to apply the definition in (2.4). Unfortunately – but also interestingly – we will see in the next section that the only continuous martingales which have finite variation are the constant ones. This shows that in order to be able to integrate with respect to general (non-constant) continuous martingales one need to develop a new theory of integration, which we discuss in Section 4.

3. Local martingales

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ be a filtered probability space.

Definition 3.1. *A filtration $(\mathcal{F}_t)_{t\geq 0}$ is said to satisfy the usual condition if*

- *$\mathcal{F}_0$ contains all $\mathbb{P}$-null sets,*
- *$(\mathcal{F}_t)_{t\geq 0}$ is right-continuous, that is for all $t \geq 0$ it holds $\mathcal{F}_t = \mathcal{F}_{t^+} := \bigcap_{s>t} \mathcal{F}_s$.*

We assume that $(\mathcal{F}_t)_{t\geq 0}$ satisfies the usual condition throughout the course.

Recall that an integrable, adapted process $X = (X_t)_{t\geq 0}$ is a martingale with respect to $(\mathcal{F}_t)_{t\geq 0}$ if for all $0 \leq s \leq t$ it holds $\mathbb{E}(X_t|\mathcal{F}_s) = X_s$ almost surely. Similarly, X is a supermartingale (resp. submartingale) if for all $0 \leq s \leq t$ it holds $\mathbb{E}(X_t|\mathcal{F}_s) \leq X_s$ (resp. $\mathbb{E}(X_t|\mathcal{F}_s) \geq X_s$) almost

surely. Moreover, a random variable $T : \Omega \to [0, \infty]$ is called a stopping time if $\{T \leq t\} \in \mathcal{F}_t$ for all $t \geq 0$. Note that if X is continuous (or càdlàg) and we let

$$\mathcal{F}_T := \{E \in \mathcal{F} : E \cap \{T \leq t\} \in \mathcal{F}_t \ \ \forall t \geq 0\}$$

then X_T is $\mathcal{F}_T$-measurable. Crucially, if X is a martingale then the stopped process $X^T := (X_{t\wedge T})_{t\geq 0}$ is again a martingale. In fact, this property can be used to characterise martingales, as shown in the next theorem.

Theorem 3.2 (Optional Stopping Theorem, OST)**.** *Let X be an adapted integrable process. Then the following are equivalent:*

(1) *X is a martingale,*
(2) *$X^T = (X_{t\wedge T})_{t\geq 0}$ is a martingale for all stopping times T,*
(3) *for all bounded stopping times $S \leq T$ it holds $\mathbb{E}(X_T|\mathcal{F}_S) = X_S$ a.s.,*
(4) *for all bounded stopping times T it holds $\mathbb{E}(X_T) = \mathbb{E}(X_0)$.*

In other words, the class of càdlàg martingales is stable under stopping. This leads us to define a slightly more general class of such processes, called local martingales.

Definition 3.3 (Local martingale)**.** *A càdlàg adapted process is called a local martingale if there exists a sequence $(T_n)_{n\geq 1}$ of stopping times with $T_n \nearrow +\infty$ almost surely, such that the stopped process X^{T_n} is a martingale for all $n \geq 1$. In this case we say that the sequence $(T_n)_{n\geq 1}$ reduces X.*

Note that any martingale is a local martingale, since any deterministic sequence $(T_n)_{n\geq 1}$ will reduce it. The following important example shows that, in fact, the class of local martingales is strictly larger.

Example 3.4. Let B be a standard Brownian motion in $\mathbb{R}^3$. Let $M_t = 1/|B_t|$. Recall from Example Sheet 3, Exercise 7.7 of Advanced Probability that:

(i) $(M_t)_{t\geq 1}$ is bounded in L^2, that is $\sup_{t\geq 1} \mathbb{E}(M_t^2) < \infty$.
(ii) $\mathbb{E}(M_t) \to 0$ as $t \to \infty$.
(iii) M is a supermartingale.

Note that M cannot be a martingale, as if it were it should have constant expectation, which would have to vanish by (ii). We now show that it is a local martingale by exhibiting a reducing sequence of stopping times.

For each n, let $T_n = \inf\{t \geq 1 : |B_t| \leq 1/n\} = \inf\{t \geq 1 : |M_t| \geq n\}$. We aim to show that $(M_t^{T_n})_{t\geq 1}$ is a martingale for all n and $T_n \nearrow \infty$ almost surely.

We begin by noting that if $n \leq M_1(\omega)$ then $T_n(\omega) = 1$ and if $n > M_1(\omega)$ then $T_n(\omega) > 1$. Moreover, since $|B_t|$ cannot hit $1/(n+1)$ without first having hit $1/n$, it follows that $T_n(\omega)$ is non-decreasing in n.

Recall from Advanced Probability that if $f \in C_b^2(\mathbb{R}^3)$ then

$$f(B_t) - f(B_0) - \frac{1}{2}\int_0^t \Delta f(B_s)ds, \quad t \geq 0,$$

is a martingale. Note that $f(x) = 1/|x|$ is harmonic in $\mathbb{R}^3 \setminus \{0\}$. To deal with the singularity at the origin, let $(f^n)_{n\geq 1}$ be a sequence of bounded $C^2(\mathbb{R}^3)$ functions such that $f^n \equiv f$ on $\{|x| \geq 1/n\}$, for all $n \geq 1$. Note that if $0 < |B_1| \leq 1/n$, then $T_n = 1$ and $M^{T_n} \equiv M_1$ is trivially a martingale. Moreover, since $B_1 \neq 0$ almost surely, we will have that $|B_1| > 1/n$ for n large enough, almost surely. Suppose therefore without loss of generality that $|B_1| > 1/n$ for all n. Then

$$f(B_{t\wedge T_n}) = f^n(B_{t\wedge T_n}) \quad \forall t \geq 1.$$

It follows that

$$\begin{aligned} M_{t\wedge T_n} &= f(B_{t\wedge T_n}) - f(B_1) + f(B_1) \\ &= \left[f(B_{t\wedge T_n}) - f(B_1) - \frac{1}{2}\int_1^{t\wedge T_n} \Delta f(B_s)ds\right] + f(B_1) \\ &= \left[f^n(B_{t\wedge T_n}) - f^n(B_1) - \frac{1}{2}\int_1^{t\wedge T_n} \Delta f^n(B_s)ds\right] + f^n(B_1), \end{aligned}$$

and so $M^{T_n} = (M_{t\wedge T_n})_{t\geq 1}$ is a martingale for all $n \geq 1$.

It only remains to show that $T_n \nearrow \infty$ almost surely as $n \to \infty$. Note that $\lim_{n\to\infty} T_n$ exists since the T_n's are non-decreasing, so we only need to show that such limit is infinite. For each $R > 0$ we let $S_R = \inf\{t \geq 1 : |B_t| > R\} = \inf\{t \geq 1 : M_t < 1/R\}$, and note that $S_R \nearrow +\infty$ almost surely as $R \to \infty$. Then clearly

$$\mathbb{P}\Big(\lim_{n\to\infty} T_n < \infty\Big) \leq \mathbb{P}\Big(\exists R : T_n < S_R \text{ for all } n\Big) = \lim_{R\to\infty}\lim_{n\to\infty} \mathbb{P}(T_n < S_R).$$

To compute the latter probability note that by the OST we have

$$\mathbb{E}(M_{T_n\wedge S_R}) = \mathbb{E}(M_1) =: \mu \in (0, \infty).$$

On the other hand, we can also rewrite the left hand side as

$$n\mathbb{P}(T_n < S_R) + \frac{1}{R}\mathbb{P}(S_R \leq T_n).$$

Using that $\mathbb{P}(S_R \leq T_n) = 1 - \mathbb{P}(T_n < S_R)$, we can solve for $\mathbb{P}(T_n < S_R)$ in the above to get that

$$\mathbb{P}(T_n < S_R) = \frac{\mu - 1/R}{n - 1/R} \to 0 \quad \text{as} \quad n \to \infty.$$

We thus conclude that $(M_t)_{t\geq 1}$ is:

- non-negative,
- a local martingale but not a martingale,
- a supermartingale, and
- L^2 bounded.

In fact, we could have concluded that M is a supermartingale from the first 2 properties, as the following result shows.

Proposition 3.5. *If X is a local martingale, and $X_t \geq 0$ for all $t \geq 0$, then X is a supermartingale.*

Proof. This follows from the conditional Fatou's lemma. Indeed, if $(T_n)_{n\geq 1}$ is a reducing sequence for X, then for any $s \leq t$ we have

$$\mathbb{E}(X_t|\mathcal{F}_s) = \mathbb{E}\Big(\liminf_{n\to\infty} X_{t\wedge T_n}|\mathcal{F}_s\Big) \leq \liminf_{n\to\infty} \mathbb{E}(X_{t\wedge T_n}|\mathcal{F}_s) = \liminf_{n\to\infty} X_{s\wedge T_n} = X_s$$

almost surely. □

Recall that our goal is to define integrals with respect to continuous martingales, such as Brownian Motion. In fact, we will be able to integrate with respect to continuous local martingales, with almost no additional effort. But why bother working with local martingales? One good reason is that, for example, this frees us from worrying about integrability, which can be difficult to check. Indeed, in Definition 3.3 the process X is only required to be integrable at stopping times. Moreover, we will often encounter stochastic processes which are only defined up to a random time τ. In this case it is not clear how to adapt the definition of martingale, since the process might fail to be defined on the whole interval $[0, \infty)$. On the other hand, it's clear what the definition of local martingale on $[0, \tau)$ should be: simply ask that $T_n \nearrow \tau$ as $n \to \infty$ in Definition 3.3.

With this in mind, we ask how big is the class of continuous local martingales with respect to the one of continuous martingales. We will then prove the main result of this section, stating that the only continuous local martingales with finite variation are the constant ones.

Definition 3.6 (Uniform integrability, UI)**.** *A set $\mathcal{X}$ of random variables is said to be uniformly integrable if*

$$\sup_{X\in\mathcal{X}} \mathbb{E}(|X|\mathbb{1}(|X| > \lambda)) \to 0 \qquad \text{as } \lambda \to \infty.$$

Remark 3.7. *Any family of uniformly bounded random variable is trivially UI. More generally, if $|X| \leq Y$ for all $X \in \mathcal{X}$ and some integrable random variable Y, then $\mathcal{X}$ is UI.*

Lemma 3.8. *If X is in $L^1(\Omega, \mathcal{F}, \mathbb{P})$, then the family*

$$\mathcal{X} = \{\mathbb{E}(X|\mathcal{G}) : \mathcal{G} \text{ is a sub } \sigma\text{-algebra}\} \qquad \text{is UI.}$$

Proof. See Example sheet 1. □

We use the above result to prove a useful criterion to determine whether a local martingale is in fact a martingale.

Proposition 3.9. *The following statements are equivalent:*

(i) *X is a martingale,*
(ii) *X is a local martingale, and for all $t \geq 0$ the family*

$$\mathcal{X}_t = \{X_T : T \text{ is a stopping time with } T \leq t\} \qquad \text{is UI.}$$

Proof. Suppose that X is a martingale. By the OST, if T is a stopping time with $T \leq t$, then we have that $X_T = \mathbb{E}(X_t \,|\, \mathcal{F}_T)$. Lemma 3.8 then implies that $\mathcal{X}_t$ is UI.

Suppose, on the other hand, that X is a local martingale and the family $\mathcal{X}_t$ is UI for all $t \geq 0$. We prove that X is a true martingale by showing that for all bounded stopping times T we have $\mathbb{E}(X_T) = \mathbb{E}(X_0)$. Indeed, let $(T_n)_{n\geq 0}$ be a reducing sequence for X. Then, for all bounded stopping times $T \leq t$,

$$\mathbb{E}(X_0) = \mathbb{E}(X_0^{T_n}) = \mathbb{E}(X_T^{T_n}) = \mathbb{E}(X_{T\wedge T_n}).$$

As $\{X_{T\wedge T_n} : n \geq 0\}$ is UI, we know from Advanced Probability that $X_{T\wedge T_n}$ converges almost surely and in L^1 as $n \to \infty$, the limit being X_T since $T_n \nearrow +\infty$. It follows that $\mathbb{E}(X_{T\wedge T_n}) \to \mathbb{E}(X_T)$ as $n \to \infty$. This shows that $\mathbb{E}(X_0) = \mathbb{E}(X_T)$ and, since T was arbitrary, that X is a martingale. □

Combining the above proposition with Remark 3.7 we obtain the following useful corollary.

Corollary 3.10. *A bounded local martingale is a true martingale. More generally, if X is a local martingale such that $|X_t| \leq Y$ for all $t \geq 0$ and some integrable random variable Y, then X is a true martingale.*

We can finally turn to the main result of this section.

Theorem 3.11. *Let X be a continuous local martingale with $X_0 = 0$, and assume that X has finite variation. Then $X \equiv 0$ almost surely.*

Proof. Let $V = (V_t)_{t\geq 0}$ denote the total variation process of X. Then $V_0 = 0$ and V is continuous, adapted and pathwise non-decreasing by Lemmas 2.2-2.5. Define the sequence of stopping times

$$T_n = \inf\{t \geq 0 : V_t = n\}, \quad n \geq 1. \tag{3.1}$$

Then $T_n \nearrow \infty$ since V is non-decreasing and finite by assumption. Moreover,

$$|X_t^{T_n}| = |X_{t\wedge T_n}| \leq |V_{t\wedge T_n}| \leq n,$$

which shows that X^{T_n} is bounded, and hence a true martingale for all $n \geq 1$. Thus $(T_n)_{n\geq 1}$ reduces X. To show that $X \equiv 0$, it clearly suffices to show that $X^{T_n} \equiv 0$ for all $n \geq 1$. Fix any $n \geq 1$ and write $Y = X^{T_n}$ to shorten the notation. Then Y is a continuous, bounded martingale with $Y_0 = 0$. By continuity, to see that $Y \equiv 0$ it suffices to show that $\mathbb{E}(Y_t^2) = 0$ for all $t \geq 0$. Fix $t > 0$ and for $N \geq 1$ write $t_k = kt/N$ for $0 \leq k \leq N$. Then we have

$$\begin{aligned}\mathbb{E}(Y_t^2) &= \mathbb{E}\bigg(\sum_{k=0}^{N-1}(Y_{t_{k+1}}^2 - Y_{t_k}^2)\bigg) = \mathbb{E}\bigg(\sum_{k=0}^{N-1}(Y_{t_{k+1}} - Y_{t_k})^2\bigg) \\ &\leq \mathbb{E}\bigg(\underbrace{\sup_{0\leq k<N}|Y_{t_{k+1}} - Y_{t_k}|}_{\leq V_t \leq n} \cdot \underbrace{\sum_{k=0}^{N-1}|Y_{t_{k+1}} - Y_{t_k}|}_{\leq V_t \leq n}\bigg),\end{aligned}$$

where the second equality follows from the orthogonality of increments, and the last from the definition of the stopping time T_n. Now, by continuity of Y we have that

$$\lim_{N\to\infty}\sup_{0\leq k<N}|Y_{t_{k+1}} - Y_{t_k}| = 0$$

almost surely. We can therefore apply the Bounded Convergence Theorem to conclude that

$$\mathbb{E}\bigg(\sup_{0\leq k<N}|Y_{t_{k+1}} - Y_{t_k}| \cdot \sum_{k=0}^{N-1}|Y_{t_{k+1}} - Y_{t_k}|\bigg) \to 0 \quad \text{as } N \to \infty,$$

and thus $\mathbb{E}(Y_t^2) = 0$ for all $t \geq 0$, which is what we wanted to show. □

Remark 3.12. *Note that:*

(1) *The continuity assumption cannot be removed, as we will see in the next section.*
(2) *The theorem shows that Brownian Motion has infinite variation, and hence the Lebesgue-Stieltjes theory of integration cannot be used to define integrals with respect to it.*

We conclude this section by observing that, although we were able to introduce an explicit reducing sequence (3.1) for X in the proof of Theorem 3.11, this was based on the total variation of X being finite, which fails to hold when dealing with non-constant continuous local martingales. The next result provides an explicit reducing sequence for this class of processes.

Proposition 3.13. *Let X be a continuous local martingale with $X_0 = 0$. For $n \geq 1$ define the stopping times*

$$T_n = \inf\{t \geq 0 : |X_t| = n\}.$$

Then the sequence $(T_n)_{n\geq 1}$ reduces X.

Note that this is the same sequence of stopping times we used to show that $M_t = 1/|B_t|$ is a local martingale in Example 3.4.

Proof. To see that $(T_n)_{n\geq 1}$ is a sequence of stopping times, note that for all $t \geq 0$ we can write

$$\{T_n \leq t\} = \Big\{ \sup_{s\in[0,t]} |X_s| \geq n \Big\} = \bigcap_{k\geq 1} \bigcup_{\substack{s\leq t \\ s\in\mathbb{Q}}} \Big\{ |X_s| > n - \frac{1}{k} \Big\} \ \in \mathcal{F}_t,$$

which shows that T_n is a stopping time for al n. We now prove that $T_n(\omega) \nearrow \infty$ for all $\omega \in \Omega$. Indeed, fix any $\omega \in \Omega$. Since $\sup_{s\in[0,t]} |X_s(\omega)| < \infty$ by continuity of X, for any $t \geq 0$ there exists $n(\omega, t)$ finite such that

$$n(\omega, t) > \sup_{s\in[0,t]} |X_s(\omega)|,$$

which implies that $T_n(\omega) > t$ for all $n \geq n(\omega, t)$, and hence that $T_n(\omega) \to +\infty$ as $n \to \infty$.

It only remains to show that $(T_n)_{n\geq 1}$ reduces X. Let $(T_n^*)_{n\geq 1}$ be a reducing sequence for X (such sequence exists by assumption, since X is a local martingale). If $T_n = T_n^*$ for all n there is nothing to prove. Otherwise, we know that $X^{T_n^*}$ is a martingale for all n, and we need to deduce that the same holds for X^{T_n}. Indeed, pick any $n \geq 1$. Then by the OST the process $X^{T_n^* \wedge T_m}$ is a martingale for all m, and hence X^{T_n} is a local martingale with reducing sequence $(T_m^*)_{m\geq 1}$. But X^{T_n} is also bounded by definition of T_n, and so it is a true martingale. Since n was arbitrary, this shows that $(T_n)_{n\geq 1}$ is a reducing sequence for X. □

4. The stochastic integral

In this section we develop a new theory of integration, which will allow us to integrate against continuous local martingales such as Brownian Motion or the process defined in Example 3.4.

Recall that a real-valued process $X = (X_t)_{t\geq 0}$ is L^2-bounded if $\sup_{t\geq 0} \|X_t\|_{L^2} < \infty$, or equivalently if $\sup_{t\geq 0} \mathbb{E}(X_t^2) < \infty$. The following two results are known to the reader from the Advanced Probability course, and they will be repeatedly used in this section.

Theorem 4.1. *Let X be a càdlàg and L^2-bounded martingale. Then there exists a random variable X_∞ such that*

$$X_t \to X_\infty \text{ almost surely and in } L^2 \text{ as } t \to \infty.$$

Moreover, $X_t = \mathbb{E}(X_\infty|\mathcal{F}_t)$ a.s. for all $t \geq 0$.

Proposition 4.2 (Doob's L^2 inequality)**.** *Let X be a càdlàg and L^2-bounded martingale. Then*

$$\mathbb{E}\Big(\sup_{t\geq 0} |X_t|^2 \Big) \leq 4\mathbb{E}(X_\infty^2).$$

Define the following sets:

$$\begin{aligned} \mathcal{M}^2 &= \big\{ L^2 \text{ bounded, càdlàg martingales} \big\}, \\ \mathcal{M}_c^2 &= \big\{ L^2 \text{ bounded, continuous martingales} \big\}, \\ \mathcal{M}_{c,loc}^2 &= \big\{ L^2 \text{ bounded, continuous local martingales} \big\}. \end{aligned} \tag{4.1}$$

Definition 4.3 (Simple process)**.** *A process $H : \Omega \times (0, \infty) \to \mathbb{R}$ is said to be a simple process if it writes as*

$$H(\omega, t) = \sum_{k=0}^{n-1} Z_k(\omega) \mathbb{1}_{(t_k, t_{k+1}]}(t)$$

for some $n \in \mathbb{N}$, $0 = t_0 < t_1 \cdots < t_n < \infty$, and Z_k bounded $\mathcal{F}_{t_k}$-measurable random variables for all k. We let $\mathcal{S}$ denote the set of simple processes.

We will start by defining integrals of simple processes against L^2 bounded, càdlàg martingales ($H \in \mathcal{S}$, $M \in \mathcal{M}^2$). To extend this definition to more general integrands we will restrict to continuous integrators ($M \in \mathcal{M}_c^2$), and equip $\mathcal{M}_c^2$ with a Hilbert space structure. We will then introduce the concept of quadratic variation, and use it to define a Hilbert space structure on a larger space of integrands (depending on the integrator), which we call $L^2(M)$. Finally, we will note that the stochastic integral defines a Hilbert space isometry between $\mathcal{M}_c^2$ and $L^2(M)$ when restricted to simple integrands, and show that asking for this property to hold for all integrands in $L^2(M)$ uniquely defines the stochastic integral on this larger space.

4.1. **Simple integrands.** In this section we define integrals of simple processes against L^2 bounded càdlàg martingales $M \in \mathcal{M}^2$. Throughout, our integrals will be over the interval $(0, t]$ and will take the value 0 at 0.

Definition 4.4. *For $H_t = \sum_{k=0}^{n-1} Z_k \mathbb{1}_{(t_k, t_{k+1}]}(t)$ simple process, and $M \in \mathcal{M}^2$, set*

$$(H \cdot M)_t := \sum_{k=0}^{n-1} Z_k (M_{t_{k+1} \wedge t} - M_{t_k \wedge t}).$$

Note that this is a continuous time version of the discrete martingale transform defined in (1.1).

The next result show that the stochastic integral of a simple process against $M \in \mathcal{M}^2$ is in $\mathcal{M}^2$.

Proposition 4.5. *If $H \in \mathcal{S}$ and $M \in \mathcal{M}^2$, then also $H \cdot M \in \mathcal{M}^2$. Moreover,*

$$\mathbb{E}[(H \cdot M)_\infty^2] = \sum_{k=0}^{n-1} \mathbb{E}[Z_k^2 (M_{t_{k+1}} - M_{t_k})^2] \leq 4 \|H\|_\infty \mathbb{E}[(M_\infty - M_0)^2]. \tag{4.2}$$

Proof. We first show that $H \cdot M$ is a martingale. For $t_k \leq s \leq t \leq t_{k+1}$, we have that $(H \cdot M)_t - (H \cdot M)_s = Z_k(M_t - M_s)$, so that

$$\mathbb{E}[(H \cdot M)_t - (H \cdot M)_s \,|\, \mathcal{F}_s] = Z_k \mathbb{E}[M_t - M_s \,|\, \mathcal{F}_s] = 0,$$

as M is a martingale. This extends to general $s \leq t$ by the tower property. Indeed, take $t_j \leq s \leq t_{j+1} \leq t_k \leq t \leq t_{k+1}$. Then we have

$$\begin{aligned}
&\mathbb{E}[(H \cdot M)_t - (H \cdot M)_s | \mathcal{F}_s] = \\
&\quad = \mathbb{E}\Big[\sum_{i=0}^{k-1} Z_i(M_{t_{i+1}} - M_{t_i}) + Z_k(M_t - M_{t_k}) - \sum_{i=0}^{j-1} Z_i(M_{t_{i+1}} - M_{t_i}) - Z_j(M_s - M_{t_j}) \Big| \mathcal{F}_s \Big] \\
&\quad = \sum_{i=j+1}^{k-1} \mathbb{E}[Z_i(M_{t_{i+1}} - M_{t_i}) | \mathcal{F}_s] + \mathbb{E}[Z_j(M_{t_{j+1}} - M_s) | \mathcal{F}_s] + \mathbb{E}[Z_k(M_t - M_{t_k}) | \mathcal{F}_s] = 0,
\end{aligned}$$

since

- $\mathbb{E}[Z_i(M_{t_{i+1}} - M_{t_i}) | \mathcal{F}_s] = \mathbb{E}[Z_i \mathbb{E}(M_{t_{i+1}} - M_{t_i} | \mathcal{F}_{t_i}) | \mathcal{F}_s] = 0$ for all $j+1 \leq i \leq k-1$,
- $\mathbb{E}[Z_j(M_{t_{j+1}} - M_s) | \mathcal{F}_s] = Z_j \mathbb{E}(M_{t_{j+1}} - M_s | \mathcal{F}_s) = 0$, and
- $E[Z_k(M_t - M_{t_k}) | \mathcal{F}_s] = \mathbb{E}[Z_k \mathbb{E}(M_t - M_{t_k} | \mathcal{F}_{t_k}) | \mathcal{F}_s] = 0$.

We thus conclude that $H \cdot M$ is a martingale. To see that $H \cdot M$ is bounded in L^2, we observe that if $j < k$ then

$$\mathbb{E}[Z_j(M_{t_{j+1}} - M_{t_j}) Z_k(M_{t_{k+1}} - M_{t_k})] = \mathbb{E}[Z_j(M_{t_{j+1}} - M_{t_j}) Z_k \mathbb{E}[M_{t_{k+1}} - M_{t_k} \,|\, \mathcal{F}_{t_k}]] = 0.$$

Therefore

$$\begin{aligned}
\mathbb{E}[(H \cdot M)_t^2] &= \mathbb{E}\left[\left(\sum_{k=0}^{n-1} Z_k(M_{t_{k+1} \wedge t} - M_{t_k \wedge t}) \right)^2 \right] \\
&= \sum_{k=0}^{n-1} \mathbb{E}\left[Z_k^2 (M_{t_{k+1} \wedge t} - M_{t_k \wedge t})^2 \right] \\
&\leq \|H\|_{L^\infty}^2 \sum_{k=0}^{n-1} \mathbb{E}[(M_{t_{k+1} \wedge t} - M_{t_k \wedge t})^2] \\
&= \|H\|_{L^\infty}^2 \mathbb{E}[(M_{t_n \wedge t} - M_0)^2] \\
&\leq 4 \|H\|_{L^\infty}^2 \mathbb{E}[(M_\infty - M_0)^2],
\end{aligned}$$

where the last line follows from Doob's L^2 inequality. Since this bound is uniform in t, we conclude that $H \cdot M$ is bounded in L^2, and so $H \cdot M \in \mathcal{M}^2$. Finally, to see that (4.2) holds, note that

$$\mathbb{E}[(H \cdot M)_\infty^2] = \lim_{t \to \infty} \mathbb{E}[(H \cdot M)_t^2] \leq \sup_{t \geq 0} \mathbb{E}[(H \cdot M)_t^2] \leq 4 \|H\|_{L^\infty}^2 \mathbb{E}[(M_\infty - M_0)^2]$$

by the previous inequality. □

In what follows it will also be useful to understand how the stochastic integral defined above behaves under stopping. The following result tells us that stopping the stochastic integral is equivalent to stopping the martingale M.

Proposition 4.6. *Let $H \in \mathcal{S}$ and $M \in \mathcal{M}^2$. Then for any stopping time T it holds*

$$H \cdot M^T = (H \cdot M)^T.$$

Proof. We have that

$$\begin{aligned}(H \cdot M^T)_t &= \sum_{k=0}^{n-1} Z_k (M^T_{t_{k+1}\wedge t} - M^T_{t_k \wedge t}) \\ &= \sum_{k=0}^{n-1} Z_k (M_{T\wedge t_{k+1}\wedge t} - M_{T\wedge t_k \wedge t}) \\ &= (H \cdot M)_{t\wedge T} = (H \cdot M)^T_t\end{aligned}$$

for any $t \geq 0$. □

4.2. **The space of integrators.** In this section we equip the space of integrators $\mathcal{M}^2_c$ with a Hilbert space structure.

For X càdlàg adapted process, define the norm

$$|\!|\!| X |\!|\!| = \|X^*\|_{L^2} \quad \text{where} \quad X^* = \sup_{t\geq 0} |X_t|,$$

and let

$$\mathcal{C}^2 = \big\{\text{càdlàg adapted processes } X \text{ with } |\!|\!| X |\!|\!| < \infty\big\}.$$

Recall from (4.1) the definition of the sets $\mathcal{M}^2$, $\mathcal{M}^2_c$, $\mathcal{M}^2_{c,loc}$, and define the following norm on $\mathcal{M}^2$

$$\|X\| := \|X_\infty\|_{L^2}.$$

This is clearly a seminorm. To see that it is a norm we have to show that $\|X\| = 0$ implies $X \equiv 0$ almost surely. Indeed, if $\|X\| = \|X_\infty\|_{L^2} = 0$ then $X_\infty = 0$ almost surely. Moreover, by Theorem 4.1 we have

$$X_t = \mathbb{E}(X_\infty | \mathcal{F}_t) = 0 \quad \text{ almost surely,}$$

for all $t \geq 0$. It then follows by continuity of X that $X \equiv 0$ almost surely.

Define further

$$\begin{aligned}\mathcal{M} &= \big\{\text{càdlàg martingales}\big\}, \\ \mathcal{M}_c &= \big\{\text{continuous martingales}\big\}, \\ \mathcal{M}_{c,loc} &= \big\{\text{continuous local martingales}\big\}.\end{aligned} \tag{4.3}$$

Proposition 4.7. *The following hold:*

(a) $(\mathcal{C}^2, |\!|\!| \cdot |\!|\!|)$ *is complete.*
(b) $\mathcal{M}^2 = \mathcal{M} \cap \mathcal{C}^2$.
(c) $(\mathcal{M}^2, \|\cdot\|)$ *is a Hilbert space and* $\mathcal{M}^2_c = \mathcal{M}_c \cap \mathcal{M}^2$ *is a closed subspace.*

(d) The map $\mathcal{M}^2 \to L^2(\mathcal{F}_\infty)$ given by $X \mapsto X_\infty$ is an isometry.

Remark 4.8. *We can identify an element of $\mathcal{M}^2$ with its terminal value and then $(\mathcal{M}^2, \|\cdot\|)$ inherits the Hilbert space structure of $(L^2(\mathcal{F}_\infty), \|\cdot\|_{L^2})$. Moreover, since $(\mathcal{M}^2_c, \|\cdot\|)$ is a closed linear subspace of $\mathcal{M}^2$ by (c), it is itself a Hilbert space: this will be our space of integrators.*

Proof. To prove (a) we have to show that any Cauchy sequence in $(\mathcal{C}^2, \|\!|\cdot|\!\|)$ converges to an element of $\mathcal{C}^2$. To this end, we will first show that a càdlàg subsequential limit exists for any such sequence, and then prove that in fact the whole sequence converges to such limit in $\|\!|\cdot|\!\|$ norm.

Let $(X^n)_{n\geq 1}$ be a Cauchy sequence in $(\mathcal{C}^2, \|\!|\cdot|\!\|)$. Then we can find a subsequence $(X^{n_k})_{k\geq 1}$ such that $\sum_k \|\!|X^{n_{k+1}} - X^{n_k}|\!\| < \infty$. It follows that

$$\left\| \sum_k \sup_{t\geq 0} |X_t^{n_{k+1}} - X_t^{n_k}| \right\|_{L^2} \leq \sum_k \|\!|X^{n_{k+1}} - X^{n_k}|\!\| < \infty.$$

Therefore $\sum_k \sup_{t\geq 0} |X_t^{n_{k+1}} - X_t^{n_k}| < \infty$ almost surely, and hence $\sup_{t\geq 0} |X_t^{n_{k+1}} - X_t^{n_k}| \to 0$ as $k \to \infty$ almost surely. This shows that $(X^{n_k}(\omega))_{k\geq 1}$ converges uniformly on $[0,\infty)$ as $k \to \infty$ to some limit, which must then be càdlàg as uniform limit of càdlàg functions.

Let X denote such limit: it remains to show that $X^n \to X$ in $\|\!|\cdot|\!\|$ norm. Indeed,

$$\begin{aligned}
\|\!|X^n - X|\!\|^2 &= \mathbb{E}\left(\sup_{t\geq 0} |X_t^n - X_t|^2 \right) \\
&\leq \mathbb{E}\left(\lim_{k\to\infty} \sup_{t\geq 0} |X_t^n - X_t^{n_k}|^2 \right) \\
&\leq \liminf_{k\to\infty} \mathbb{E}\left(\sup_{t\geq 0} |X_t^n - X_t^{n_k}|^2 \right) \quad \text{(Fatou's lemma)} \\
&= \liminf_{k\to\infty} \|\!|X^n - X^{n_k}|\!\|^2 \to 0 \quad \text{as} \quad n \to \infty
\end{aligned}$$

as (X^n) is Cauchy. This proves (a).

To see (b) we show that both inclusions hold. Suppose that $X \in \mathcal{C}^2 \cap \mathcal{M}$. Then $\|\!|X|\!\| < \infty$, and so

$$\sup_{t\geq 0} \|X_t\|_{L^2} \leq \left\| \sup_{t\geq 0} |X_t| \right\|_{L^2} = \|\!|X|\!\|.$$

Therefore $X \in \mathcal{M}^2$. On the other hand, if $X \in \mathcal{M}^2$, by Doob's L^2-inequality we find

$$\|\!|X|\!\| \leq 2\|X_\infty\|_{L^2} < \infty,$$

so $X \in \mathcal{C}^2 \cap \mathcal{M}$. Hence $\mathcal{M}^2 = \mathcal{M} \cap \mathcal{C}^2$, which proves (b).

We now prove (c).

Note that $(X,Y) \mapsto \mathbb{E}(X_\infty Y_\infty)$ defines an inner product on $\mathcal{M}^2$. For $X \in \mathcal{M}^2$, we have just shown that

$$\|X\| \leq |\!|\!| X |\!|\!| \leq 2\|X\|,$$

so $\|\cdot\|$ and $|\!|\!|\cdot|\!|\!|$ are equivalent norms. Therefore, in order to show that $(\mathcal{M}^2, \|\cdot\|)$ is complete we can equivalently show that $(\mathcal{M}^2, |\!|\!|\cdot|\!|\!|)$ is complete. By (a), it suffices to show that $\mathcal{M}^2$ is closed in $(\mathcal{C}^2, |\!|\!|\cdot|\!|\!|)$. To see this, let $(X^n)_{n\geq 1}$ be a sequence in $\mathcal{M}^2$ such that $|\!|\!| X^n - X |\!|\!| \to 0$ as $n \to \infty$ for some X. Then X is càdlàg , adapted and bounded in L^2. To see that X is also a martingale, note that for $s < t$

$$\begin{aligned}
\big\|\mathbb{E}(X_t \,|\, \mathcal{F}_s) - X_s\big\|_{L^2} &\leq \big\|\mathbb{E}(X_t - X_t^n \,|\, \mathcal{F}_s) + X_s^n - X_s\big\|_{L^2} \\
&\leq \big\|\mathbb{E}(X_t - X_t^n \,|\, \mathcal{F}_s)\big\|_{L^2} + \big\|X_s^n - X_s\big\|_{L^2} \quad \text{(Minkowski's inequality)} \\
&\leq \big\|X_t - X_t^n\big\|_{L^2} + \big\|X_s^n - X_s\big\|_{L^2} \\
&\leq 2|\!|\!| X^n - X |\!|\!| \to 0 \quad \text{as} \quad n \to \infty.
\end{aligned}$$

Therefore $X \in \mathcal{M}^2$, which is therefore closed. Exactly the same argument applies to show that $\mathcal{M}_c^2$ is a closed subspace of $(\mathcal{M}^2, \|\cdot\|)$. This proves (c).

Part (d) follows from the definition. □

In this section we have equipped our space of integrators with a Hilbert space structure. In the next section we do the same for a suitable space of integrands.

4.3. **The space of integrands.** The crucial tool that will allow us to equip the space of integrands with a Hilbert space structure is the so-called quadratic variation, that we now discuss in detail.

Definition 4.9 (Ucp convergence)**.** *Let $(X^n)_{n\geq 1}$ be a sequence of processes. We say that $X^n \to X$ uniformly on compacts in probability (ucp) if for every $\epsilon, t > 0$ we have that*

$$\mathbb{P}\Big(\sup_{s\leq t} |X_s^n - X_s| > \epsilon\Big) \to 0 \quad \textit{as} \quad n \to \infty.$$

The following result defines the quadratic variation of a continuous local martingale.

Theorem 4.10 (Quadratic variation)**.** *Let M be a continuous local martingale. Then there exists a unique (up to indistinguishability) continuous, adapted and non-decreasing process $[M]$ such that $[M]_0 = 0$ and $M^2 - [M]$ is a continuous, local martingale. Moreover, if we define*

$$[M]_t^n = \sum_{k=0}^{\lceil 2^n t\rceil - 1} (M_{(k+1)2^{-n}} - M_{k2^{-n}})^2$$

then $[M]^n \to [M]$ ucp as $n \to \infty$. The process $[M]$ is called the quadratic variation of M.

Proof of Theorem 4.10. By replacing M with $M_t - M_0$ if necessary, we may assume without loss of generality that $M_0 = 0$.

Step 1: Uniqueness. If A and A' are two increasing processes satisfying the conditions of the theorem, then we can write

$$A_t - A'_t = (M_t^2 - A'_t) - (M_t^2 - A_t).$$

The left hand side is a continuous process of bounded variation as it is given by the difference of two non-decreasing functions while the right hand side is a continuous local martingale as it is given by the difference of two continuous local martingales. Since a continuous local martingale starting from 0 with bounded variation is almost surely equal to 0, it follows that $A - A' \equiv 0$. That is, $A \equiv A'$, as desired.

Step 2: Existence, M bounded. We shall first consider the case that M is a bounded, continuous martingale. Then we have that $M \in \mathcal{M}_c^2$. We first build the quadratic variation process on compact time sets $[0, T]$, for T deterministic and finite. Fix such $T > 0$ and let

$$H_t^n = \sum_{k=0}^{\lceil 2^n T \rceil - 1} M_{k2^{-n}} \mathbb{1}_{(k2^{-n}, (k+1)2^{-n}]}(t).$$

Note that H^n approximates M^T for large n. Then, since H^n is a simple process for all n, we have that

$$X_t^n = (H^n \cdot M)_t = \sum_{k=0}^{\lceil 2^n T \rceil - 1} M_{k2^{-n}} (M_{(k+1)2^{-n} \wedge t} - M_{k2^{-n} \wedge t})$$

is an L^2-bounded continuous martingale, that is $X^n \in \mathcal{M}_c^2$ for all n. We next show that the sequence $(X^n)_{n \geq 1}$ is Cauchy in $\| \cdot \|$ norm, and therefore it has a limit in $\mathcal{M}_c^2$. Pick $n \geq m \geq 1$ and write H in place of $H^n - H^m$ to shorten the notation. Then

$$X^n - X^m = (H^n - H^m) \cdot M = H \cdot M,$$

and so

$$\begin{aligned}
\|X^n - X^m\|^2 &= \mathbb{E}[(H\cdot M)_\infty^2] = \mathbb{E}[(H\cdot M)_T^2] \\
&= \mathbb{E}\Big[\Big(\sum_{k=0}^{\lceil 2^n T\rceil -1} H_{k2^{-n}}(M_{(k+1)2^{-n}} - M_{k2^{-n}})\Big)^2\Big] \\
&= \mathbb{E}\Big[\sum_{k=0}^{\lceil 2^n T\rceil -1} H_{k2^{-n}}^2(M_{(k+1)2^{-n}} - M_{k2^{-n}})^2\Big] \\
&\le \mathbb{E}\Big[\sup_{t\in[0,T]}|H_t|^2 \sum_{k=0}^{\lceil 2^n T\rceil -1}(M_{(k+1)2^{-n}} - M_{k2^{-n}})^2\Big] \\
&\le \mathbb{E}\Big[\Big(\sup_{t\in[0,T]}|H_t|^2\Big)^2\Big]^{1/2}\mathbb{E}\Big[\Big(\sum_{k=0}^{\lceil 2^n T\rceil -1}(M_{(k+1)2^{-n}} - M_{k2^{-n}})^2\Big)^2\Big]^{1/2} \\
&= \mathbb{E}\Big[\sup_{t\in[0,T]}|H_t^n - H_t^m|^4\Big]^{1/2}\mathbb{E}\Big[\Big(\sum_{k=0}^{\lceil 2^n T\rceil -1}(M_{(k+1)2^{-n}} - M_{k2^{-n}})^2\Big)^2\Big]^{1/2},
\end{aligned}$$

where we have used Cauchy-Schwarz in the last inequality. Note that the first term is bounded by

$$\sup_{t\in[0,T]}|H_t^n - H_t^m|^4 \le 16\|M\|_{L^\infty}^4$$

since M is bounded, and

$$\sup_{t\in[0,T]}|H_t^n - H_t^m|^4 \to 0 \text{ almost surely as } n,m\to\infty$$

by uniform continuity of M on $[0,T]$. It therefore follows from the Bounded Convergence Theorem that

$$\mathbb{E}\Big[\sup_{t\in[0,T]}|H_t^n - H_t^m|^4\Big]^{1/2} \to 0$$

as $n,m\to\infty$. We claim that the second term is bounded since M is.

Lemma 4.11. *Let $M\in\mathcal{M}$ be bounded. Then, for any $N\ge 1$ and $0=t_0\le t_1\le\ldots\le t_N<\infty$ we have that*

$$\mathbb{E}\Big[\Big(\sum_{k=0}^{N-1}(M_{t_{k+1}} - M_{t_k})^2\Big)^2\Big] \le 48\|M\|_{L^\infty}^4.$$

Proof. Note that

$$\begin{aligned}
&\mathbb{E}\Big[\Big(\sum_{k=0}^{N-1}(M_{t_{k+1}} - M_{t_k})^2\Big)^2\Big] = \\
&\qquad = \sum_{k=0}^{N-1}\mathbb{E}[(M_{t_{k+1}} - M_{t_k})^4] + 2\sum_{k=0}^{N-1}\mathbb{E}\Big[(M_{t_{k+1}} - M_{t_k})^2\sum_{j=k+1}^{N-1}(M_{t_{j+1}} - M_{t_j})^2\Big].
\end{aligned} \tag{4.4}$$

For each fixed k, we have

$$\begin{aligned}\mathbb{E}\Big[(M_{t_{k+1}}-M_{t_k})^2\sum_{j=k+1}^{N-1}(M_{t_{j+1}}-M_{t_j})^2\Big] &= \\ &= \mathbb{E}\Big[(M_{t_{k+1}}-M_{t_k})^2\,\mathbb{E}\Big(\sum_{j=k+1}^{N-1}(M_{t_{j+1}}-M_{t_j})^2\Big|\mathcal{F}_{t_{k+1}}\Big)\Big] \\ &= \mathbb{E}\Big[(M_{t_{k+1}}-M_{t_k})^2\,\mathbb{E}\Big[\Big(\sum_{j=k+1}^{N-1}(M_{t_{j+1}}-M_{t_j})\Big)^2\Big|\mathcal{F}_{t_{k+1}}\Big)\Big] \\ &= \mathbb{E}\big[(M_{t_{k+1}}-M_{t_k})^2\mathbb{E}[(M_{t_N}-M_{t_{k+1}})^2|\mathcal{F}_{t_{k+1}}]\big] \\ &= \mathbb{E}\big[(M_{t_{k+1}}-M_{t_k})^2(M_{t_N}-M_{t_{k+1}})^2\big]\end{aligned}$$

Inserting this into (4.4) we then get

$$\begin{aligned}\mathbb{E}\Big[\Big(\sum_{k=0}^{N-1}(M_{t_{k+1}}-M_{t_k})^2\Big)^2\Big] &\le \\ &\le \mathbb{E}\Big[\Big(\sup_{0\le j\le N-1}|M_{t_{j+1}}-M_{t_j}|^2+2\sup_{0\le j\le N-1}|M_{t_N}-M_{t_j}|^2\Big)\sum_{k=0}^{N-1}(M_{t_{k+1}}-M_{t_k})^2\Big] \\ &\le 12\|M\|_{L^\infty}^2\,\mathbb{E}\Big[\sum_{k=0}^{N-1}(M_{t_{k+1}}-M_{t_k})^2\Big] \\ &= 12\|M\|_{L^\infty}^2\,\mathbb{E}[(M_{t_N}-M_{t_0})^2] \\ &\le 48\|M\|_{L^\infty}^4.\end{aligned}$$

□

Going back to the proof of the theorem, this implies that

$$\|X^n-X^m\|^2\to 0\ \text{ as } n,m\to\infty,$$

and so $(X^n)_{n\ge1}$ is Cauchy in $(\mathcal{M}_c^2,\|\cdot\|)$. It follows that there exists $Y\in\mathcal{M}_c^2$ such that $X^n\to Y$ in $\|\cdot\|$ norm as $n\to\infty$.

Now, for any n and $1\le k\le\lceil 2^nT\rceil$, we can write

$$M_{k2^{-n}}^2-2X_{k2^{-n}}^n=\sum_{j=0}^{k-1}(M_{(j+1)2^{-n}}-M_{j2^{-n}})^2=[M]_{k2^{-n}}^n.$$

Consequently, for each $n\ge1$ the process M^2-2X^n is non-decreasing when restricted to the set of times $\{k2^{-n}:1\le k\le\lceil 2^nT\rceil\}$. To show that the same holds for the limiting process M^2-2Y it suffices to prove that $X^n\to Y$ almost surely along a subsequence, which is true by the equivalence of the norms $\|\cdot\|$ and $|\!|\!|\cdot|\!|\!|$.

Set

$$[M]_t = M_t^2 - 2Y_t.$$

Then $[M] = ([M]_t)_{t\le T}$ is a continuous, adapted non-decreasing process and $M^2 - [M] = 2Y$ is a martingale on $[0,T]$.

We can extend this definition to all times in $[0,\infty)$ by applying the above for each $T = k \in \mathbb{N}$. By uniqueness, we note that the process obtained with $T = k$ must be the restriction to $[0,k]$ of the process obtained with $T = k+1$.

Finally we show that $[M]^n \to [M]$ ucp as $n \to \infty$. To this end, note that the convergence $X^n \to Y$ in $(\mathcal{M}_c^2, \|\cdot\|)$ implies, by the equivalence of $\|\cdot\|$ and $\|\!|\cdot|\!\|$, that $\sup_{0\le t\le T} |X_t^n - Y_t| \to 0$ in L^2 as $n \to \infty$, and so in particular

$$\sup_{0\le t\le T} |X_t^n - Y_t| \to 0 \ \text{ in probability as } n \to \infty. \tag{4.5}$$

Now,

$$[M]_t^n = M^2_{2^{-n}\lfloor 2^n t\rfloor} - 2X^n_{2^{-n}\lfloor 2^n t\rfloor}$$

and so

$$\begin{aligned}\sup_{0\le t\le T} \big|[M]_t - [M]_t^n\big| \le{}& \sup_{0\le t\le T} \big|M^2_{2^{-n}\lfloor 2^n t\rfloor} - M_t^2\big| \\ &+ 2\sup_{0\le t\le T} \big|X^n_{2^{-n}\lfloor 2^n t\rfloor} - Y_{2^{-n}\lfloor 2^n t\rfloor}\big| \\ &+ 2\sup_{0\le t\le T} \big|Y_{2^{-n}\lfloor 2^n t\rfloor} - Y_t\big|.\end{aligned}$$

Each term in the right hand side converges to 0 in probability as $n \to \infty$. Indeed, this follows for the first and third terms by uniform continuity of M^2 and Y on $[0,T]$, and for the second term by (4.5). Thus $[M]^n \to [M]$ ucp as $n \to \infty$.

Step 3: Existence, $M \in \mathcal{M}_{c,loc}$. We now obtain existence for the general case $M \in \mathcal{M}_{c,loc}$ by mean of a localisation argument. For each $n \ge 1$, define

$$T_n = \inf\{t \ge 0 : |M_t| \ge n\}.$$

Then $(T_n)_{n\ge1}$ reduces M and moreover M^{T_n} is a bounded martingale by construction for all n. Thus we can apply Step 2 to conclude that there exists a unique continuous, adapted and non-decreasing process $[M^{T_n}]$ on $[0,\infty)$ such that $[M^{T_n}]_0 = 0$ and $(M^{T_n})^2 - [M^{T_n}] \in \mathcal{M}_{c,\mathrm{loc}}$. Write $A^n = [M^{T_n}]$ for simplicity. By uniqueness, $(A^{n+1}_{t\wedge T_n})_{t\ge0}$ and $(A^n_t)_{t\ge0}$ must be indistinguishable. We thus let A be the non-decreasing process such that $A_{t\wedge T_n} = A_t^n$ for all n. By construction, $M^2_{t\wedge T_n} - A_{t\wedge T_n}$ is a martingale for each n. Therefore $M^2 - A$ is a local martingale with reducing sequence $(T_n)_{n\ge1}$, and we can take $[M]_t = A_t$.

Finally, to see that $[M]^n \to [M]$ ucp as $n \to \infty$, note that $[M^{T_k}]^n \to [M^{T_k}]$ as $n \to \infty$ ucp by Step 2, for all k, that is for all $\epsilon > 0$ it holds

$$\mathbb{P}\bigg(\sup_{t\in[0,T]} \big|[M^{T_k}]^n_t - [M^{T_k}]_t\big| > \epsilon \bigg) \to 0 \quad \text{as } n \to \infty. \tag{4.6}$$

Observing that on the event $\{T_k > T\}$ we have $[M]^n_t = [M^{T_k}]^n_t$ and $[M]_t = [M^{T_k}]_t$ for all $t \le T$, we have that

$$\mathbb{P}\bigg(\sup_{t\in[0,T]} \big|[M]^n_t - [M]_t\big| > \epsilon \bigg) \le \mathbb{P}(T_k \le T) + \mathbb{P}\bigg(\sup_{t\in[0,T]} \big|[M^{T_k}]^n_t - [M^{T_k}]_t\big| > \epsilon \bigg) \to 0$$

as $n \to \infty$, since $\mathbb{P}(T_k \le T) \to 0$ as $k \to \infty$, and the second term in the r.h.s. converges to 0 as $n \to \infty$ for all k by (4.6).

Example 4.12. Let B be a standard Brownian motion. Then $(B_t^2 - t)_{t\ge 0}$ is a martingale. Therefore $[B]_t = t$. As we will see later, the fact that $[B]_t = t$ singles out the standard Brownian motion among continuous local martingales. This is the so-called Lévy characterization of Brownian motion.

The following result will be useful later on, and it shows that the a priori local martingale $M^2 - [M]$ is in fact a true UI martingale if $M \in \mathcal{M}_c^2$.

Theorem 4.13. *If $M \in \mathcal{M}_c^2$, then $M^2 - [M]$ is a uniformly integrable martingale.*

Proof. We show that the martingale is dominated by an integrable random variable.

Let $S_n = \inf\{t \ge 0 : [M]_t \ge n\}$ for $n \ge 1$. Then $S_n \nearrow +\infty$ and S_n is a stopping time for all n, with $[M]_{t\wedge S_n} \le n$. Therefore the local martingale $M^2_{t\wedge S_n} - [M]_{t\wedge S_n}$ is dominated by the random variable $n + \sup_{t\ge 0} M_t^2$, which is integrable by Doob's L^2 inequality. It then follows from Corollary 3.10 that $M^2_{t\wedge S_n} - [M]_{t\wedge S_n}$ is a true martingale. The OST then gives

$$\mathbb{E}[[M]_{t\wedge S_n}] = \mathbb{E}[M^2_{t\wedge S_n}].$$

Letting $t \to \infty$ and using the Monotone Convergence Theorem on the l.h.s. and the Dominated Convergence Theorem on the r.h.s., we conclude that

$$\mathbb{E}[[M]_{S_n}] = \mathbb{E}[M^2_{S_n}].$$

Sending $n \to \infty$ and using the same theorems to exchange limits and expectation, we find that

$$\mathbb{E}[[M]_\infty] = \mathbb{E}[M^2_\infty] < \infty,$$

thus showing that $[M]_\infty$ is integrable. Moreover,

$$|M_t^2 - [M]_t| \le \sup_{t\ge 0} M_t^2 + [M]_\infty \qquad \forall t \ge 0,$$

and the r.h.s. is integrable, so we conclude that $M_t^2 - [M]_t$ is a true UI martingale. □

4.3.1. *The space $L^2(M)$.* We can finally define our space of integrands. Recall that $\mathcal{P}$ denotes the previsible σ-algebra on $\Omega \times (0,\infty)$. Given $M \in \mathcal{M}_c^2$, we define a measure μ on $\mathcal{P}$ by setting

$$\mu(E \times (s,t]) = \mathbb{E}\big(\mathbb{1}(E)([M]_t - [M]_s)\big) \quad \text{for} \quad s \le t, \quad E \in \mathcal{F}_s.$$

Since $\mathcal{P}$ is generated by a π-system of events of the form $E \times (s,t]$ for E, s, t as above, this uniquely specifies μ. Note that this is equivalent to defining

$$\mu(d\omega \otimes dt) := d[M](\omega, dt)\mathbb{P}(d\omega),$$

where, for fixed $\omega \in \Omega$, the measure $d[M](\omega,\cdot)$ is the Lebesgue-Stieltjes measure associated to the non-decreasing function $[M](\omega)$. Then for a previsible process $H \ge 0$ we have that

$$\int_{\Omega\times(0,\infty)} H d\mu = \mathbb{E}\Big(\int_0^\infty H_s d[M]_s\Big).$$

Definition 4.14. *Let $L^2(M) := L^2(\Omega \times (0,\infty), \mathcal{P}, \mu)$ and write*

$$\|H\|_{L^2(M)} = \|H\|_M := \left[\mathbb{E}\Big(\int_0^\infty H_s^2 d[M]_s\Big)\right]^{1/2}.$$

Then $L^2(M)$ is the set of previsible processes H such that $\|H\|_M < \infty$. This is clearly a Hilbert space, and it will be our space of integrands.

Remark 4.15. *Note that the space of integrands $(L^2(M), \|\cdot\|_M)$ depends on the integrator $M \in \mathcal{M}_c^2$, since the measure μ depends on it. On the other hand, recalling that $\mathcal{S}$ denotes the set of simple processes, we have that $\mathcal{S} \subseteq L^2(M)$ for all $M \in \mathcal{M}_c^2$.*

4.4. **Itô integrals.** Up to now, we have identified two Hilbert spaces, that of integrators $(\mathcal{M}_c^2, \|\cdot\|)$ and that of integrands $(L^2(M), \|\cdot\|_M)$. For a subset $\mathcal{S}$ of $L^2(M)$, that of simple processes of the form $H_t = \sum_{k=0}^{n-1} Z_k \mathbb{1}_{(t_k, t_{k+1}]}(t)$, we have defined the integral $H \cdot M$ by setting

$$(H \cdot M)_t = \sum_{k=0}^{n-1} Z_k (M_{t_{k+1}\wedge t} - M_{t_k \wedge t}) \ \in \mathcal{M}_c^2.$$

We now argue that for any $M \in \mathcal{M}_c^2$ the map $H \mapsto H \cdot M$ provides an isometry between $(L^2(M), \|\cdot\|_M)$ and $(\mathcal{M}_c^2, \|\cdot\|)$ when restricted to simple processes $\mathcal{S} \subseteq L^2(M)$. To see this, we have to show that $\|H \cdot M\| = \|H\|_M$ for all $H \in \mathcal{S}$, $M \in \mathcal{M}_c^2$. Indeed, recall from (4.2) that

$$\|H \cdot M\|^2 = \big\|(H \cdot M)_\infty\big\|_{L^2}^2 = \sum_{k=0}^{n-1} \mathbb{E}\big[Z_k^2 (M_{t_{k+1}} - M_{t_k})^2\big].$$

Moreover, by Theorem 4.13 the process $M^2 - [M]$ is a true martingale, so that

$$\begin{aligned}\mathbb{E}\big[Z_k^2(M_{t_{k+1}} - M_{t_k})^2\big] &= \mathbb{E}\big[Z_k^2\,\mathbb{E}\big((M_{t_{k+1}} - M_{t_k})^2 \,|\, \mathcal{F}_{t_k}\big)\big] \\ &= \mathbb{E}\big[Z_k^2\,\mathbb{E}\big(M_{t_{k+1}}^2 - M_{t_k}^2 \,|\, \mathcal{F}_{t_k}\big)\big] \\ &= \mathbb{E}\big[Z_k^2\,\mathbb{E}\big([M]_{t_{k+1}} - [M]_{t_k} \,|\, \mathcal{F}_{t_k}\big)\big] \\ &= \mathbb{E}\big[Z_k^2\,([M]_{t_{k+1}} - [M]_{t_k})\big].\end{aligned}$$

Therefore

$$\big\|H \cdot M\big\|^2 = \mathbb{E}\bigg(\sum_{k=0}^{n-1} Z_k^2\,([M]_{t_{k+1}} - [M]_{t_k})\bigg) = \mathbb{E}\bigg(\int_0^\infty H_s^2 d[M]_s\bigg) = \|H\|_M^2, \tag{4.7}$$

thus showing that the stochastic integral provides an isometry between $L^2(M)$ and $\mathcal{M}_c^2$ when restricted to simple processes. The following theorem is the main result of this section, and it tells us that this property uniquely defines the stochastic integral on the whole $L^2(M)$.

Theorem 4.16 (Itô's isometry)**.** *There exists a unique isometry $I\colon L^2(M) \to \mathcal{M}_c^2$ such that $I(H) = H \cdot M$ for all $H \in \mathcal{S}$.*

Definition 4.17. *For any $M \in \mathcal{M}_c^2$ and $H \in L^2(M)$, let*

$$H \cdot M := I(H),$$

where I is the unique isometry from $(L^2(M), \|\cdot\|_M)$ to $(\mathcal{M}_c^2, \|\cdot\|)$ of Theorem 4.16.

We now proceed with the proof of Theorem 4.16 above. To start with, we show that simple processes are dense in $L^2(M)$, so that there is hope to extend the definition of the stochastic integral from $\mathcal{S}$ to the whole $L^2(M)$ uniquely.

Lemma 4.18. *Let ν be a finite measure on $\mathcal{P}$, the previsible σ-algebra. Then $\mathcal{S}$ is a dense subspace of $L^2(\mathcal{P}, \nu)$. In particular, taking $\nu = \mu$ we have that $\mathcal{S}$ is a dense subspace of $L^2(M)$ for any $M \in \mathcal{M}_c^2$.*

Proof. It is certainly the case that $\mathcal{S} \subseteq L^2(\mathcal{P}, \nu)$. Denote by $\overline{\mathcal{S}}$ the closure of $\mathcal{S}$ in $L^2(\mathcal{P}, \nu)$. We aim to show that $\overline{\mathcal{S}} = L^2(\mathcal{P}, \nu)$. Let $\mathcal{A} = \{A \in \mathcal{P} : \mathbb{1}_A \in \overline{\mathcal{S}}\}$. Then clearly $\mathcal{A} \subseteq \mathcal{P}$. To see the reverse inclusion, observe that $\mathcal{A}$ is a d-system containing the π-system $\{E \times (s,t] : E \in \mathcal{F}_s,\ s \le t\}$, which generates $\mathcal{P}$. So, by Dynkin's lemma, we have that $\mathcal{A} \supseteq \mathcal{P}$ and hence $\mathcal{A} = \mathcal{P}$. This shows that $\overline{\mathcal{S}}$ contains indicator functions of all previsible processes. The result then follows because finite linear combinations of such indicators are dense in $L^2(\mathcal{P}, \nu)$. □

We can now prove Itô's isometry.

Proof of Theorem 4.16. Take any $H \in L^2(M)$. Then by Lemma 4.18 there exists a sequence $(H^n)_{n\geq 1}$ in $\mathcal{S} \subseteq L^2(M)$ such that $\|H^n - H\|_M \to 0$ as $n \to \infty$, that is

$$\mathbb{E}\bigg(\int_0^\infty (H^n_s - H_s)^2 d[M]_s \bigg) \to 0 \quad \text{as} \quad n \to \infty.$$

In particular, $(H^n)_{n\geq 1}$ is Cauchy in $\|\cdot\|_M$ norm. We argue that this implies that $(I(H^n))_{n\geq 1}$ is Cauchy in $\|\cdot\|$ norm. Indeed, by linearity of the integral for simple processes we have

$$\|I(H^n) - I(H^m)\| = \|H^n \cdot M - H^m \cdot M\| = \|(H^n - H^m) \cdot M\| = \|H^n - H^m\|_M \to 0$$

as $n, m \to \infty$, where the last equality follows from (4.7). Thus $I(H^n) = H^n \cdot M$ converges in $\|\cdot\|$ norm to an element of $\mathcal{M}^2_c$, that we define to be $I(H)$. Note that this definition does not depend on the choice of the approximating sequence. Indeed, if $(K^n)_{n\geq 1}$ is another sequence of simple processes converging to H in $\|\cdot\|_M$ norm, then

$$\|I(H^n) - I(K^n)\| = \|H^n \cdot M - K^n \cdot M\| = \|H^n - K^n\|_M \leq \|H^n - H\|_M + \|K^n - H\|_M \to 0$$

as $n \to \infty$, thus showing that the limits of $H^n \cdot M$ and $K^n \cdot M$ must be indistinguishable. This uniquely extends I to the whole $L^2(M)$. It only remains to show that the resulting map is an isometry from $L^2(M)$ to $\mathcal{M}^2_c$. Indeed, since convergence in norm implies convergence of the norms by the reverse triangle inequality[2], we have

$$\|H \cdot M\| = \lim_{n\to\infty} \|H^n \cdot M\| = \lim_{n\to\infty} \|H^n\|_M = \|H\|_M$$

for all $H \in L^2(M)$, as required. □

We write

$$I(H)_t = (H \cdot M)_t = \int_0^t H_s dM_s.$$

The process $H \cdot M$ is Itô's stochastic integral of H with respect to M.

Remark 4.19. *Recall from the introduction that we were seeking to define $\int H dM$ in such a way that if H^n is a sequence of simple processes converging to H, then $\int H^n dM \to \int H dM$. The above definition matches this requirement by construction, where the convergence $H^n \to H$ is to be intended in $\|\cdot\|_M$ norm, and the one of the integrals $\int H^n dM \to H dM$ is to be intended in $\|\cdot\|$ norm.*

4.5. **Extensions.** We now extend the definition of stochastic integral to a larger class of integrators and integrands. To this end, we first need to understand how the stochastic integral behaves under stopping.

Proposition 4.20. *Let $M \in \mathcal{M}^2_c$ and $H \in L^2(M)$. Let T be a stopping time. Then*

$$(H \cdot M)^T = (H \mathbb{1}_{(0,T]}) \cdot M = H \cdot (M^T).$$

[2] $|\|x\| - \|y\|| \leq \|x - y\|$.

Proof. It is easy to check that $H \in L^2(M)$ implies that also $H\mathbb{1}_{(0,T]} \in L^2(M)$ and $H \in L^2(M^T)$ so that all the integrals above are well defined. We split the proof into several steps.

Case 1. Fix $H \in \mathcal{S}$, $M \in \mathcal{M}_c^2$, T taking only finitely many values. Then one can check directly from the definition that $H\mathbb{1}_{(0,T]} \in \mathcal{S}$ and $(H \cdot M)^T = (H\mathbb{1}_{(0,T]}) \cdot M = H \cdot (M^T)$.

Case 2. Fix $H \in \mathcal{S}$, $M \in \mathcal{M}_c^2$, T a general stopping time. Then we have already shown that $(H \cdot M)^T = H \cdot (M^T)$ in Proposition 4.6. To see that $(H \cdot M)^T = (H\mathbb{1}_{(0,T]}) \cdot M$ we reduce to Case 1 by approximating T. For $n, m \geq 1$, we let $T_{n,m} = (2^{-n}\lceil 2^n T \rceil) \wedge m$. Then $T_{n,m}$ takes on only finitely many values and $T_{n,m} \searrow T \wedge m$ as $n \to \infty$. Thus,

$$\|H\mathbb{1}_{(0,T_{n,m}]} - H\mathbb{1}_{(0,T\wedge m]}\|_M^2 = \mathbb{E}\bigg(\int_0^\infty H_t^2 \mathbb{1}_{(T\wedge m, T_{n,m}]} d[M]_t \bigg) \to 0 \quad \text{as} \quad n \to \infty$$

by the Dominated Convergence Theorem. Since convergence of the integrands implies convergence of the integrals, we get that

$$(H\mathbb{1}_{(0,T_{n,m}]}) \cdot M \to (H \cdot \mathbb{1}_{(0,T\wedge m]}) \cdot M \qquad \text{in } \mathcal{M}_c^2 \text{ as } n \to \infty.$$

On the other hand, the l.h.s. equals $(H \cdot M)^{T_{n,m}}$ by Case 1, and

$$(H \cdot M)^{T_{n,m}} \to (H \cdot M)^{T\wedge m} \qquad \text{pointwise almost surely as } n \to \infty$$

by continuity of $H \cdot M$. It then follows from uniqueness of the limit[3] that

$$(H \cdot \mathbb{1}_{(0,T\wedge m]}) \cdot M = (H \cdot M)^{T\wedge m}$$

for all $m \geq 1$. Sending now $m \to \infty$ and using the same argument as above we conclude that $(H\mathbb{1}_{(0,T]}) \cdot M = (H \cdot M)^T$.

Case 3. $H \in L^2(M)$, $M \in \mathcal{M}_c^2$, T a general stopping time. Let $(H^n)_{n\geq 1}$ be a sequence in $\mathcal{S}$ such that $H^n \to H$ in $L^2(M)$. Then

$$\begin{aligned}
\|(H^n \cdot M)^T - (H \cdot M)^T\| &= \|(H^n \cdot M)_T - (H \cdot M)_T\|_{L^2} \\
&\leq \Big\| \sup_{t\geq 0} |(H^n \cdot M)_t - (H \cdot M)_t| \Big\|_{L^2} \\
&\leq 2\|(H^n \cdot M)_\infty - (H \cdot M)_\infty\|_{L^2} \qquad \text{(Doob's } L^2 \text{ inequality)} \\
&= \|(H^n - H) \cdot M\| = \|H^n - H\|_M \to 0
\end{aligned}$$

as $n \to \infty$, that is

$$(H^n \cdot M)^T \to (H \cdot M)^T \qquad \text{in } \mathcal{M}_c^2. \tag{4.8}$$

On the other hand,

$$\|H^n\mathbb{1}_{(0,T]} - H\mathbb{1}_{(0,T]}\|_M^2 = \mathbb{E}\bigg(\int_0^T (H_t^n - H_t)^2 d[M]_t \bigg) \leq \|H^n - H\|_M^2 \to 0 \quad \text{as} \quad n \to \infty.$$

[3]Recall from the proof of Proposition 4.7 that convergence in $\|\cdot\|$ is equivalent to convergence in $\|\!|\cdot|\!\|$, which in turn implies almost sure uniform convergence of a subsequence.

Thus

$$(H^n \mathbb{1}_{(0,T]}) \cdot M \to (H \mathbb{1}_{(0,T]}) \cdot M \qquad \text{in } \mathcal{M}_c^2.$$

Since $(H^n \cdot M)^T = (H^n \mathbb{1}_{(0,T]}) \cdot M$ for all n, this shows that $(H \cdot M)^T = (H \mathbb{1}_{(0,T]}) \cdot M$. Finally, to see that $(H \cdot M)^T = H \cdot (M^T)$ it suffices to show that if $H^n \to H$ in $L^2(M)$ then also $H^n \to H$ in $L^2(M^T)$. Indeed,

$$\begin{aligned} \|H^n - H\|_{M^T}^2 &= \mathbb{E}\bigg(\int_0^\infty (H_s^n - H_s)^2 d[M^T]_s \bigg) \\ &= \mathbb{E}\bigg(\int_0^T (H_s^n - H_s)^2 d[M]_s \bigg) \\ &\leq \|H^n - H\|_M^2 \to 0 \quad \text{as} \quad n \to \infty. \end{aligned}$$

Thus $H^n \cdot (M^T) \to H \cdot (M^T)$ in $\mathcal{M}_c^2$. This together with (4.8) gives $(H \cdot M)^T = H \cdot (M^T)$. □

We use Proposition 4.20 to extend the definition of Itô's integral.

Definition 4.21 (Locally bounded process)**.** *We say that a previsible process H is* locally bounded *if there exists a sequence $(S_n)_{n\geq 1}$ of stopping times with $S_n \nearrow \infty$ almost surely as $n \to \infty$ such that $H \mathbb{1}_{(0,S_n]}$ is bounded for all n.*

Definition 4.22 (Itô's integral, H locally bounded, $M \in \mathcal{M}_{c,loc}$)**.** *Let H be a locally bounded previsible process such that $H \mathbb{1}_{(0,S_n]}$ is bounded for all n, for $(S_n)_{n\geq 1}$ sequence of stopping times with $S_n \nearrow \infty$ almost surely. Let moreover M be a continuous local martingale with reducing sequence $(S_n')_{n\geq 1}$ given by $S_n' = \inf\{t \geq 0 : |M_t| \geq n\}$, so that $M^{S_n'} \in \mathcal{M}_c^2$ for all n. Set $T_n := S_n \wedge S_n'$ for all $n \geq 1$, and define*

$$(H \cdot M)_t := ((H \mathbb{1}_{(0,T_n]}) \cdot M^{T_n})_t \qquad \text{for } t \leq T_n. \tag{4.9}$$

Note that the r.h.s. in the above definition is well defined, since if $M \in \mathcal{M}_c^2$ then any bounded previsible process is in $L^2(M)$, and clearly $M^{T_n} = (M^{S_n'})^{T_n} \in \mathcal{M}_c^2$. Moreover, it is easy to check that the definition of the integral does not depend on the choice of the reducing sequence $(S_n)_{n\geq 1}$ for H, and it is consistent with our previous definitions of Itô's integral.

Proposition 4.23. *Let $M \in \mathcal{M}_{c,loc}$ and let H be a locally bounded previsible process. Then $H \cdot M \in \mathcal{M}_{c,loc}$ and $(T_n)_{n\geq 1}$ defined as in (4.9) is a reducing sequence. Moreover, for any T stopping time we have*

$$(H \cdot M)^T = (H \mathbb{1}_{(0,T]}) \cdot M = H \cdot (M^T).$$

Proof. The fact that $H \cdot M \in \mathcal{M}_{c,loc}$ with reducing sequence $(T_n)_{n\geq 1}$ follows from the definition, since the r.h.s. of (4.9) is in $\mathcal{M}_c^2$ for all n. Moreover,

$$\begin{aligned}(H \cdot M)^T &= \lim_{n\to\infty} ((H\mathbb{1}_{(0,T_n]}) \cdot M^{T_n})^T \\ &= \lim_{n\to\infty} ((H\mathbb{1}_{(0,T_n]}\mathbb{1}_{(0,T]}) \cdot M^{T_n}) \\ &= (H\mathbb{1}_{(0,T]}) \cdot M\end{aligned}$$

where the limits hold pointwise almost surely, and the second equality follows from Proposition 4.20. The same argument shows that $(H \cdot M)^T = H \cdot (M^T)$. □

The next result allows us to express the quadratic variation of the integral in terms of the quadratic variation of the integrator.

Proposition 4.24. *Let $M \in \mathcal{M}_{c,loc}$ and let H be a locally bounded previsible process. Then*

$$[H \cdot M] = H^2 \cdot [M].$$

Proof. Assume first that H, M are uniformly bounded. Then we have that

$$\begin{aligned}\mathbb{E}[(H \cdot M)_T^2] &= \mathbb{E}[((H\mathbb{1}_{(0,T]}) \cdot M)_\infty^2] \\ &= \mathbb{E}[((H^2\mathbb{1}_{(0,T]}) \cdot [M])_\infty] \qquad \text{(Itô's isometry)} \\ &= \mathbb{E}[(H^2 \cdot [M])_T]\end{aligned}$$

for all bounded stopping times T. By the OST, we thus have that $(H \cdot M)^2 - H^2 \cdot [M]$ is a continuous martingale. On the other hand, the unique non-decreasing, adapted, continuous process X such that $(H \cdot M)^2 - X$ is a martingale is $[H \cdot M]$. Therefore it must be $[H \cdot M] = H^2 \cdot [M]$.

We now treat the general case H locally bounded by mean of a *localization argument*. Let $(T_n)_{n\geq 1}$ be a sequence of stopping times such that $H\mathbb{1}_{(0,T_n]}$ is uniformly bounded for all n. Then we have that

$$\begin{aligned}[H \cdot M] &= \lim_{n\to\infty} [H \cdot M]^{T_n} && \text{(pointwise, almost surely)} \\ &= \lim_{n\to\infty} [(H \cdot M)^{T_n}] \\ &= \lim_{n\to\infty} [(H\mathbb{1}_{(0,T_n]}) \cdot M] \\ &= \lim_{n\to\infty} (H\mathbb{1}_{(0,T_n]})^2 \cdot [M] && (H\mathbb{1}_{(0,T_n]} \text{ uniformly bounded}) \\ &= H^2 \cdot [M] && \text{(pointwise, almost surely)}\end{aligned}$$

where the last equality holds by the Monotone Convergence Theorem. □

If H is a locally bounded previsible process, and $M \in \mathcal{M}_{c,loc}$, then we have seen that $H \cdot M \in \mathcal{M}_{c,loc}$. It follows that we can iterate the integral.

Proposition 4.25 (Iterated integral)**.** *Let $M \in \mathcal{M}_{c,loc}$ and let H, K be locally bounded previsible processes. Then*

$$H \cdot (K \cdot M) = (HK) \cdot M. \tag{4.10}$$

Proof. The fact that (4.10) holds for H, K simple is an elementary exercise. For general H, K, assume first that both H and K are uniformly bounded. Let us first show that both integrals are well defined. We have

$$\begin{aligned}
\|H\|^2_{L^2(K\cdot M)} &= \mathbb{E}[(H^2 \cdot [K \cdot M])_\infty] = \mathbb{E}[(H^2 \cdot (K^2 \cdot [M]))_\infty] \\
&= \mathbb{E}[((HK)^2 \cdot [M])_\infty] = \|HK\|^2_{L^2(M)} \\
&\leq \min\left\{ \|H\|^2_{L^\infty} \|K\|^2_{L^2(M)}; \|K\|^2_{L^\infty} \|H\|^2_{L^2(M)} \right\},
\end{aligned}$$

where the third equality follows from the corresponding property for the Lebesgue-Stieltjes integral. Thus both $H \in L^2(K \cdot M)$ and $HK \in L^2(M)$, and the integrals are well defined. Note that incidentally we have shown that

$$\|H\|_{L^2(K\cdot M)} = \|HK\|_{L^2(M)}, \tag{4.11}$$

a property that we will use later on. Let $(H^n)_{n\geq 1}$ and $(K^n)_{n\geq 1}$ be sequences of simple processes converging to H and K respectively in $L^2(M)$, and assume that $\|H^n\|_{L^\infty}$ and $\|K^n\|_{L^\infty}$ are uniformly bounded over all $n \geq 1$ without loss of generality. We have already observed that $H^n \cdot (K^n \cdot M) = (H^n K^n) \cdot M$. Moreover,

$$\begin{aligned}
\|H^n \cdot (K^n \cdot M) - H \cdot (K \cdot M)\| &\leq \|(H^n - H) \cdot (K^n \cdot M)\| + \|H \cdot ((K^n - K) \cdot M)\| \\
&= \|H^n - H\|_{L^2(K^n \cdot M)} + \|H\|_{L^2((K^n - K)\cdot M)} \qquad \text{(Itô's isometry)} \\
&= \|(H^n - H)K^n\|_{L^2(M)} + \|H(K^n - K)\|_{L^2(M)} \\
&\leq \|H^n - H\|_{L^2(M)} \|K^n\|_{L^\infty} + \|H\|_{L^\infty} \|K^n - K\|_{L^2(M)} \to 0
\end{aligned}$$

as $n \to \infty$, where the second equality follows from (4.11). A similar argument implies that $(H^n K^n) \cdot M \to (HK) \cdot M$ in $\mathcal{M}^2_c$ as $n \to \infty$, thus showing that $H \cdot (K \cdot M) = (HK) \cdot M$. Finally, for H, K locally bounded we again use a *localization argument*. Let $(T_n)_{n\geq 1}$ be a sequence of stopping times such that both $H\mathbb{1}_{(0,T_n]}$ and $K\mathbb{1}_{(0,T_n]}$ are bounded for all n (it is always possible to construct such sequence by setting $T_n = S_n \wedge S'_n$, where $(S_n)_{n\geq 1}$ and $(S'_n)_{n\geq 1}$ are sequences of stopping times making H and K uniformly bounded respectively). Then by the previous point we have

$$(H\mathbb{1}_{(0,T_n]}) \cdot ((K\mathbb{1}_{(0,T_n]}) \cdot M) = (HK\mathbb{1}_{(0,T_n]}) \cdot M$$

for all n. Moreover, $(K\mathbb{1}_{(0,T_n]}) \cdot M = (K \cdot M)^{T_n}$, from which

$$(H\mathbb{1}_{(0,T_n]}) \cdot ((K\mathbb{1}_{(0,T_n]}) \cdot M) = (H\mathbb{1}_{(0,T_n]}) \cdot (K \cdot M) = (H \cdot (K \cdot M))^{T_n} \to H \cdot (K \cdot M)$$

pointwise almost surely as $n \to \infty$. Finally,

$$(HK\mathbb{1}_{(0,T_n]}) \cdot M = ((HK) \cdot M)^{T_n} \to (HK) \cdot M$$

pointwise almost surely as $n \to \infty$. We thus conclude that $H \cdot (K \cdot M) = (HK) \cdot M$. □

Remark 4.26. *In both the proof of Proposition 4.24 and the one of Proposition 4.25 we have used a so-called localization argument to reduce the case of locally bounded integrands to the one of uniformly bounded integrands. This is a standard procedure that we have chosen to spell out in the two proofs above, but will be left for the reader to check in the rest of these notes.*

4.5.1. *Semimartingales.* As a last extension of the Itô integral we enlarge the class of integrators to the so-called semimartingales.

Definition 4.27 (Semimartingale)**.** *A continuous adapted process X is called a semimartingale if it can be written in the form*

$$X = X_0 + M + A$$

where M is a continuous local martingale, A is a process of finite variation and $M_0 = A_0 = 0$. This is called the Doob-Meyer decomposition of X.

For $X = X_0 + M + A$ continuous semimartingale define its quadratic variation to be that of its martingale part, i.e. $[X] := [M]$. This definition is justified by the fact that

$$\sum_{k=0}^{\lceil 2^n t \rceil - 1} (X_{(k+1)2^{-n}} - X_{k2^{-n}})^2 \to [X]_t \qquad \text{ucp as } n \to \infty,$$

which is easy to check. Then we can easily extend the definition of Itô's integral by linearity, since we know how to integrate both with respect to M (Itô's theory) and with respect to A (Lebesgue-Stieltjes theory).

Definition 4.28. *For H locally bounded previsible process, and $X = X_0 + M + A$ continuous semimartingale, define*

$$H \cdot X := H \cdot M + H \cdot A.$$

Note that $H \cdot X$ defined above is again a continuous semimartingale, since $H \cdot M \in \mathcal{M}_{c,loc}$ and $H \cdot A$ is of finite variation.

The next result shows that, under the additional assumption that H is left-continuous, the integral can be obtained as the limit of its Riemann sum approximations.

Proposition 4.29. *Let X be a continuous semimartingale and H be a locally bounded left-continuous process which is adapted. Then*

$$\sum_{k=0}^{\lfloor 2^n t \rfloor - 1} H_{k2^{-n}}(X_{(k+1)2^{-n}} - X_{k2^{-n}}) \to (H \cdot X)_t \quad \textit{ucp as } n \to \infty. \tag{4.12}$$

Proof. To start with, note that the integral $H \cdot X$ is well defined, as H is previsible since left continuous and adapted. We can treat the local martingale and finite variation parts separately. The latter is Problem 2 on Example Sheet 1, so this leaves just the martingale part. By applying a localization argument, we can reduce to the case $M \in \mathcal{M}_c^2$ and H bounded. Moreover, since convergence in $\|\cdot\|$ norm implies ucp convergence, it suffices to show that the convergence in (4.12) holds in $\|\cdot\|$ norm. To this end, let $H_t^n = H_{2^{-n}\lfloor 2^n t\rfloor}$. Then H^n is simple for all n, and by the left-continuity of H we have that $H_t^n \to H_t$ almost surely as $n \to \infty$. Moreover,

$$(H^n \cdot M)_t = \sum_{k=0}^{\lfloor 2^n t\rfloor - 1} H_{k2^{-n}}\big(M_{(k+1)2^{-n}} - M_{k2^{-n}}\big) + H_{2^{-n}\lfloor 2^n t\rfloor}(M_t - M_{2^{-n}\lfloor 2^n t\rfloor}).$$

Since H is bounded and $|M_t - M_{2^{-n}\lfloor 2^n t\rfloor}| \to 0$ almost surely as $n \to \infty$ by the continuity of M, the Dominated Convergence Theorem implies that the second term in the right hand side above converges to 0 in $\|\cdot\|$ norm as $n \to \infty$. It thus suffices to show that the remaining term converges to the right limit, that is to show that $H^n \cdot M \to H \cdot M$ in $\|\cdot\|$ norm as $n \to \infty$. By Itô's isometry, this is equivalent to showing that $H^n \to H$ in $\|\cdot\|_M$ norm. Indeed, we have

$$\|H^n - H\|_M^2 = \mathbb{E}\left(\int_0^\infty (H_t^n - H_t)^2 d[M]_t\right) \to 0 \quad \text{as} \quad n \to \infty$$

by the Bounded Convergence Theorem. □

This concludes our construction of the Itô's integral, which we now briefly summarize.

Summary of the stochastic integral

Step 1. $H \in \mathcal{S}$, $M \in \mathcal{M}_c^2$. If $H_t = \sum_{k=0}^{n-1} Z_k \mathbb{1}_{(t_k, t_{k+1}]}(t)$ where Z_k is a bounded, $\mathcal{F}_{t_k}$-measurable random variable for each k, define

$$(H \cdot M)_t = \sum_{k=0}^{n-1} Z_k (M_{t_{k+1}\wedge t} - M_{t_k \wedge t}).$$

Then $H \cdot M \in \mathcal{M}_c^2$.

Step 2. Equip $\mathcal{M}_c^2$ with a Hilbert space structure with norm $\|X\| := \|X_\infty\|_{L^2}$.

Step 3. Establish the existence of $[M]$ for all $M \in \mathcal{M}_{c,loc}$. This is the unique, adapted, continuous, non-decreasing process $[M]$ such that $M^2 - [M] \in \mathcal{M}_{c,loc}$.

Step 4. For $M \in \mathcal{M}_c^2$, use $[M]$ to define the Hilbert space $(L^2(M), \|\cdot\|_M)$ with

$$\|H\|_M = \left(\mathbb{E}\Big(\int_0^\infty H_s^2 d[M]_s\Big)\right)^{1/2}.$$

Step 5. Extend the integral to $H \in L^2(M)$, $M \in \mathcal{M}_c^2$ by requiring that

$$\|H \cdot M\| = \|H\|_M \qquad \text{(Itô's isometry)}$$

for all $H \in L^2(M)$. Then $H \cdot M \in \mathcal{M}_c^2$.

Step 4. Extend to H locally bounded previsible and $M \in \mathcal{M}_{c,loc}$ by setting

$$(H \cdot M)_t = ((H\mathbb{1}_{(0,T_n]}) \cdot M^{T_n})_t \quad \text{for} \quad t \leq T_n.$$

Then $H \cdot M \in \mathcal{M}_{c,loc}$.

Step 5. Extend to H locally bounded previsible and $X = X_0 + M + A$ continuous semimartingale by setting

$$H \cdot X = H \cdot M + H \cdot A.$$

Then $H \cdot X$ is again a continuous semimartingale.

5. Stochastic calculus

In this section we develop the usual tools of calculus for the Itô's integral. We start with some preliminary results on the so called covariation of two local martingales. This will then be used to introduce and prove the celebrated Itô's formula, which can be thought of as the generalization of the chain rule to the Itô setting. Finally, we discuss a different notion of stochastic integral called Stratonovich integral, for which the classical rules of calculus apply.

5.1. **Covariation.** It is easy to see that the quadratic variation defines a quadratic form. We now define the associated bilinear form, called covariation.

Definition 5.1 (Covariation)**.** *For $M, N \in \mathcal{M}_{c,loc}$, define the covariation of M and N, denoted $[M, N]$, by setting*

$$[M, N] := \frac{1}{4}([M + N] - [M - N]) \qquad \textit{(Polarization identity).} \tag{5.1}$$

Note that $[M, M] = [M]$.

Theorem 5.2. *Let $M, N \in \mathcal{M}_{c,loc}$. Then the following hold:*

(a) $[M, N]$ is the unique (up to indistinguishability) continuous, adapted, finite variation process such that $[M, N]_0 = 0$ and $MN - [M, N] \in \mathcal{M}_{c,loc}$.

(b) For $n \geq 1$, define

$$[M, N]_t^n := \sum_{k=0}^{\lceil 2^n t \rceil - 1} (M_{(k+1)2^{-n}} - M_{k2^{-n}})(N_{(k+1)2^{-n}} - N_{k2^{-n}}).$$

Then $[M, N]_t^n \to [M, N]_t$ ucp as $n \to \infty$.

(c) If $M, N \in \mathcal{M}_c^2$, then $MN - [M, N]$ is a UI martingale.

(d) For H locally bounded and previsible, it holds

$$[H \cdot M, N] + [M, H \cdot N] = 2H \cdot [M, N].$$

Note that $[M, N]$ is a symmetric bilinear form, as one can easily see from (a).

Proof of Theorem 5.2. Note that $MN = \frac{1}{4}((M+N)^2 - (M-N)^2)$, so

$$MN - [M,N] = \frac{1}{4}\big(\underbrace{(M+N)^2 - [M+N]}_{\in \mathcal{M}_{c,loc}} \underbrace{-(M-N)^2 + [M-N]}_{\in \mathcal{M}_{c,loc}}\big),$$

which shows that $MN - [M,N] \in \mathcal{M}_{c,loc}$. The covariation process $[M,N]$ is continuous, adapted and of finite variation since difference of continuous, adapted non-decreasing processes. Finally, uniqueness follows from the same argument used to show uniqueness of the quadratic variation. This proves (a).

Note further that

$$[M,N]^n_t = \frac{1}{4}\left([M+N]^n_t - [M-N]^n_t\right).$$

Thus (b)– (c) follow from the equivalent properties for quadratic variation.

For (d), we note that

$$[H \cdot (M \pm N)] = H^2 \cdot [M \pm N].$$

Plugging this into (5.1), we have that

$$[H \cdot M, H \cdot N] = H^2 \cdot [M,N].$$

Moreover,

$$\begin{aligned}(H+1)^2 \cdot [M,N] &= [(H+1)\cdot M, (H+1)\cdot N] \\ &= [H \cdot M + M, H \cdot N + N] \\ &= [H \cdot M, H \cdot N] + [M, H \cdot N] + [H \cdot M, N] + [M,N],\end{aligned}$$

and

$$\begin{aligned}(H+1)^2 \cdot [M,N] &= (H^2 + 2H + 1) \cdot [M,N] \\ &= H^2 \cdot [M,N] + 2H \cdot [M,N] + [M,N].\end{aligned}$$

By matching up terms, we see that

$$2H \cdot [M,N] = [M, H \cdot N] + [H \cdot M, N],$$

as required. □

The next result tells us how the covariation behaves under integration, and it will be very useful later on.

Proposition 5.3 (Kunita-Watanabe identity)**.** *Let $M, N \in \mathcal{M}_{c,loc}$ and H a locally bounded, previsible process. Then*

$$[H \cdot M, N] = H \cdot [M,N].$$

Proof. It suffices to show that $[H \cdot M, N] = [M, H \cdot N]$, since then the result follows from (d). We have that $(H \cdot M)N - [H \cdot M, N] \in \mathcal{M}_{c,loc}$ and $M(H \cdot N) - [M, H \cdot N] \in \mathcal{M}_{c,loc}$. Suppose we could show that $(H \cdot M)N - M(H \cdot N) \in \mathcal{M}_{c,loc}$. Then $[H \cdot M, N] - [M, H \cdot N] \in \mathcal{M}_{c,loc}$. But this last process can be expressed as sums and differences of quadratic variations and, as quadratic variations are non-decreasing, this implies that $[H \cdot M, N] - [M, H \cdot N]$ is of finite variation. Hence $[H \cdot M, N] - [M, H \cdot N]$ is a continuous local martingale starting from 0 and of finite variation, which implies that it must be indistinguishable from the 0 process. Thus we conclude that $[H \cdot M, N] = [M, H \cdot N]$.

It only remains to show that $(H \cdot M)N - M(H \cdot N) \in \mathcal{M}_{c,loc}$. By localization, we may assume that $M, N \in \mathcal{M}_c^2$ and that H is bounded. By the OST, it suffices to show that

$$\mathbb{E}((H \cdot M)_T N_T) = \mathbb{E}(M_T (H \cdot N)_T)$$

for all bounded stopping times T. But the l.h.s. above equals $\mathbb{E}((H \cdot M^T)_\infty N_\infty^T)$, and the r.h.s. equals $\mathbb{E}(M_\infty^T (H \cdot N^T)_\infty)$, and $M^T, N^T \in \mathcal{M}_c^2$. It therefore suffices to show that

$$\mathbb{E}((H \cdot M)_\infty N_\infty) = \mathbb{E}(M_\infty (H \cdot N)_\infty) \tag{5.2}$$

for all $M, N \in \mathcal{M}_c^2$ and H bounded. Consider first the case that $H = Z\mathbb{1}_{(s,t]}$ where Z is bounded and $\mathcal{F}_s$-measurable. Then

$$\begin{aligned}
\mathbb{E}((H \cdot M)_\infty N_\infty) &= \mathbb{E}(Z(M_t - M_s)N_\infty) \\
&= \mathbb{E}(ZM_t \mathbb{E}(N_\infty \,|\, \mathcal{F}_t) - ZM_s \mathbb{E}(N_\infty \,|\, \mathcal{F}_s)) \quad \text{(tower property)} \\
&= \mathbb{E}(Z(M_t N_t - M_s N_s)).
\end{aligned}$$

The same argument implies that

$$\mathbb{E}(M_\infty (H \cdot N)_\infty) = \mathbb{E}(Z(M_t N_t - M_s N_s))$$

which proves (5.2) for $H = Z\mathbb{1}_{(s,t]}$.

By linearity, this extends to all $H \in \mathcal{S}$. If H is bounded, then we can find[4] a sequence $(H^n)_{n \geq 1}$ of simple processes such that $H^n \to H$ in both $L^2(M)$ and $L^2(N)$. Therefore $H^n \cdot M \to H \cdot M$ and $H^n \cdot N \to H \cdot N$ in $\|\cdot\|$ norm, i.e. $(H^n \cdot M)_\infty \to (H \cdot M)_\infty$ and $(H^n \cdot N)_\infty \to (H \cdot N)_\infty$ in L^2 as $n \to \infty$. It follows that

$$\|((H^n \cdot M)_\infty - (H \cdot M)_\infty) N_\infty\|_{L^1}^2 \leq \|(H^n \cdot M)_\infty - (H \cdot M)_\infty\|_{L^2}^2 \|N_\infty\|_{L^2}^2 \to 0,$$

which implies convergence of the expectation, i.e. $\mathbb{E}((H^n \cdot M)_\infty N_\infty) \to \mathbb{E}((H \cdot M)_\infty N_\infty)$. Similarly one shows that $\mathbb{E}(M_\infty (H^n \cdot N)_\infty) \to \mathbb{E}(M_\infty (H \cdot N)_\infty)$, which concludes the proof. □

Definition 5.4. *For X, Y continuous semimartingales we define their covariation $[X, Y]$ to be the covariation of their respective martingale parts in the Doob-Meyer decomposition.*

[4]This can be always insured by taking, for example, $\mu = d[M] + d[N]$ as measure to define the space of integrands.

This definition is justified by the fact that

$$[X,Y]^n_t := \sum_{k=0}^{\lceil 2^n t\rceil -1} (X_{(k+1)2^{-n}} - X_{k2^{-n}})(Y_{(k+1)2^{-n}} - Y_{k2^{-n}}) \to [X,Y]_t \qquad \text{ucp as } n\to\infty. \tag{5.3}$$

Note that the Kunita-Watanabe identity also holds for continuous semimartingales.

An important property of the covariation is that two independent semimartingales have zero covariation. However, having zero covariation is not a sufficient condition for independence, in the same way two uncorrelated random variables are not necessarily independent.

Proposition 5.5. *Let X, Y be independent*[5] *continuous semimartingales. Then $[X,Y]=0$.*

Proof. See Example sheet 2. □

5.2. **Ito's formula.** We now move to the celebrated Itô's formula. To start with, we prove a particular case of it, which can be thought of as the generalization of integration by parts to stochastic integrals.

Theorem 5.6 (Integration by parts)**.** *Let X, Y be continuous semimartingales. Then*

$$X_tY_t - X_0Y_0 = \int_0^t X_s dY_s + \int_0^t Y_s dX_s + [X,Y]_t.$$

Proof. Note that the above integrals are well defined, since any continuous adapted process is locally bounded and previsible. Since both sides are continuous, it suffices to prove the result for t of the form $t = m2^{-n}$ for $m,n\in\mathbb{N}$. Fix t of this form. Note that for $s\le t$ we have that

$$X_tY_t - X_sY_s = X_s(Y_t-Y_s) + Y_s(X_t-X_s) + (X_t-X_s)(Y_t-Y_s).$$

Applying this for $j\ge n$, we find

$$\begin{aligned} X_tY_t - X_0Y_0 &= \sum_{k=0}^{m2^{j-n}-1} (X_{(k+1)2^{-j}}Y_{(k+1)2^{-j}} - X_{k2^{-j}}Y_{k2^{-j}}) \\ &= \sum_{k=0}^{m2^{j-n}-1} \Big(X_{k2^{-j}}(Y_{(k+1)2^{-j}} - Y_{k2^{-j}}) + Y_{k2^{-j}}(X_{(k+1)2^{-j}} - X_{k2^{-j}}) \\ &\qquad\qquad + (X_{(k+1)2^{-j}} - X_{k2^{-j}})(Y_{(k+1)2^{-j}} - Y_{k2^{-j}}) \Big) \\ &\to (X\cdot Y)_t + (Y\cdot X)_t + [X,Y]_t \quad \text{ucp as} \quad n\to\infty, \end{aligned}$$

where convergence follows from (4.12) and (5.3). □

[5]Recall that X, Y are independent if $\sigma(X_t, t\ge 0)$ and $\sigma(Y_t, t\ge 0)$ are independent.

Note that the covariation term does not appear in the usual integration by parts formula for Lebesgue-Stieltjes integrals. This term, sometimes referred to as Itô's correction term, will vanish if either X or Y is of finite variation, or if X and Y are independent (cf. Proposition 5.5).

Theorem 5.7 (Itô's formula)**.** *Let $X^1, \dots, X^d$ be continuous semimartingales and let $X = (X^1, \dots, X^d)$. Let $f \colon \mathbb{R}^d \to \mathbb{R}$ be C^2. Then*

$$f(X_t) = f(X_0) + \sum_{i=1}^{d} \int_0^t \frac{\partial f}{\partial x_i}(X_s) dX_s^i + \frac{1}{2} \sum_{i,j=1}^{d} \int_0^t \frac{\partial^2 f}{\partial x_i \partial x_j}(X_s) d[X^i, X^j]_s.$$

Remark 5.8. *Note that Theorem 5.7 implies that $f(X_t)$ is a semimartingale since each of the summands in Itô's formula is a semimartingale.*

We will often apply Itô's formula for $d = 1$, that is

$$f(X_t) = f(X_0) + \int_0^t f'(X_s) dX_s + \frac{1}{2} \int_0^t f''(X_s) d[X]_s.$$

One can derive this formula using Taylor expansion. Indeed, we have that

$$\begin{aligned} f(X_t) &= f(X_0) + \sum_{k=0}^{\lfloor 2^n t \rfloor - 1} \big(f(X_{(k+1)2^{-n}}) - f(X_{k2^{-n}}) \big) + \big(f(X_t) - f(X_{2^{-n}\lfloor 2^n t \rfloor}) \big) \\ &= f(X_0) + \sum_{k=0}^{\lfloor 2^n t \rfloor - 1} f'(X_{k2^{-n}})(X_{(k+1)2^{-n}} - X_{k2^{-n}}) + \frac{1}{2} \sum_{k=0}^{\lfloor 2^n t \rfloor - 1} f''(X_{k2^{-n}})(X_{(k+1)2^{-n}} - X_{k2^{-n}})^2 + \text{ error} \\ &\to f(X_0) + \int_0^t f'(X_s) dX_s + \frac{1}{2} \int_0^t f''(X_s) d[X]_s \quad \text{ucp as} \quad n \to \infty. \end{aligned}$$

This is Itô's formula for $d = 1$. We will not take this route to prove Itô's formula, since the error term turns out to be difficult to deal with.

Proof of Theorem 5.7. We are going to give the proof of Itô's formula for $d = 1$. We leave the proof for $d > 1$ as an exercise. Write $X = X_0 + M + A$ where A has corresponding total variation process V. Let

$$T_r := \inf\{t \geq 0 : |X_t| + V_t + [M]_t > r\}.$$

Then $(T_r)_{r \geq 1}$ is a sequence of stopping times with $T_r \nearrow +\infty$ almost surely as $r \to \infty$. It suffices to prove the formula for arbitrary $f \in C^2(\mathbb{R})$ in the time-interval $[0, T_r]$. Let $\mathcal{A}$ denote the subset of $C^2(\mathbb{R})$ where the formula holds: we aim to show that $\mathcal{A} = C^2(\mathbb{R})$. To this end, we will show that:

(a) $\mathcal{A}$ contains the functions $f(x) \equiv 1$ and $f(x) = x$.
(b) $\mathcal{A}$ is a vector space.

(c) $\mathcal{A}$ is in fact an algebra. That is, if $f, g \in \mathcal{A}$ then $fg \in \mathcal{A}$.

(d) If $(f_n)_{n\geq 1}$ is a sequence in $\mathcal{A}$ with $f_n \to f$ in $C^2(B_r)$ for each $r > 0$, then $f \in \mathcal{A}$. Here, $B_r = \{x : |x| < r\}$ and $f_n \to f$ in $C^2(B_r)$ means that with

$$\Delta_{n,r} = \max\Big\{ \sup_{x\in B_r} |f_n(x) - f(x)|,\ \sup_{x\in B_r} |f_n'(x) - f'(x)|,\ \sup_{x\in B_r} |f_n''(x) - f''(x)| \Big\}$$

we have that $\Delta_{n,r} \to 0$ as $n \to \infty$ with r fixed.

Note that (a), (b), (c) together imply that all polynomials are in $\mathcal{A}$. By Weierstrass' approximation theorem, polynomials are dense in $C^2(B_r)$ for each $r > 0$, so (d) implies that $\mathcal{A} = C^2(\mathbb{R})$.

The fact that (a) and (b) hold is easy to check.

Proof of (c) Suppose that $f, g \in \mathcal{A}$. Set $F_t = f(X_t)$ and $G_t = g(X_t)$. Since Itô's formula holds for f and g, F and G must be continuous semimartingales. By integration by parts, we find

$$F_tG_t - F_0G_0 = \int_0^t F_s dG_s + \int_0^t G_s dF_s + [F, G]_t. \tag{5.4}$$

Using that $H \cdot (K \cdot M) = (HK) \cdot M$ and Itô's formula for g, we also have that

$$\begin{aligned} \int_0^t F_s dG_s &= \int_0^t f(X_s) d\bigg(\int_0^s g'(X_u) dX_u + \frac{1}{2} \int_0^u g''(X_u) d[X]_u \bigg) \\ &= \int_0^t f(X_s) g'(X_s) dX_s + \frac{1}{2} \int_0^t f(X_s) g''(X_s) d[X]_s. \end{aligned} \tag{5.5}$$

Similarly one shows that

$$\int_0^t G_s dF_s = \int_0^t f'(X_s) g(X_s) dX_s + \frac{1}{2} \int_0^t f''(X_s) g(X_s) d[X]_s. \tag{5.6}$$

Finally, by the Kunita-Watanabe identity, we have that $[H \cdot M, N] = H \cdot [M, N]$, hence

$$[F, G]_t = [f(X), g(X)]_t = [f'(X) \cdot X, g'(X) \cdot X]_t = \int_0^t f'(X_s) g'(X_s) d[X]_s. \tag{5.7}$$

Inserting (5.5)-(5.6)-(5.7) into (5.4) implies that Itô's formula holds for fg. That is, $fg \in \mathcal{A}$.

Proof of (d) Suppose that $(f_n)_{n\geq 1}$ is a sequence in $\mathcal{A}$ and we have that $f_n \to f$ in $C^2(B_r)$ for all $r > 0$. We want to show that Itô's formula holds for f, that is $f \in \mathcal{A}$. Since $f_n \in \mathcal{A}$ for all n, we have that

$$\begin{aligned} f_n(X_t) &= f_n(X_0) + \int_0^t f_n'(X_s) dX_s + \frac{1}{2} \int_0^t f_n''(X_s) d[X]_s \\ &= f_n(X_0) + \int_0^t f_n'(X_s) dA_s + \int_0^t f_n'(X_s) dM_s + \frac{1}{2} \int_0^t f_n''(X_s) d[M]_s. \end{aligned}$$

We show that each term above converges to the corresponding one with f in place of f_n, thus proving Itô's formula for f.

Let us start from the finite variation part. We have

$$\int_0^{t\wedge T_r} |f_n'(X_s) - f'(X_s)|dV_s + \frac{1}{2}\int_0^{t\wedge T_r} |f_n''(X_s) - f''(X_s)|d[M]_s \\ \leq \Delta_{n,r} V_{t\wedge T_r} + \frac{1}{2}\Delta_{n,r}[M]_{t\wedge T_r} \leq r\Delta_{n,r} \to 0 \quad \text{as} \quad n \to \infty.$$

Consequently,

$$\int_0^{t\wedge T_r} f_n'(X_s)dA_s + \frac{1}{2}\int_0^{t\wedge T_r} f_n''(X_s)d[M]_s \to \int_0^{t\wedge T_r} f'(X_s)dA_s + \frac{1}{2}\int_0^{t\wedge T_r} f''(X_s)d[M]_s$$

uniformly in t, almost surely. Moreover, for the martingale part we have that $M^{T_r} \in \mathcal{M}_c^2$, from which

$$\|(f_n'(X)\cdot M)^{T_r} - (f'(X)\cdot M)^{T_r}\|^2 = \mathbb{E}\left(\int_0^{T_r} (f_n'(X_s) - f'(X_s))^2 d[M]_s\right) \\ \leq \Delta_{n,r}^2\, \mathbb{E}([M]_{T_r}) \leq r\Delta_{n,r}^2 \to 0 \quad \text{as} \quad n \to \infty.$$

Therefore $(f_n'(X)\cdot M)^{T_r} \to (f'(X)\cdot M)^{T_r}$ in $\mathcal{M}_c^2$ as $n \to \infty$.

Hence, for any r, we can pass to the limit in Itô's formula to obtain

$$f(X_{t\wedge T_r}) = f(X_0) + \int_0^{t\wedge T_r} f'(X_s)dX_s + \frac{1}{2}\int_0^{t\wedge T_r} f''(X_s)d[X]_s.$$

□

Example 5.9. Let $X = B$ where B is a standard Brownian motion in $\mathbb{R}$ and let $f(x) = x^2$. Applying Itô's formula to $f(B)$, we then have that

$$B_t^2 = 2\int_0^t B_s dB_s + t.$$

Rearranging, we obtain

$$B_t^2 - t = 2\int_0^t B_s dB_s \in \mathcal{M}_{c,loc}.$$

Example 5.10. Let $f\colon \mathbb{R}_+ \times \mathbb{R}^d \to \mathbb{R}$ be $C^{1,2}$ and let $X_t = (t, B_t^1, \ldots, B_t^d)$ where $B^1, \ldots, B^d$ are independent standard Brownian motions. Then, by Itô's formula, we have that

$$f(t, B_t) - f(0, B_0) - \int_0^t \left(\frac{\partial}{\partial t} + \frac{1}{2}\Delta\right) f(s, B_s)ds = \sum_{i=1}^d \int_0^t \frac{\partial}{\partial x_i} f(s, B_s)dB_s^i \in \mathcal{M}_{c,loc}.$$

In particular, if f does not depend on t and f is harmonic in x, then $f(B_t) \in \mathcal{M}_{c,loc}$. If furthermore f is bounded, then $f(B_t)$ is a true bounded martingale.

5.3. **Stratonovich integral.** Let X, Y be continuous semimartingales. We define the *Stratonovich integral* as

$$\int_0^t X_s \partial Y_s = \int_0^t X_s dY_s + \frac{1}{2}[X,Y]_t.$$

Here, the integral on the right hand side is an Itô integral. Using the Riemann sum approximations for the terms on the right hand side, we have that

$$\sum_{k=0}^{\lfloor 2^n t \rfloor - 1} \left(\frac{X_{(k+1)2^{-n}} + X_{k2^{-n}}}{2} \right) (Y_{(k+1)2^{-n}} - Y_{k2^{-n}}) \to \int_0^t X_s \partial Y_s \quad \text{ucp as} \quad n \to \infty.$$

Proposition 5.11. *Let $X^1, \dots, X^d$ be continuous semimartingales and let $f \colon \mathbb{R}^d \to \mathbb{R}$ be C^3. Then*

$$f(X_t) = f(X_0) + \sum_{i=1}^d \int_0^t \frac{\partial f}{\partial x_i}(X_s) \partial X_s^i. \tag{5.8}$$

In particular, the integration by parts formula is given by

$$X_t Y_t - X_0 Y_0 = \int_0^t X_s \partial Y_s + \int_0^t Y_s \partial X_s. \tag{5.9}$$

Note that Proposition 5.11 implies that the Stratonovich integral satisfies the usual rules of calculus. On the other hand, the price to pay for this property is that the Stratonovich integral with respect to a local martingale is not necessarily a local martingale. Indeed, for example,

$$\int_0^t B_s \partial B_s = \int_0^t B_s dB_s + \frac{1}{2}t = \frac{1}{2}B_t^2 \ \notin \mathcal{M}_{c,loc}.$$

Proof of Proposition 5.11. We will treat the case that $d = 1$ and leave the case $d \geq 2$ as an exercise. By Itô's formula, we have that

$$f(X_t) = f(X_0) + \int_0^t f'(X_s) dX_s + \frac{1}{2} \int_0^t f''(X_s) d[X]_s \tag{5.10}$$

$$f'(X_t) = f'(X_0) + \int_0^t f''(X_s) dX_s + \frac{1}{2} \int_0^t f'''(X_s) d[X]_s. \tag{5.11}$$

By the Kunita-Watanabe identity and (5.11), we have that $[f'(X), X] = [\int_0^t f''(X_s) dX_s, X] = f''(X) \cdot [X]$. Combining this with (5.10) and the definition of the Stratonovich integral, we then have that

$$f(X_t) = f(X_0) + \int_0^t f'(X_s) \partial X_s.$$

□

Remark 5.12. *Note that in Proposition 5.11, we require more regularity of the function f in order to obtain* (5.8) *in comparison to Itô's formula. We also emphasize that for the Stratonovich integral, the integrand must be a semimartingale. This is also in contrast to the case of Itô's formula, in which case the integrand need only be locally bounded and previsible.*

5.4. **Shorthand notation.** We now record some shorthand notation which is common for stochastic calculus.

- If $Z_t = Z_0 + \int_0^t H_s dX_s$ write $dZ_t = H_t dX_t$.
- If $Z_t = Z_0 + \int_0^t H_s \partial X_s$ write $\partial Z_t = H_t \partial X_t$.
- If $Z_t = [X,Y]_t = \int_0^t d[X,Y]_s$ we write $dZ_t = dX_t dY_t$.

We then have the following computational rules:

- $H_t d(K_t dX_t) = (H_t K_t) dX_t$ (iterated integral),
- $H_t(dX_t dY_t) = d(H_t dX_t) dY_t$ (Kunita-Watanabe),
- $d(X_t Y_t) = X_t dY_t + Y_t dX_t + dX_t dY_t$ (integration by parts),
- $df(X_t) = \sum_{i=1}^d \frac{\partial f}{\partial x_i}(X_t) dX_t^i + \frac{1}{2}\sum_{i,j=1}^d \frac{\partial^2 f}{\partial x_i \partial x_j}(X_t) dX_t^i dX_t^j$ (Itô's formula).

The relationship between the Itô and Stratonovich integrals is that $\partial Z_t = Y_t \partial X_t$ if and only if $dZ_t = Y_t dX_t + \frac{1}{2} dX_t dY_t$.

6. Applications

6.1. **Brownian motion.** Throughout, we will work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ where $(\mathcal{F}_t)_{t\geq 0}$ satisfies the usual conditions.

Theorem 6.1 (Lévy's characterization of Brownian motion)**.** *Let $X^1, \dots, X^d$ be continuous local martingales and set $X = (X^1, \dots, X^d)$. Suppose that $X_0 = 0$ and that $[X^i, X^j]_t = \delta_{ij} t$ for all i, j and $t \geq 0$. Then X is a standard Brownian motion on $\mathbb{R}^d$.*

Proof. It suffices to show that, for all $0 \leq s \leq t < \infty$, we have that $X_t - X_s$ is independent of $\mathcal{F}_s$ and it is distributed as a $N(0, (t-s) I_d)$ random variable, where I_d denotes the $d \times d$ identity matrix. This is equivalent to showing that

$$\mathbb{E}(e^{i(\theta, X_t - X_s)} \,|\, \mathcal{F}_s) = \exp\left(-\frac{1}{2}|\theta|^2 (t-s)\right), \quad \text{for all} \quad \theta \in \mathbb{R}^d,$$

where $(\cdot,\cdot)$ denotes the scalar product in $\mathbb{R}^d$, and $|\theta|^2 = (\theta, \theta)$.

Fix $\theta \in \mathbb{R}^d$ and set $Y_t = (\theta, X_t) = \sum_{j=1}^d \theta^j X_t^j$. Then $Y \in \mathcal{M}_{c,loc}$ as $\mathcal{M}_{c,loc}$ is a vector space. By assumption, we have that

$$[Y]_t = [Y,Y]_t = \Big[\sum_{j=1}^d \theta^j X^j, \sum_{k=1}^d \theta^k X^k\Big]_t = \sum_{j,k=1}^d \theta^j \theta^k [X^j, X^k]_t = |\theta|^2 t.$$

Consider

$$Z_t := \exp(iY_t + \tfrac{1}{2}[Y]_t) = \exp(i(\theta, X_t) + \tfrac{1}{2}|\theta|^2 t).$$

By Itô's formula with $X = iY + \frac{1}{2}[Y]$ and $f(x) = e^x$, we have that

$$dZ_t = Z_t(idY_t + \tfrac{1}{2}d[Y]_t) - \tfrac{1}{2}Z_t d[Y]_t = iZ_t dY_t.$$

Consequently, $Z \in \mathcal{M}_{c,loc}$. Moreover, Z is bounded on $[0,t]$ for each $t \geq 0$, hence $Z \in \mathcal{M}_c$ by Proposition 3.9. Therefore $\mathbb{E}(Z_t \,|\, \mathcal{F}_s) = Z_s$, which implies that

$$\mathbb{E}(\exp(i(\theta, X_t - X_s)) \,|\, \mathcal{F}_s) = \exp(-\tfrac{1}{2}|\theta|^2(t-s)).$$

□

The following result is a powerful consequence of Lévy's characterization of Brownian motion.

Theorem 6.2 (Dubins-Schwarz theorem)**.** *Let $M \in \mathcal{M}_{c,loc}$ with $M_0 = 0$ and $[M]_\infty = \infty$. Set $\tau_s := \inf\{t \geq 0 : [M]_t > s\}$, $B_s := M_{\tau_s}$, and $\mathcal{G}_s := \mathcal{F}_{\tau_s}$. Then τ_s is an $(\mathcal{F}_t)$-stopping time and $[M]_{\tau_s} = s$ for all $s \geq 0$. Moreover, B is a $(\mathcal{G}_s)$-Brownian motion and*

$$M_t = B_{[M]_t}.$$

Remark 6.3. *Theorem 6.2 states that any continuous local martingale starting from 0 can be represented as a stochastic time-change of Brownian motion. Thus, in some sense, Brownian motion is the "most general" continuous local martingale.*

Proof of Theorem 6.2. Since $[M]$ is continuous and adapted, τ_s is a stopping time for each $s \geq 0$. The hypothesis $[M]_\infty = \infty$ implies that $\tau_s < \infty$ for all $s \geq 0$. Note that $(\mathcal{G}_s)$ is a filtration, since if S, T are two stopping times with $S \leq T$ almost surely, and $A \in \mathcal{F}_S$, then

$$A \cap \{T \leq t\} = (A \cap \{S \leq t\}) \cap \{T \leq t\} \in \mathcal{F}_t,$$

so $A \in \mathcal{F}_T$ and hence $\mathcal{F}_S \subseteq \mathcal{F}_T$. Applying this with $S = \tau_r$ and $T = \tau_s$ gives that $\mathcal{G}_r \subseteq \mathcal{G}_s$ for all $r \leq s$.

To see that B is adapted to $(\mathcal{G}_s)_{s\geq 0}$ we have to show that B_{τ_s} is $\mathcal{F}_{\tau_s}$-measurable for all $s \geq 0$. Indeed, this follows from the general fact that if X is a càdlàg adapted process and T is a stopping time, then $X_T \mathbb{1}_{\{T<\infty\}}$ is $\mathcal{F}_T$-measurable (see the Advanced Probability course).

To finish the proof, we will show that B is continuous and that it is a local martingale with $[B]_t = t$. The result will then follow from Lévy's characterization of Brownian motion.

Step 1: B is continuous. We first note that $s \mapsto \tau_s$ is non-decreasing and càdlàg . Therefore B is càdlàg . To prove the continuity of B, we must therefore show that $B_{s^-} = B_s$ for all $s \geq 0$. Equivalently, $M_{\tau_{s^-}} = M_{\tau_s}$ for all $s \geq 0$, with $\tau_{s^-} = \inf\{t \geq 0 : [M]_t = s\}$. If $\tau_s = \tau_{s^-}$ there is nothing to show. If, on the other hand, $\tau_s > \tau_{s^-}$ then $[M]$ must be constant on the time

interval $[\tau_{s^-}, \tau_s]$. It thus suffices to prove that if $[M]$ is constant on a given interval then M is also constant on the same interval.

By applying localization, we may assume that $M \in \mathcal{M}_c^2$. It suffices to show that for all $q \in \mathbb{Q}$, $q > 0$, with $S_q = \inf\{t > q : [M]_t > [M]_q\}$ we have that M is constant on $[q, S_q]$. We know that $M^2 - [M]$ is a UI martingale as $M \in \mathcal{M}_c^2$. By the OST, we have that

$$\mathbb{E}(M_{S_q}^2 - [M]_{S_q} \,|\, \mathcal{F}_q) = M_q^2 - [M]_q.$$

Since $[M]_{S_q} = [M]_q$ and M is a martingale, we also have that

$$\mathbb{E}((M_{S_q} - M_q)^2 \,|\, \mathcal{F}_q) = \mathbb{E}(M_{S_q}^2 - M_q^2 | \mathcal{F}_q) = \mathbb{E}[[M]_{S_q} - [M]_q \,|\, \mathcal{F}_q] = 0.$$

Therefore M is almost surely constant on $[q, S_q]$, hence B is continuous.

Step 2: B is a Brownian motion. Fix $s > 0$. Then we know that $[M^{\tau_s}]_\infty = [M]_{\tau_s} = s$. Consequently, $M^{\tau_s} \in \mathcal{M}_c^2$ since $\mathbb{E}([M^{\tau_s}]_\infty) < \infty$. Therefore $(M^2 - [M])^{\tau_s}$ is a UI martingale. By the OST, we have for $0 \le r < s < \infty$ that

$$\begin{aligned}\mathbb{E}(B_s \,|\, \mathcal{G}_r) &= \mathbb{E}(M_\infty^{\tau_s} \,|\, \mathcal{F}_{\tau_r}) = M_{\tau_r} = B_r \\ \mathbb{E}(B_s^2 - s \,|\, \mathcal{G}_r) &= \mathbb{E}((M^2 - [M])_\infty^{\tau_s} \,|\, \mathcal{F}_{\tau_r}) = M_{\tau_r}^2 - [M]_{\tau_r} = B_r^2 - r.\end{aligned}$$

Combining, we have that $B \in \mathcal{M}_c$ with $[B]_t = t$ for all $t \ge 0$. Thus by Theorem 6.1, we have that B is a $(\mathcal{G}_t)_{t\ge 0}$-Brownian motion. □

Here is an extension to Dubins-Schwarz theorem that we state without proof.

Theorem 6.4. *Let $M \in \mathcal{M}_{c,loc}$ with $M_0 = 0$. Then define a new process B on the same probability space by setting*

$$B_s = \begin{cases} M_{\tau_s}, & s \le [M]_\infty, \\ M_\infty + \beta_{s - [M]_\infty}, & s \ge [M]_\infty \end{cases}$$

where β is a standard Brownian motion defined on the same probability space, independent of everything else. Then B is a standard Brownian motion, and $M_t = B_{[M]_t}$ for all $t \ge 0$.

Example 6.5. Let B be a Brownian motion, and h be a deterministic measurable function in $L^2([0,\infty))$. Let $M_t := \int_0^t h(s) dB_s$. Then $M_0 = 0$, $M \in \mathcal{M}_{c,loc}$ and $[M]_t = \int_0^t h^2(s) ds$. Then

$$M_\infty \overset{(d)}{=} B_{\int_0^\infty h^2(s) ds} \sim N(0, \|h\|_{L^2}^2).$$

Example 6.6. Let $M \in \mathcal{M}_{c,loc}$. Then

$$\{[M]_\infty < \infty\} = \Big\{ \lim_{t\to\infty} M_t \text{ exists finite} \Big\}.$$

Conversely,

$$\{[M]_\infty = \infty\} = \Big\{ \liminf_{t\to\infty} M_t = -\infty, \ \limsup_{t\to\infty} M_t = +\infty \Big\}.$$

6.2. **Exponential martingales.** Let $M \in \mathcal{M}_{c,loc}$ with $M_0 = 0$. Set $Z_t = \exp(M_t - \frac{1}{2}[M]_t)$. By Itô's formula applied to $X = M - \frac{1}{2}[M]$ and $f(x) = e^x$, we have that

$$dZ_t = Z_t(dM_t - \tfrac{1}{2}d[M]_t) + \tfrac{1}{2}Z_t d[M]_t = Z_t dM_t.$$

Consequently, $Z \in \mathcal{M}_{c,loc}$ as it is the integral of Z with respect to an element of $\mathcal{M}_{c,loc}$. We call Z the stochastic exponential of M.

Definition 6.7 (Exponential martingale)**.** *Let $M \in \mathcal{M}_{c,loc}$ with $M_0 = 0$. Define the process $\mathcal{E}(M)_t$ by setting*

$$\mathcal{E}(M)_t = \exp(M_t - \tfrac{1}{2}[M]_t).$$

Then $\mathcal{E}(M)$ is called the stochastic exponential of (or exponential martingale associated with) M. Note that $\mathcal{E}(M) \in \mathcal{M}_{c,loc}$, and it satisfies $d\mathcal{E}(M)_t = \mathcal{E}(M)_t dM_t$.

The next result shows that under some additional assumptions on M the stochastic exponential is a true UI martingale.

Proposition 6.8. *Let $M \in \mathcal{M}_{c,loc}$ with $M_0 = 0$. Suppose that $[M]$ is uniformly bounded. Then $\mathcal{E}(M)$ is a UI martingale.*

To prove the above result we will use a powerful inequality which tells us that if the quadratic variation of a continuous local martingale is small then the martingale cannot be too large.

Proposition 6.9. *Let $M \in \mathcal{M}_{c,loc}$ with $M_0 = 0$. Then for all $\epsilon, \delta > 0$ we have that*

$$\mathbb{P}\left(\sup_{t\geq 0} M_t \geq \epsilon,\ [M]_\infty \leq \delta \right) \leq \exp\left(-\frac{\epsilon^2}{2\delta} \right).$$

Proof. Fix $\epsilon > 0$ and set $T = \inf\{t \geq 0 : M_t \geq \epsilon\}$. Fix $\theta > 0$ and set $Z_t = \exp(\theta M_t^T - \frac{1}{2}\theta^2 [M]_t^T)$. Then $Z_t \in \mathcal{M}_{c,loc}$ since $Z = \mathcal{E}(\theta M^T)$ and $\theta M^T \in \mathcal{M}_{c,loc}$. Moreover, $|Z_t| \leq e^{\theta\epsilon}$ for all $t \geq 0$ by construction, so Z is in fact a bounded martingale. Therefore $\mathbb{E}(Z_\infty) = Z_0 = 1$. For each $\delta > 0$, we have that

$$\begin{aligned}
\mathbb{P}\left(\sup_{t\geq 0} M_t \geq \epsilon,\ [M]_\infty \leq \delta \right) &= \mathbb{P}\left(\sup_{t\geq 0} e^{\theta M_t^T} \geq e^{\theta\epsilon},\ [M]_\infty \leq \delta \right) \\
&\leq \mathbb{P}\Big(\sup_{t\geq 0} Z_t \geq e^{\theta\epsilon - \theta^2\delta/2} \Big) \\
&\leq \exp(-\theta\epsilon + \theta^2\delta/2) \qquad \text{(Doob's maximal inequality).}
\end{aligned}$$

The result follows by optimizing this bound over θ. □

We can now prove Proposition 6.8.

Proof of Proposition 6.8. We show that $\mathcal{E}(M)$ is uniformly bounded by an integrable random variable, which implies the result. Note that

$$\sup_{t\geq 0}\mathcal{E}(M)_t \leq \exp\Big(\sup_{t\geq 0} M_t\Big)$$

as $[M]_t \geq 0$ for all $t \geq 0$. It suffices to show that the r.h.s. is integrable. Let c be such that $[M]_\infty \leq c$. By Proposition 6.9, we have that

$$\mathbb{P}\bigg(\sup_{t\geq 0} M_t \geq \epsilon\bigg) = \mathbb{P}\bigg(\sup_{t\geq 0} M_t \geq \epsilon,\ [M]_\infty \leq c\bigg) \leq \exp\left(-\frac{\epsilon^2}{2c}\right).$$

It follows that

$$\begin{aligned}\mathbb{E}\Big(\exp\Big(\sup_{t\geq 0} M_t\Big)\Big) &= \int_0^\infty \mathbb{P}\Big(\exp\Big(\sup_{t\geq 0} M_t\Big) \geq \lambda\Big)d\lambda \\ &= \int_0^\infty \mathbb{P}\Big(\sup_{t\geq 0} M_t \geq \log\lambda\Big)d\lambda \\ &\leq 1 + \int_1^\infty \exp\left(-\frac{(\log\lambda)^2}{2c}\right)d\lambda < \infty.\end{aligned}$$

Therefore $\mathcal{E}(M)$ is UI, as desired. □

Proposition 6.8 provides one circumstance in which the hypotheses of the following theorem are satisfied.

Suppose that $\mathbb{P}$ and $\mathbb{Q}$ are two probability measures on $(\Omega, \mathcal{F})$. Then we say that $\mathbb{Q}$ is *absolutely continuous* with respect to $\mathbb{P}$ and write $\mathbb{Q} \ll \mathbb{P}$ if for any $A \in \mathcal{F}$, we have that $\mathbb{P}(A) = 0$ implies $\mathbb{Q}(A) = 0$.

Theorem 6.10 (Girsanov's theorem)**.** *Let $M \in \mathcal{M}_{c,loc}$ be such that $M_0 = 0$. Suppose that $Z = \mathcal{E}(M)$ is a UI martingale. Then we can define a new probability measure $\widetilde{\mathbb{P}} \ll \mathbb{P}$ on $(\Omega, \mathcal{F})$ by setting $\widetilde{\mathbb{P}}(A) = \mathbb{E}(Z_\infty \mathbb{1}(A))$ for $A \in \mathcal{F}$. If $X \in \mathcal{M}_{c,loc}(\mathbb{P})$, then $X - [X, M] \in \mathcal{M}_{c,loc}(\widetilde{\mathbb{P}})$.*

Proof. Since Z is UI, we know that Z_∞ exists with $Z_\infty \geq 0$ and $\mathbb{E}(Z_\infty) = \mathbb{E}(Z_0) = 1$. It is not difficult to see that $\widetilde{\mathbb{P}}$ is a probability measure with $\widetilde{\mathbb{P}} \ll \mathbb{P}$. Let $X \in \mathcal{M}_{c,loc}(\mathbb{P})$ and set $T_n = \inf\{t \geq 0 : |X_t - [X, M]_t| \geq n\}$. As $X - [X, M]$ is continuous, we have that $\mathbb{P}(T_n \nearrow \infty) = 1$. By absolute continuity, we therefore have that $\widetilde{\mathbb{P}}(T_n \nearrow \infty) = 1$. So, to show that $Y = X - [X, M] \in \mathcal{M}_{c,loc}(\widetilde{\mathbb{P}})$, it suffices to show that $Y^{T_n} = X^{T_n} - [X^{T_n}, M] \in \mathcal{M}_c(\widetilde{\mathbb{P}})$ for all n. Write X and Y in place of X^{T_n} and Y^{T_n} in what follows.

By integration by parts, we have that

$$\begin{aligned}d(Z_tY_t) &= Y_t dZ_t + Z_t dY_t + dY_t dZ_t \\ &= Y_t dZ_t + Z_t dX_t - Z_t dX_t dM_t + dX_t dZ_t \\ &= Y_t dZ_t + Z_t dX_t,\end{aligned}$$

where we have used that $dZ_t = Z_t dM_t$. Consequently, $ZY \in \mathcal{M}_{c,loc}(\mathbb{P})$. Also, $\{Z_T : T \leq t \text{ is a stopping time}\}$ is UI for all t. Since Y is bounded, we also have that $\{Z_T Y_T : T \leq t \text{ is a stopping time}\}$ is UI for all t as well. Hence, $ZY \in \mathcal{M}_c(\mathbb{P})$. But, for $s \leq t$, we have that

$$\widetilde{\mathbb{E}}(Y_t - Y_s \,|\, \mathcal{F}_s) = \mathbb{E}(Z_\infty Y_t - Z_\infty Y_s \,|\, \mathcal{F}_s) = \mathbb{E}(Z_t Y_t - Z_s Y_s \,|\, \mathcal{F}_s) = 0,$$

as $ZY \in \mathcal{M}_c(\mathbb{P})$. Therefore $Y \in \mathcal{M}_c(\widetilde{\mathbb{P}})$, as required. □

Remark 6.11. *Note that the quadratic variation does not change when performing a change of measures from $\mathbb{P}$ to $\widetilde{\mathbb{P}}$ (cf. Example Sheet 4).*

Corollary 6.12. *Let B be a standard Brownian motion under $\mathbb{P}$ and let $M \in \mathcal{M}_{c,loc}$ with $M_0 = 0$. Suppose that $Z = \mathcal{E}(M)$ is a UI martingale and $\widetilde{\mathbb{P}}(A) = \mathbb{E}(Z_\infty \mathbb{1}_A)$ for $A \in \mathcal{F}$. Then $\widetilde{B} = B - [B, M]$ is a $\widetilde{\mathbb{P}}$-Brownian motion.*

Proof. Since $\widetilde{B}$ is a continuous $\widetilde{\mathbb{P}}$-local martingale by Girsanov's theorem and has $[\widetilde{B}]_t = [B]_t = t$, it follows from the Lévy characterization that $\widetilde{B}$ must be a $\widetilde{\mathbb{P}}$-Brownian motion. □

7. Stochastic differential equations

In ths section we turn to the study of Stochastic Differential Equations, SDEs in short, of the form

$$dX_t = b(X_t)dt + \sigma(X_t)dB_t.$$

Recall from the introduction that these arise from Ordinary Differential equations by perturbation.

7.1. General definitions.

Let $\mathscr{M}^{d\times m}(\mathbb{R})$ denote the set of $d \times m$ matrices with real entries. Suppose that $\sigma\colon \mathbb{R}^d \to \mathscr{M}^{d\times m}(\mathbb{R})$ and $b\colon \mathbb{R}^d \to \mathbb{R}^d$ are measurable functions which are bounded on compact sets. We write $\sigma(x) = (\sigma_{ij}(x))$ and $b(x) = (b_i(x))$. Consider the equation

$$dX_t = \sigma(X_t)dB_t + b(X_t)dt. \tag{7.1}$$

We can write this equation component-wise as

$$dX_t^i = \sum_{j=1}^m \sigma_{ij}(X_t)dB_t^j + b_i(X_t)dt.$$

A solution to (7.1) consists of:

- A filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ where $(\mathcal{F}_t)_{t\geq 0}$ satisfies the usual conditions.
- An $(\mathcal{F}_t)$-Brownian motion $B = (B^1, \ldots, B^m)$ in $\mathbb{R}^m$.
- An $(\mathcal{F}_t)$-adapted continuous process $X = (X^1, \ldots, X^d)$ in $\mathbb{R}^d$ such that

$$X_t = X_0 + \int_0^t \sigma(X_s)dB_s + \int_0^t b(X_s)ds.$$

When, in addition, $X_0 = x \in \mathbb{R}^d$, we say that X is a solution started from x.

There are several different notions of existence and uniqueness for solutions to SDEs.

We say that an SDE has a *weak solution* if for all $x \in \mathbb{R}^d$, there exists a solution to the equation started from x. There is *uniqueness in law* if all solutions to the SDE started from x have the same distribution. There is *pathwise uniqueness* if, when we fix $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ and B, then any two solutions X and X' such that $X_0 = X_0'$ are indistinguishable, that is $\mathbb{P}(X_t = X_t' \text{ for all } t \geq 0) = 1$. We say that a solution X of a SDE started from x is a *strong solution* if X is adapted to the filtration generated by B.

In the case of a weak solution, the filtration $(\mathcal{F}_t)_{t\geq 0}$ is not necessarily the filtration generated by B. In other words, it is not necessarily true that $X_t(\omega)$ is a measurable function of $(B_s(\omega) : s \leq t)$ and x.

Example 7.1. It is possible to have the existence of a weak solution and uniqueness in law without pathwise uniqueness.

Suppose that β is a Brownian motion in $\mathbb{R}$ with $\beta_0 = x$. Set $B_t = \int_0^t \operatorname{sgn}(\beta_s) d\beta_s$. We take the convention that $\operatorname{sgn}(x) = \mathbb{1}_{(0,\infty)}(x) - \mathbb{1}_{(-\infty,0]}(x)$. Note that $(\operatorname{sgn}(\beta_t))_{t\geq 0}$ is previsible since left-continuous, so that the integral is well-defined. Note that

$$x + \int_0^t \operatorname{sgn}(\beta_s) dB_s = x + \int_0^t (\operatorname{sgn}(\beta_s))^2 d\beta_s = x + \int_0^t d\beta_s = \beta_t.$$

Consequently, β solves the SDE

$$\begin{cases} dX_t = \operatorname{sgn}(X_t) dB_t \\ X_0 = x. \end{cases} \tag{7.2}$$

Therefore (7.2) has a weak solution. By the Lévy characterization of Brownian motion, any solution to this SDE is a Brownian motion. Therefore we have uniqueness in law.

On the other hand, there is not pathwise uniqueness. To see this take $x = 0$: we show that both β and $-\beta$ are solutions to (7.2). Indeed,

$$-\beta_t = -\int_0^t \operatorname{sgn}(\beta_s) dB_s = \int_0^t \operatorname{sgn}(-\beta_s) dB_s + 2\int_0^t \mathbb{1}(\beta_s = 0) dB_s.$$

The second term in the r.h.s. is a continuous local martingale with quadratic variation $4\int_0^t \mathbb{1}(\beta_s = 0) ds = 0$, since the zero set of Brownian motion has zero Lebesgue measure by Fubini. It follows that $\int_0^t \mathbb{1}(\beta_s = 0) dB_s$ is indistinguishable from the zero martingale. We thus conclude that both β and $-\beta$ are solutions (with the same probability space and Brownian motion) to (7.2), thus contradicting pathwise uniqueness.

7.2. **Lipschitz coefficients.** We now turn to the question of sufficient conditions for existence and uniqueness of solutions to (7.1).

Recall that for $U \subseteq \mathbb{R}^d$ and $f\colon U \to \mathbb{R}^n$, we say that f is *Lipschitz* if there exists $K < \infty$ such that $|f(x) - f(y)| \leq K|x - y|$ for all $x, y \in U$. For $d, m \geq 1$ integers, let $\mathscr{M}^{d\times m}(\mathbb{R})$ denote the space of $d \times m$ matrices with real entries. We equip this space with the following Frobenius norm: if $A \in \mathscr{M}^{d\times m}(\mathbb{R})$ with $A = (a_{ij})$, define

$$|A| := \left(\sum_{i=1}^{d}\sum_{j=1}^{m} a_{ij}^2\right)^{1/2}.$$

Now, if $f\colon U \to \mathscr{M}^{d\times m}(\mathbb{R})$, then we say that f Lipschitz if there exists $K < \infty$ such that $|f(x) - f(y)| \leq K|x - y|$ for all $x, y \in U$.

We establish the existence and uniqueness of solutions to SDEs when the coefficients σ and b are Lipschitz.

Theorem 7.2 (Existence and uniqueness of solutions)**.** *Suppose that $\sigma\colon \mathbb{R}^d \to \mathscr{M}^{d\times m}(\mathbb{R})$ and $b\colon \mathbb{R}^d \to \mathbb{R}^d$ are Lipschitz. Then there is pathwise uniqueness for the SDE*

$$dX_t = \sigma(X_t)dB_t + b(X_t)dt. \tag{7.3}$$

Moreover, for each filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F})_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions and each $(\mathcal{F}_t)$-Brownian motion B, there exists for each $x \in \mathbb{R}^d$ a strong solution starting from x.

In our proof the existence of solutions to SDEs with Lipschitz coefficients, we will make use of the following theorem which we state without proof.

Theorem 7.3 (Contraction mapping theorem)**.** *Let (X, d) be a complete metric space.*

(a) Suppose that $F\colon X \to X$ is a contraction, i.e. there exists $r < 1$ such that $d(F(x), F(y)) \leq r\, d(x, y)$ for all $x, y \in X$. Then F has a unique fixed point.
(b) Suppose that $F\colon X \to X$ is such that there exists $r < 1$ and $n \in \mathbb{N}$ with $d(F^n(x), F^n(y)) < r\, d(x, y)$ for all $x, y \in X$. Then F has a unique fixed point.

Uniqueness will on the other hand follow as an application of Gronwall's inequality, that we now recall.

Lemma 7.4 (Gronwall's lemma)**.** *Let $T > 0$ and let f be a non-negative, bounded, measurable function on $[0, T]$. Suppose that there exists $a, b \geq 0$ such that*

$$f(t) \leq a + b\int_0^t f(s)ds \quad \text{for all} \quad t \in [0, T].$$

Then

$$f(t) \leq ae^{bt} \quad \text{for all} \quad t \in [0, T].$$

Proof. See Example sheet 3. □

We can now prove the existence and uniqueness result.

Proof of Theorem 7.2. We will give the proof in the case that $d, m = 1$ and leave the general case as an exercise. Let $K > 0$ be such that

$$|\sigma(x) - \sigma(y)| \leq K|x - y| \quad \text{and} \quad |b(x) - b(y)| \leq K|x - y| \quad \text{for all} \quad x, y \in \mathbb{R}.$$

Pathwise uniqueness.

Suppose that X, X' are solutions to the SDE with the same probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ and Brownian motion B, and with $X_0 = X'_0$. We will show that they must be indistinguishable, that is $X_t = X'_t$ for all $t \geq 0$ almost surely. Fix $M > 0$ and let

$$\tau = \inf\{t \geq 0 : |X_t| \vee |X'_t| \geq M\}.$$

Then we have that

$$X_{t\wedge\tau} = X_0 + \int_0^{t\wedge\tau} \sigma(X_s)dB_s + \int_0^{t\wedge\tau} b(X_s)ds$$
$$X'_{t\wedge\tau} = X_0 + \int_0^{t\wedge\tau} \sigma(X'_s)dB_s + \int_0^{t\wedge\tau} b(X'_s)ds.$$

Fix $T > 0$. If $t \in [0, T]$, then

$$\begin{aligned}
\mathbb{E}((X_{t\wedge\tau} - X'_{t\wedge\tau})^2) &\leq 2\mathbb{E}\left(\left(\int_0^{t\wedge\tau} (\sigma(X_s) - \sigma(X'_s))dB_s\right)^2\right) + 2\mathbb{E}\left(\left(\int_0^{t\wedge\tau} (b(X_s) - b(X'_s))ds\right)^2\right) \\
&\leq 2\mathbb{E}\left(\int_0^{t\wedge\tau} (\sigma(X_s) - \sigma(X'_s))^2 ds\right) + 2T\mathbb{E}\left(\int_0^{t\wedge\tau} (b(X_s) - b(X'_s))^2 ds\right) \\
&\qquad \text{(by Itô's isometry and Cauchy-Schwarz)} \\
&\leq 2K^2(1 + T)\mathbb{E}\left(\int_0^{t\wedge\tau} (X_s - X'_s)^2 ds\right) \\
&\leq 2K^2(1 + T)\int_0^t \mathbb{E}((X_{s\wedge\tau} - X'_{s\wedge\tau})^2)ds.
\end{aligned}$$

Let $f(t) = \mathbb{E}((X_{t\wedge\tau} - X'_{t\wedge\tau})^2)$. Then f is non-negative and bounded by $4M^2$, and we have shown that

$$f(t) \leq 2K^2(1 + T)\int_0^t f(s)ds \quad \text{for all} \quad t \in [0, T].$$

This is the right setting for applying Gronwall's lemma, which gives $f(t) = 0$ for all $t \in [0, T]$. This implies that $X_{t\wedge\tau} = X'_{t\wedge\tau}$ for all $t \in [0, T]$, almost surely. Since M and T were arbitrary, we conclude that $X_t = X'_t$ for all $t \geq 0$, almost surely.

Existence of a strong solution.

Fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ and a standard $(\mathcal{F}_t)$-Brownian motion B. Let $(\mathcal{F}_t^B)$ be the filtration generated by B, and observe that $\mathcal{F}_t^B \subseteq \mathcal{F}_t$ for all $t \geq 0$.

For any $T > 0$, we write $\mathcal{C}_T$ for the set of continuous, adapted processes $X\colon [0,T] \to \mathbb{R}$ such that $\|\!|X|\!\|_T := \| \sup_{t\leq T} |X_t| \|_2 < \infty$. Recall that $\mathcal{C}_T$ is complete. Write further $\mathcal{C}$ for the set of continuous, adapted processes $X\colon [0,T] \to \mathbb{R}$ such that $\|\!|X|\!\|_T < \infty$ for all $T > 0$.

For each $T > 0$, we construct a strong solution (on the space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ with $(\mathcal{F}_t)$-Brownian motion B) as the fixed point of a contraction map F from $\mathcal{C}_T$ to itself. We will then "glue" these solutions to obtain a solution defined on the whole $[0, \infty)$.

Fix $x \in \mathbb{R}$. Using the Lipschitz property of σ and b, we note that for all $y \in \mathbb{R}$ we have that

$$|\sigma(y)| = |\sigma(y) - \sigma(0) + \sigma(0)| \leq |\sigma(0)| + |\sigma(y) - \sigma(0)| \leq |\sigma(0)| + K|y|. \tag{7.4}$$

We similarly have that

$$|b(y)| \leq |b(0)| + K|y|. \tag{7.5}$$

Fix $T > 0$ and $X \in \mathcal{C}_T$. Let $M_t = \int_0^t \sigma(X_s) dB_s$, $0 \leq t \leq T$. Then $[M]_T = \int_0^T \sigma^2(X_s) ds$. Consequently, by (7.4) we have that

$$\mathbb{E}([M]_T) \leq 2T(|\sigma(0)|^2 + K^2 \|\!|X|\!\|_T^2) < \infty.$$

It thus follows that $(M_t)_{t\in[0,T]}$ is an L^2-bounded martingale. Doob's L^2 inequality then implies that

$$\mathbb{E}\left(\sup_{0\leq t\leq T} \left| \int_0^t \sigma(X_s) dB_s \right|^2 \right) \leq 8T \left(|\sigma(0)|^2 + K^2 \|\!|X|\!\|_T^2 \right) < \infty.$$

Using (7.5) and the Cauchy-Schwarz inequality, we also have that

$$\mathbb{E}\left(\sup_{0\leq t\leq T} \left| \int_0^t b(X_s) ds \right|^2 \right) \leq T\mathbb{E}\left(\int_0^T b^2(X_s) ds \right) \leq 2T^2 \left(|b(0)|^2 + K \|\!|X|\!\|_T^2 \right) < \infty.$$

Combining, we see that the map F defined on $\mathcal{C}_T$ by

$$F(X)_t = x + \int_0^t \sigma(X_s) dB_s + \int_0^t b(X_s) ds, \quad t \leq T$$

takes values in $\mathcal{C}_T$. Note that for $X, Y \in \mathcal{C}_T$, similar arguments to those above imply that

$$\begin{aligned} \|\!|F(X) - F(Y)|\!\|_t^2 &\leq 2K^2(4+T) \int_0^t \|\!|X - Y|\!\|_s^2 ds \\ &=: C_T \int_0^t \|\!|X - Y|\!\|_s^2 ds. \end{aligned}$$

Thus by iterating, we have that

$$\begin{aligned} \|\!| F^{(n)}(X) - F^{(n)}(Y) |\!\|_T^2 &\leq C_T^n \int_0^T \int_0^{t_1} \int_0^{t_2} \cdots \int_0^{t_{n-1}} \|\!| X - Y |\!\|_{t_n}^2 dt_1 dt_2 \dots dt_n \\ &\leq \frac{C_T^n T^n}{n!} \|\!| X - Y |\!\|_T^2. \end{aligned} \tag{7.6}$$

Note that we can make n sufficiently large so that $C_T^n T^n / n! < 1$. The contraction mapping theorem therefore implies that there exists a unique fixed point $X^{(T)} \in \mathcal{C}_T$ of F.

By pathwise uniqueness, we have that $X_t^{(T)} = X_t^{(T')}$ for all $t \leq T \wedge T'$, almost surely. We thus define X by setting $X_t = X_t^{(N)}$ whenever $t \leq N$ and $N \in \mathbb{N}$. Then X is the pathwise unique solution to the SDE started from x. To see that X is a strong solution, it remains to prove that it is $(\mathcal{F}_t^B)$-adapted. To this end, we show that for all $T \geq 0$ the process $(X_t)_{t \leq T}$ is a limit of $(\mathcal{F}_t^B)$-adapted processes.

Define a sequence $(Y^n)_{n \geq 0}$ in $\mathcal{C}_T$ by setting $Y^0 = x$ and $Y^n = F(Y^{n-1})$ for each $n \geq 1$. Then Y^n is $(\mathcal{F}_t^B)_{t \geq 0}$ adapted for all n. Since $X = F^{(n)}(X)$ for all $n \geq 0$, it follows from (7.6) that

$$\|\!| X - Y^n |\!\|_T^2 = \|\!| F^{(n)}(X) - F^{(n)}(x) |\!\|_T^2 \leq \frac{C_T^n T^n}{n!} \|\!| X - x |\!\|_T^2.$$

Thus $Y^n \to X$ in $\mathcal{C}_T$, and hence there exists a subsequence $(Y^{n_k})_{k \geq 1}$ such that $Y^{n_k} \to X$ uniformly on $[0, T]$, almost surely. In particular, $(X_t)_{t \leq T}$ is the almost sure limit of $(\mathcal{F}_t^B)$-adapted processes, and it is therefore $(\mathcal{F}_t^B)$-adapted. Since T was arbitrary, this shows that $X = (X_t)_{t \geq 0}$ is $(\mathcal{F}_t^B)$-adapted, and thus a strong solution. □

Remark 7.5. *Note that the above proof also shows that the pathwise unique strong solution to the SDE* (7.3) *belongs to $\mathcal{C}_T$ for all $T > 0$.*

Proposition 7.6. *Under the hypotheses of Theorem 7.2, there is uniqueness in law for the SDE $dX_t = \sigma(X_t) dB_t + b(X_t) dt$.*

Proof. See Example sheet 3. □

Example 7.7 (Ornstein-Uhlenbeck process)**.** Fix $\lambda \in \mathbb{R}$ and consider the SDE in $\mathbb{R}^2$

$$\begin{aligned} dV_t &= dB_t - \lambda V_t dt, \quad V_0 = v_0 \\ dX_t &= V_t dt, \quad X_0 = x_0. \end{aligned}$$

When $\lambda > 0$, this SDE models the movement of a pollen grain on the surface of a liquid. X represents the x-coordinate of the position and V is the velocity in the x-direction. The term $-\lambda V$ is the "damping" due to the viscosity of the liquid. Whenever $|V|$ gets large, the system acts to reduce $|V|$.

Theorem 7.2 implies that there exists a unique strong solution to this SDE started at each x_0, v_0. This is a rare example of an SDE that we can solve explicitly. By Itô's formula, we have that

$$d(e^{\lambda t}V_t) = e^{\lambda t}dV_t + \lambda e^{\lambda t}V_t dt = e^{\lambda t}dB_t$$

and so

$$e^{\lambda t}V_t = v_0 + \int_0^t e^{\lambda s}dB_s.$$

Rearranging, we find

$$V_t = e^{-\lambda t}v_0 + \int_0^t e^{\lambda(s-t)}dB_s.$$

Note that V_t has the normal distribution with mean $e^{-\lambda t}v_0$ and variance $(1-e^{-2\lambda t})/2\lambda$. If $\lambda > 0$, then $V_t \overset{(d)}{\to} \mathcal{N}(0,(2\lambda)^{-1})$ as $t \to \infty$. In particular, $\mathcal{N}(0,(2\lambda)^{-1})$ is the stationary distribution for V in the sense that if $V_0 \sim \mathcal{N}(0,(2\lambda)^{-1})$ then $V_t \sim \mathcal{N}(0,(2\lambda)^{-1})$ for all $t \geq 0$.

7.3. **Local solutions.** A *locally defined process* $(X,\mathcal{T})$ is a stopping time $\mathcal{T}$ together with a map

$$X\colon \{(\omega,t) \in \Omega \times [0,\infty) : t < \mathcal{T}(\omega)\} \to \mathbb{R}.$$

It is càdlàg if the map $t \mapsto X_t(\omega)$ from $[0,\mathcal{T}(\omega))$ to $\mathbb{R}$ is càdlàg for all $\omega \in \Omega$. Let

$$\Omega_t = \{\omega \in \Omega : t < \mathcal{T}(\omega)\}.$$

Then $(X,\mathcal{T})$ is *adapted* if $X_t\colon \Omega_t \to \mathbb{R}$ is $\mathcal{F}_t$-measurable for all t. We say that $(X,\mathcal{T})$ is a *locally defined local martingale* if there exist stopping times $T_n \nearrow \mathcal{T}$ such that X^{T_n} is a martingale for all n. We say that (H,η) is a *locally defined locally bounded previsible process* if there exist stopping times $S_n \nearrow \eta$ such that $H\mathbb{1}_{(0,S_n]}$ is bounded and previsible for all n. Then we can define $(H\cdot X, \mathcal{T}\wedge\eta)$ by setting

$$(H\cdot X)_t^{S_n\wedge T_n} := \big((H\mathbb{1}_{(0,S_n\wedge T_n]})\cdot X^{S_n\wedge T_n}\big)_t, \quad \text{for all } n.$$

Proposition 7.8 (Local Itô's formula)**.** *Let $X^1,\dots,X^d$ be continuous semimartingales and let $X = (X^1,\dots,X^d)$. Let $U \subseteq \mathbb{R}^d$ be open and let $f\colon U \to \mathbb{R}^d$ be C^2. Set $\mathcal{T} = \inf\{t \geq 0 : X_t \notin U\}$. Then for all $t < \mathcal{T}$, we have that*

$$f(X_t) = f(X_0) + \sum_{i=1}^d \int_0^t \frac{\partial f}{\partial x_i}(X_s)dX_s^i + \frac{1}{2}\sum_{i,j=1}^d \int_0^t \frac{\partial^2 f}{\partial x_i \partial x_j}(X_s)d[X^i,X^j]_s.$$

Proof. This follows by applying Itô's formula to X^{T_n} where $T_n = \inf\{t \geq 0 : \operatorname{dist}(X_t, U^c) \leq 1/n\}$, and observing that $T_n \nearrow \mathcal{T}$ as $n \to \infty$. □

Example 7.9. Let $X = B$ where B is a standard Brownian motion starting from 1 with $d = 1$, $U = (0, \infty)$, and $f(x) = \sqrt{x}$. Then Proposition 7.8 implies that

$$\sqrt{B_t} = 1 + \frac{1}{2}\int_0^t B_s^{-1/2} dB_s - \frac{1}{8}\int_0^t B_s^{-3/2} ds$$

for $t < \mathcal{T} = \inf\{t \geq 0 : B_t = 0\}$.

Suppose that $U \subseteq \mathbb{R}^d$ is open and $\sigma \colon U \to \mathscr{M}^{d\times m}(\mathbb{R})$ and $b \colon U \to \mathbb{R}^d$ are measurable functions which are bounded on compact sets. A *local solution* to the SDE consists of:

- A filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions.
- An $(\mathcal{F}_t)$-Brownian motion B in $\mathbb{R}^m$.
- A continuous $(\mathcal{F}_t)$-adapted locally defined process $(X, \mathcal{T})$ with $X \in \mathbb{R}^d$ such that $X_t = X_0 + \int_0^t \sigma(X_s) dB_s + \int_0^t b(X_s) ds$ for all $t < \mathcal{T}$.

We say that $(X, \mathcal{T})$ is *maximal* if for any other local solution (Y, η) on the same space such that $X_t = Y_t$ for all $t < \mathcal{T} \wedge \eta$, we have that $\eta \leq \mathcal{T}$.

7.4. **Locally Lipschitz coefficients.** Suppose that $U \subseteq \mathbb{R}^d$ is open. Then a function $f \colon U \to \mathbb{R}^d$ is called *locally Lipschitz* if for each compact set $C \subseteq U$ we have that $f|_C$ is Lipschitz.

Theorem 7.10. *Suppose that $U \subseteq \mathbb{R}^d$ is open and $\sigma \colon U \to \mathscr{M}^{d\times m}(\mathbb{R})$ and $b \colon U \to \mathbb{R}^d$ are locally Lipschitz. Then for all $x \in U$ the SDE $dX_t = \sigma(X_t) dB_t + b(X_t) dt$ has a pathwise unique maximal local solution $(X, \mathcal{T})$ started from x. Moreover, for all compact sets $C \subseteq U$, on the event $\{\mathcal{T} < \infty\}$ we have that $\sup\{t < \mathcal{T} : X_t \in C\} < \mathcal{T}$.*

Note that the last assertion tells us that if the solution blows up in finite time, then at least it will not return to any compact set infinitely often as it approaches that time.

Lemma 7.11. *Let $U \subseteq \mathbb{R}^d$ be open, $C \subseteq U$ compact. Then:*

(1) there exists a C^∞ function $\varphi \colon \mathbb{R}^d \to [0, 1]$ such that $\varphi|_C \equiv 1$ and $\varphi|_{U^c} \equiv 0$.
(2) given a locally Lipschitz function $f \colon U \to \mathbb{R}$, there exists a (globally) Lipschitz function $g \colon \mathbb{R}^d \to \mathbb{R}$ such that $f|_C = g|_C$.

Proof. We leave the first part as an exercise. For the second part, we take φ as in the first part and then set $g = f\varphi$. □

Proof of Theorem 7.10. We will assume for simplicity that $d = m = 1$. Fix $C \subseteq U$ compact. By Lemma 7.11 we can find (globally) Lipschitz functions $\widetilde{\sigma}$, $\widetilde{b}$ on $\mathbb{R}$ such that $\widetilde{\sigma}|_C = \sigma|_C$ and $\widetilde{b}|_C = b|_C$. By Theorem 7.2, there exists a pathwise unique strong solution $\widetilde{X}$ to the SDE

$$\begin{cases} d\widetilde{X}_t = \widetilde{\sigma}(\widetilde{X}_t) dB_t + \widetilde{b}(\widetilde{X}_t) dt, \\ \widetilde{X}_0 = x. \end{cases}$$

Set $T = \inf\{t \geq 0 : \widetilde{X}_t \notin C\}$ and denote by X the restriction of $\widetilde{X}$ to $[0, T)$. Then (X, T) is a local solution in C and if $T < \infty$ then X_{T^-} exists in $U \setminus C^\circ$. Suppose that (X, T) and (Y, S) are both local solutions in C. Consider

$$f(t) = \mathbb{E}\bigg(\sup_{s \leq S \wedge T \wedge t} |X_s - Y_s|^2 \bigg).$$

Then using that b and σ are Lipschitz in C and applying Gronwall's lemma, as in the proof of Theorem 7.2, one shows that $f \equiv 0$, and thus $X_t = Y_t$ for all $t \leq S \wedge T$, almost surely.

Now to build a local solution to the original SDE we approximate U by growing compact sets C_n and "glue" together the corresponding local solutions.

Take a sequence of compact sets $(C_n)_{n \geq 1}$ increasing to U and construct for each n a local solution (X^n, T_n) in C_n by the above procedure. Then if $T_n < \infty$, we have that $X^n_{T_n^-} \in U \setminus C_n^\circ$. Since the times T_n are increasing, we have $T_n \nearrow \mathcal{T} := \sup_n T_n$. Then we can define a local solution $(X, \mathcal{T})$ to the original SDE by setting $X_t = X^n_t$ whenever $t < T_n$, where the definition is well posed since $X^n_t = X^{n+1}_t$ for all $t < T_n$ and all $n \geq 1$.

We will now show that $(X, \mathcal{T})$ is a maximal solution. Suppose that (Y, η) is another local solution on the same probability space and, for each n, set[6] $S_n = \inf\{t \leq \eta : Y_t \notin C_n\} \wedge \eta$. Then by the uniqueness of the solution on each C_n, we have that $X_t = Y_t$ for all $t < T_n \wedge S_n$, and therefore $S_n \leq T_n$. As $n \to \infty$, we have that $S_n \nearrow \eta$ and $T_n \nearrow \mathcal{T}$, hence $\eta \leq \mathcal{T}$ and $X_t = Y_t$ for all $t \leq \eta$.

We now prove the final assertion of the theorem. Suppose that C_1, C_2 are compact sets in U with $C_1 \subseteq C_2^\circ \subseteq C_2 \subseteq U$ and let $\varphi \colon U \to \mathbb{R}$ be a C^∞ function with $\varphi|_{C_1} \equiv 1$ and $\varphi|_{(C_2^\circ)^c} \equiv 0$. We then set

$$\begin{aligned} R_0 &= \inf\{t \geq 0 : X_t \notin C_2\} \\ S_n &= \inf\{t \in [R_{n-1}, \mathcal{T}) : X_t \in C_1\} \wedge \mathcal{T} \\ R_n &= \inf\{t \in [S_n, \mathcal{T}) : X_t \notin C_2\} \wedge \mathcal{T}. \end{aligned}$$

Let N be the number of crossings that X makes from C_2 to C_1. On $\{\mathcal{T} \leq t, N \geq n\}$ we have that

$$\begin{aligned} -n &= \sum_{k=1}^n \big(\varphi(X_{R_k}) - \varphi(X_{S_k})\big) \\ &= \int_0^t \sum_{k=1}^n \mathbb{1}_{(S_k, R_k]}(s) \Big(\varphi'(X_s) dX_s + \frac{1}{2} \varphi''(X_s) d[X]_s \Big) \\ &= \int_0^t \big(H^n_s dB_s + K^n_s ds \big) := Z^n_t \end{aligned}$$

[6]Here we are using the convention that $\inf \emptyset = +\infty$.

where H^n and K^n are previsible and bounded uniformly in n. This implies that

$$n^2 \mathbb{1}_{\{\mathcal{T} \le t, N \ge n\}} \le (Z_t^n)^2$$

and hence

$$\mathbb{P}\big(\mathcal{T} \le t, N \ge n\big) \le \frac{1}{n^2} \mathbb{E}\big((Z_t^n)^2\big).$$

Note that $\mathbb{E}((Z_t^n)^2) \le C < \infty$ where C is independent of n as H^n and K^n are bounded uniformly in n and Z_t^n is defined by integrating H^n, K^n over a time-interval which does not depend on n. Thus $\mathbb{P}\big(\mathcal{T} \le t, N \ge n\big) \le C/n^2$, which is summable, so by Borel-Cantelli we see that $\mathbb{P}(\mathcal{T} \le t, N = \infty) = 0$. Therefore, on the event $\{\mathcal{T} < \infty\}$, X eventually stops coming back to C_1, which shows that $\sup\{t < \mathcal{T} : X_t \in C_1\} < \mathcal{T}$. □

Example 7.12 (Bessel processes)**.** Fix $\nu \in \mathbb{R}$. Consider the SDE in $U = (0, \infty)$ given by

$$dX_t = dB_t + \left(\frac{\nu - 1}{2X_t}\right) dt, \quad X_0 = x_0 \in U.$$

Theorem 7.10 implies that there exists a unique maximal local solution $(X, \mathcal{T})$ in U with $\mathcal{T} = \inf\{t \ge 0 : X_t = 0\}$. We call $(X, \mathcal{T})$ a *Bessel process of dimension* ν.

Suppose that $\nu \in \mathbb{N}$ and let β be a Brownian motion in $\mathbb{R}^\nu$ with $|\beta_0| = x_0 > 0$. Set $Y_t = |\beta_t|$ and $\eta = \inf\{t \ge 0 : \beta_t = 0\}$. By the local Itô formula, we have that

$$dY_t = \frac{(\beta_t, d\beta_t)}{|\beta_t|} + \left(\frac{\nu - 1}{2|\beta_t|}\right) dt, \quad t < \eta.$$

Consider $W_t = \int_0^t \frac{(\beta_s, d\beta_s)}{|\beta|_s} ds$ for $t \ge 0$. Then $W \in \mathcal{M}_{c,loc}$ and

$$d[W]_t = \frac{1}{|\beta_t|^2} \sum_{i,j=1}^{\nu} \beta_t^i \beta_t^j d[\beta^i, \beta^j]_t = dt.$$

Therefore, by Lévy's characterization of Brownian motion, we have that W is a standard Brownian motion in $\mathbb{R}$. It follows that Y solves the SDE

$$dY_t = dW_t + \left(\frac{\nu - 1}{2Y_t}\right) dt, \quad t < \eta.$$

Remark 7.13. *This shows that a Bessel process of dimension $\nu \in \mathbb{N}_+$ describes the evolution of the Euclidean norm of a ν-dimensional Brownian motion, up to the first time it hits* 0.

8. Diffusion processes

Suppose that we are given bounded, measurable functions $a \colon \mathbb{R}^d \to \mathscr{M}^{d \times d}(\mathbb{R})$ and $b \colon \mathbb{R}^d \to \mathbb{R}^d$ with a symmetric (i.e., each for each $x \in \mathbb{R}^d$ the matrix $a(x)$ satisfies $a(x) = (a(x))^T$). For $f \in C_b^2(\mathbb{R}^d)$ (i.e., the space of $C^2(\mathbb{R}^d)$ bounded functions with bounded derivatives), we set

$$Lf(x) = \frac{1}{2} \sum_{i,j=1}^{d} a_{ij}(x) \frac{\partial^2 f}{\partial x_i \partial x_j}(x) + \sum_{i=1}^{d} b_i(x) \frac{\partial f}{\partial x_i}(x). \tag{8.1}$$

Let X be a continuous, adapted process in $\mathbb{R}^d$. We say that X is an *L-diffusion* if for all $f \in C_b^2(\mathbb{R}^d)$, we have that

$$M_t^f := f(X_t) - f(X_0) - \int_0^t Lf(X_s)ds$$

is a martingale. We call a the *diffusivity* and b the *drift*. If a and b are clear from the context, we will sometimes refer to an L-diffusion as an (a, b)-diffusion.

Example 8.1. Suppose that σ and b are constant and $a = \sigma\sigma^T$. Let B be a standard Brownian motion in $\mathbb{R}^d$. Then the process $X_t = \sigma B_t + bt$ is an (a, b)-diffusion.

One special case of this is when $\sigma = I$ and $b = 0$. Then $X_t = B_t$ is an L-diffusion with $L = \frac{1}{2}\Delta$.

The next result tells us how to construct L-diffusions from SDEs.

Proposition 8.2. *Suppose that X is a solution to the SDE*

$$dX_t = \sigma(X_t)dB_t + b(X_t)dt. \tag{8.2}$$

Let $f \in C_b^{1,2}(\mathbb{R}_+ \times \mathbb{R}^d)$ (i.e., the space of bounded functions which are C^1 in the first variable and C^2 in the remaining d-variables, with bounded derivatives). Then the process

$$M_t^f = f(t, X_t) - f(0, X_0) - \int_0^t \left(\frac{\partial}{\partial s} + L\right) f(s, X_s)ds$$

is a continuous local martingale, where $a = \sigma\sigma^T$ and L is as in (8.1)*. In particular, if σ, b are bounded, then X is an L-diffusion.*

Proof. See Example Sheet 3. □

Proposition 8.2 tells us that, in particular, if X solves (8.2) with bounded and Lipschitz b, σ, then X is a $(\sigma\sigma^T, b)$-diffusion. This raises the question of which a can be written as $a = \sigma\sigma^T$ for σ as above.

We suppose that a, b are Lipschitz and bounded and that a is *uniformly positive definite* (UPD), that is there exists $\epsilon > 0$ such that

$$(\xi, a(x)\xi) \geq \epsilon^2|\xi|^2 \quad \text{for all } x, \xi \in \mathbb{R}^d. \tag{8.3}$$

Then there exists a Lipschitz map $\sigma\colon \mathbb{R}^d \to \mathbb{R}^d$ with $\sigma\sigma^T = a$. Indeed, if $d = 1$ then we can take $\sigma = \sqrt{a}$. For $d \geq 2$, we can write $a(x) = u(x)\lambda(x)(u(x))^T$ where $\lambda(x)$ is the diagonal matrix of eigenvalues of $a(x)$ and $u(x)$ is an orthogonal matrix having the corresponding eigenvectors as columns. So, in this case, we can take $\sigma(x) = u(x)\sqrt{\lambda(x)}(u(x))^T$. The fact that such σ is again Lipschitz follows from the differentiability of the square root function on the set of UPD matrices, and we take it for granted. For such b and σ, Theorem 7.2 implies that the SDE $dX_t = \sigma(X_t)dB_t + b(X_t)dt$ has a unique strong solution, which is then an (a, b)-diffusion. Thus

we can use SDEs to build (a, b) diffusions for all a, b Lipschitz, bounded and such that a is UPD.

The next result tells us that restarting an L-diffusion at a finite stopping time gives again an L-diffusion.

Proposition 8.3. *Let X be an L-diffusion and T a finite stopping time. Set $\widetilde{X}_t = X_{T+t}$ and $\widetilde{\mathcal{F}}_t = \mathcal{F}_{T+t}$. Then $\widetilde{X}$ is an L-diffusion with respect to $(\widetilde{\mathcal{F}}_t)_{t\geq 0}$.*

Proof. Consider for $f \in C_b^2(\mathbb{R}^d)$ the process

$$\widetilde{M}_t^f = f(\widetilde{X}_t) - f(\widetilde{X}_0) - \int_0^t Lf(\widetilde{X}_s)ds.$$

Then $\widetilde{M}_t^f$ is $\widetilde{\mathcal{F}}_t$-adapted and integrable. For $A \in \widetilde{\mathcal{F}}_s$ and $n \geq 0$, we note that

$$\begin{aligned}\mathbb{E}\big((\widetilde{M}_t^f - \widetilde{M}_s^f)\mathbb{1}_{A\cap\{T\leq n\}}\big) &= \mathbb{E}\big((M_{T+t}^f - M_{T+s}^f)\mathbb{1}_{A\cap\{T\leq n\}}\big)\\ &= \mathbb{E}\big((M_{(T\wedge n)+t}^f - M_{(T\wedge n)+s}^f)\mathbb{1}_A\big) = 0\end{aligned}$$

by the OST. Letting $n \to \infty$ and using the Dominated Convergence Theorem, we see that $\widetilde{M}^f$ is a martingale. □

Lemma 8.4. *Let X be an L-diffusion. Then for all $f \in C_b^{1,2}(\mathbb{R}_+ \times \mathbb{R}^d)$, the process*

$$M_t^f = f(t, X_t) - f(0, X_0) - \int_0^t \left(\frac{\partial}{\partial s} + L\right) f(s, X_s)ds$$

is a martingale.

Proof. We fix $T > 0$ and consider

$$Z_n = \sup_{\substack{0\leq s\leq t\leq T\\ t-s\leq 1/n}} |\dot{f}(s, X_t) - \dot{f}(s, X_s)| + \sup_{\substack{0\leq s\leq t\leq T\\ t-s\leq 1/n}} |Lf(s, X_t) - Lf(t, X_t)|.$$

Then Z_n is bounded and $Z_n \to 0$ as $n \to \infty$ by uniform continuity of continuous functions on compact sets. Consequently, $\mathbb{E}(Z_n) \to 0$ as $n \to \infty$ by the Bounded Convergence Theorem. We note that:

$$\begin{aligned}M_t^f - M_s^f = &\left(f(t, X_t) - f(s, X_t) - \int_s^t \dot{f}(r, X_t)dr\right) +\\ &\left(f(s, X_t) - f(s, X_s) - \int_s^t Lf(s, X_r)dr\right) +\\ &\int_s^t \big(\dot{f}(r, X_t) - \dot{f}(r, X_r)\big)dr + \int_s^t \big(Lf(s, X_r) - Lf(r, X_r)\big)dr.\end{aligned}$$

Choose $s_0 \le s_1 \le \cdots \le s_m$ with $s_0 = s$ and $s_m = t$ and $s_{k+1} - s_k \le 1/n$. As the first line of our formula for $M_t^f - M_s^f$ is zero and the second line has zero conditional expectation given $\mathcal{F}_s$ (as X is an L-diffusion), it follows that

$$\mathbb{E}\left(\left|\mathbb{E}\big(M^f_{s_{k+1}} - M^f_{s_k} \,|\, \mathcal{F}_s\big)\right|\right) \le (s_{k+1} - s_k)\,\mathbb{E}(Z_n).$$

Combining this with the triangle inequality, we thus see that

$$\mathbb{E}\left(\left|\mathbb{E}\big(M^f_{t} - M^f_{s} \,|\, \mathcal{F}_s\big)\right|\right) \le (t - s)\,\mathbb{E}(Z_n).$$

As the right hand side goes to zero as $n \to \infty$, it follows that $\mathbb{E}(M_t^f \,|\, \mathcal{F}_s) = M_s^f$, as desired. □

8.1. **Dirichlet and Cauchy problems.** Assume that a, b are Lipschitz and that a is UPD, i.e. $(\xi, a(x)\xi) \ge \epsilon|\xi|^2$ for all $x, \xi \in \mathbb{R}^d$. Let D be an bounded, open subset of $\mathbb{R}^d$ with smooth boundary ∂D.

8.1.1. *Dirichlet problems.* We shall assume the following result from the theory of PDEs.

Theorem 8.5 (Dirichlet problem)**.** *For all $f \in C(\partial D)$, $\varphi \in C(\overline{D})$, there exists a unique function $u \in C(\overline{D}) \cap C^2(D)$ such that*

$$\begin{cases} Lu + \varphi = 0 & \text{on } D \\ u = f & \text{on } \partial D. \end{cases}$$

Moreover, there exist continuous functions $m\colon D \times \partial D \to (0, \infty)$ and $g\colon \{(x, y) \in D \times D : x \ne y\} \to (0, \infty)$ such that for all f and φ as above we have that

$$u(x) = \int_D g(x, y)\varphi(y)dy + \int_{\partial D} f(y)m(x, y)\lambda(dy). \tag{8.4}$$

We call g the *Green kernel* and $m(x, y)\lambda(dy)$ the *harmonic measure* on ∂D starting from x.

Theorem 8.6. *Suppose that $u \in C(\overline{D}) \cap C^2(D)$ satisfies*

$$\begin{cases} Lu + \varphi = 0 & \text{on } D \\ u = f & \text{on } \partial D, \end{cases}$$

with $f \in C(\partial D)$, $\varphi \in C(\overline{D})$. Then for any L-diffusion X starting from $x \in D$, we have that

$$u(x) = \mathbb{E}_x\left(\int_0^T \varphi(X_s)ds + f(X_T)\right)$$

where $T = \inf\{t \ge 0 : X_t \notin D\}$. In particular, for all Borel sets $A \subseteq D$ and $B \subseteq \partial D$, we have that

$$\mathbb{E}_x\left(\int_0^T \mathbb{1}(X_s \in A)ds\right) = \int_A g(x, y)dy \tag{8.5}$$

$$\mathbb{P}_x(X_T \in B) = \int_B m(x, y)\lambda(dy). \tag{8.6}$$

Proof. Fix $n \geq 1$ and set $T_n = \inf\{t \geq 0 : X_t \notin D_n\}$ where $D_n = \{x \in D : \operatorname{dist}(x, D^c) > 1/n\}$ and consider the process

$$M_t = u(X_{t\wedge T_n}) - u(X_0) + \int_0^{t\wedge T_n} \varphi(X_s)ds.$$

There exists $\widetilde{u} \in C_b^2(\mathbb{R}^d)$ with $u = \widetilde{u}$ on D_n. Then $M = (\tilde{M})^{T_n}$ where

$$\tilde{M}_t = \widetilde{u}(X_t) - \widetilde{u}(X_0) - \int_0^t L\widetilde{u}(X_s)ds.$$

Now, since X is an L-diffusion we have that $M^{\tilde{u}}$ is a martingale, so by the OST we conclude that M is a martingale. Hence,

$$u(x) = \mathbb{E}_x\left(u(X_{t\wedge T_n}) + \int_0^{t\wedge T_n} \varphi(X_s)ds\right). \tag{8.7}$$

In order to send $n \to \infty$ in both terms, we first show that $\mathbb{E}_x(T) < \infty$. To this end, take $\varphi \equiv 1$ and $f \equiv 0$, and let $u^{1,0}$ denote the solution to the associated Dirichlet problem. Then (8.7) holds for $u^{1,0}$, and so

$$\mathbb{E}_x(T_n \wedge t) = u^{1,0}(x) - \mathbb{E}_x(u^{1,0}(X_{t\wedge T_n})).$$

Since $u^{1,0}$ is bounded and $T_n \nearrow T$, we deduce by the Monotone Convergence Theorem (with $t = n \to \infty$) that $\mathbb{E}_x(T) < \infty$.

Returning to the general case, we can let $t \to \infty$ and $n \to \infty$ in (8.7). Then $t \wedge T_n \nearrow T$ almost surely. Since u is continuous on $\overline{D}$, we have that $u(X_{t\wedge T_n}) \to f(X_T)$. Finally, u is also bounded on $\overline{D}$ since continuous function on a compact set, so by the Bounded Convergence Theorem we find

$$\mathbb{E}_x(u(X_{t\wedge T_n})) \to \mathbb{E}_x(f(X_T))$$

as $t, n \to \infty$. Moreover,

$$\mathbb{E}_x\left(\int_0^T |\varphi(X_s)|ds\right) \leq \|\varphi\|_\infty \mathbb{E}_x(T) < \infty.$$

Thus by the Dominated Convergence Theorem we find

$$\mathbb{E}_x\left(\int_0^{t\wedge T_n} \varphi(X_s)ds\right) \to \mathbb{E}_x\left(\int_0^T \varphi(X_s)ds\right).$$

Consequently,

$$u(x) = \mathbb{E}_x\left(f(X_T) + \int_0^T \varphi(X_s)ds\right).$$

Finally, combining this with (8.4) we have shown that for all $\varphi \in C(\overline{D})$ and $f \in C(\partial D)$ it holds

$$\mathbb{E}_x\left(f(X_T) + \int_0^T \varphi(X_s)ds\right) = \int_{\partial D} m(x,y)f(y)\lambda(dy) + \int_D g(x,y)\varphi(y)dy.$$

Taking $f \equiv 0$ and $\varphi_n \to \mathbb{1}_A$ with $\varphi_n \in C(\overline{D})$ we get (8.5). Similarly, taking $\varphi \equiv 0$ and $f_n \to \mathbb{1}_B$ with $f_n \in C(\partial D)$ yields (8.6). □

8.1.2. *Cauchy problems.* We shall assume the following result from the theory of PDEs.

Theorem 8.7 (Cauchy problem)**.** *For each $f \in C_b^2(\mathbb{R}^d)$, there exists a unique $u \in C_b^{1,2}(\mathbb{R}_+ \times \mathbb{R}^d)$ such that*

$$\begin{cases} \dfrac{\partial u}{\partial t} = Lu \quad \text{on } \mathbb{R}_+ \times \mathbb{R}^d \\ u(0,\cdot) = f \quad \text{on } \mathbb{R}^d. \end{cases}$$

Moreover, there exists a continuous function $p\colon (0,\infty) \times \mathbb{R}^d \times \mathbb{R}^d \to (0,\infty)$ such that

$$u(t,x) = \int_{\mathbb{R}^d} p(t,x,y) f(y) dy$$

for all $(t,x) \in \mathbb{R}_+ \times \mathbb{R}^d$.

In the setting of Theorem 8.7, the function p is called the *heat kernel*.

Theorem 8.8. *Assume that $f \in C_b^2(\mathbb{R}^d)$. Let $u \in C_b^{1,2}(\mathbb{R}_+ \times \mathbb{R}^d)$ satisfy*

$$\begin{cases} \dfrac{\partial u}{\partial t} = Lu \quad \text{on } \mathbb{R}_+ \times \mathbb{R}^d \\ u(0,\cdot) = f \quad \text{on } \mathbb{R}^d. \end{cases}$$

Then, for any L-diffusion X, for all $t \in \mathbb{R}_+$, $x \in \mathbb{R}^d$, $s \le t$, we have that $\mathbb{E}(f(X_t) \,|\, \mathcal{F}_s) = u(t-s, X_s)$ almost surely. In particular,

$$\mathbb{E}_x(f(X_t)) = u(t,x) = \int_{\mathbb{R}^d} p(t,x,y) f(y) dy.$$

Moreover, under $\mathbb{P}_x$ the finite dimensional distributions of X are given by

$$\mathbb{P}_x(X_{t_1} \in dx_1, \dots, X_{t_n} \in dx_n) = p(t_1, x_0, x_1) \cdots p(t_n - t_{n-1}, X_{t_{n-1}}, X_{t_n}) dx_1 \cdots dx_n$$

for $0 < t_1 < \cdots < t_n$ and $x_1, \dots, x_n \in \mathbb{R}^d$.

Remark 8.9. *In the language of Markov processes, the above theorem tells us that X is a Markov process with transition density function p.*

Proof. Fix $t \in (0,\infty)$ and consider the function $g(s,x) = u(t-s,x)$ for $s \le t$. Note that

$$\left(\frac{\partial}{\partial s} + L\right) g(s,x) = -\dot{u}(t-s,x) + Lu(t-s,x) = 0.$$

Consequently, $M_s^g = g(s, X_s) - g(0, X_0)$ is a martingale. This implies that $\mathbb{E}(M_t^g | \mathcal{F}_s) = M_s^g$ almost surely, i.e. $\mathbb{E}(f(X_t) \,|\, \mathcal{F}_s) = u(t-s, X_s)$. Taking $s = 0$ and $X_0 = x$ implies that $\mathbb{E}(f(X_t)) = u(t,x)$. This proves the first part of the theorem.

For the second part of the theorem, we set

$$P_t f(x) = \int_{\mathbb{R}^d} p(t,x,y) f(y) dy = u(t,x).$$

By the uniqueness of solutions to the Cauchy problem, we have that $P_s(P_t f) = P_{s+t} f$. We claim by induction on n that

$$\mathbb{E}_{x_0}\left(\prod_{i=1}^n f_i(X_{t_i})\right) = \int_{(\mathbb{R}^d)^n} p(t_1, x_0, x_1) f_1(x_1) \cdots p(t_n - t_{n-1}, x_{n-1}, x_n) f_n(x_n) dx_1 \cdots dx_n.$$

For the induction step, we use that

$$\begin{aligned}\mathbb{E}_{x_0}\left(\prod_{i=1}^n f_i(X_{t_i}) \,\middle|\, \mathcal{F}_{t_{n-1}}\right) &= \left(\prod_{i=1}^{n-1} f_i(X_{t_i})\right) \mathbb{E}(f_n(X_{t_n}) \,|\, \mathcal{F}_{t_{n-1}}) \\ &= \left(\prod_{i=1}^{n-1} f_i(X_{t_i})\right) P_{t_n - t_{n-1}} f_n(X_{t_{n-1}}),\end{aligned}$$

and then apply the $n-1$ case. □

Theorem 8.10 (Feynman-Kac)**.** *Let $f \in C_b^2(\mathbb{R}^d)$ and $V \in C_b(\mathbb{R}^d)$. Suppose that $u \in C_b^{1,2}(\mathbb{R}_+ \times \mathbb{R}^d)$ satisfies the PDE*

$$\frac{\partial u}{\partial t} = \frac{1}{2}\Delta u + Vu \quad \text{on} \quad \mathbb{R}_+ \times \mathbb{R}^d \quad \text{and} \quad u(0,\cdot) = f \quad \text{on} \quad \mathbb{R}^d.$$

Then for all $t \in \mathbb{R}_+$, $x \in \mathbb{R}^d$ we have that

$$u(t,x) = \mathbb{E}_x\left(f(B_t) \exp\left(\int_0^t V(B_s) ds\right)\right)$$

where B is a standard Brownian motion.

Proof. Let

$$E_t = \exp\left(\int_0^t V(B_s) ds\right).$$

Fix $T \in (0,\infty)$ and set $M_t = u(T-t, B_t) E_t$. By Itô's formula, we have that

$$dM_t = E_t \sum_{i=1}^d \frac{\partial u}{\partial x_i}(T-t, B_t) dB_t^i + \underbrace{\left(\frac{1}{2}\Delta u(T-t, B_t) - \dot{u}(T-t, B_t) + u(T-t, B_t) V(B_t)\right)}_{=0} E_t dt.$$

So, M is a local martingale which is uniformly bounded on $[0,T]$ as $u \in C_b^{1,2}(\mathbb{R}_+ \times \mathbb{R}^d)$ and V is bounded on $[0,T]$. Therefore M is a martingale. Hence,

$$u(T,x) = M_0 = \mathbb{E}_x(M_T) = \mathbb{E}_x(f(B_T) E_T),$$

as desired. □

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